



北京邮电大学

Beijing University of Posts and Telecommunications

# Discrete Mathematics and Its Applications

Eighth Edition

Kenneth H. Rosen

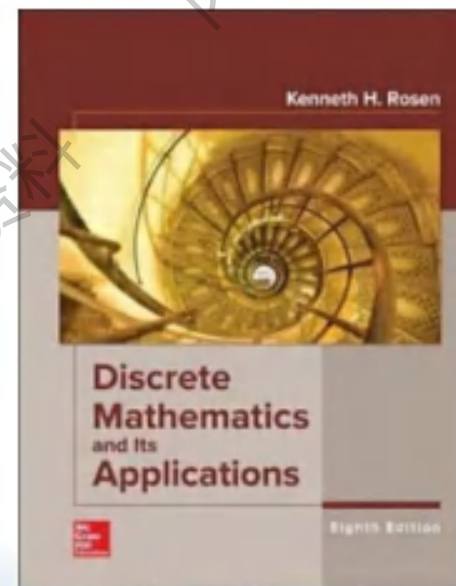
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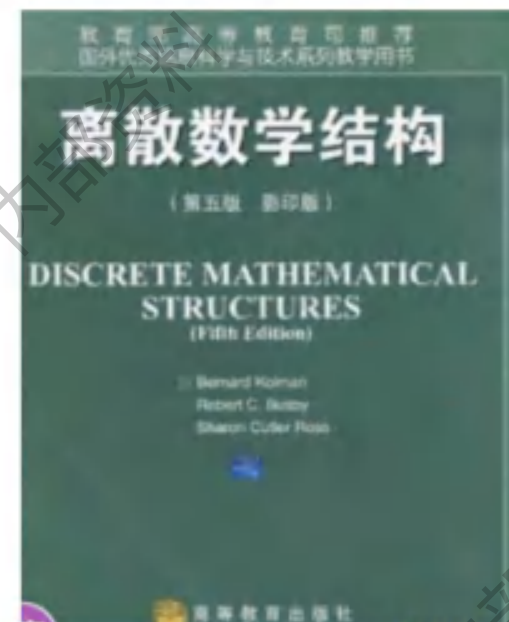
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# 本学期教学安排

- 9 Relations
- DMS-9 Semigroups and Groups P319-356
- DMS-11 Groups and Coding P401-423
- 8 Advanced Counting Techniques
- 10 Graphs
- 11 Trees



# Relations 关系

Rosen 8<sup>th</sup> ed., ch. 9



# 9 Relations

- 9.1 Relations and Their Properties

关系及关系性质

- 9.2  $n$ -ary Relations and Their Applications

$n$ 元关系及应用

- 9.3 Representing Relations

关系的表示

- 9.4 Closures of Relations

关系闭包

- 9.5 Equivalence Relations

等价关系

- 9.6 Partial Orderings

偏序关系



# 9.1 Relations and Their Properties

## 关系及关系性质

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- Introduction
- Functions as relations
- Relations on Set
- Properties of Relations
- Combining Relations

# Ordered pair 有序对

- An **ordered pair** (有序对或序偶)  $(a, b)$  is a listing of the objects  $a$  and  $b$  in a prescribed order.

第1元素

第2元素

- $(a, b) \neq (b, a) \quad (a \neq b)$
- $(a, b) = (u, v) \Leftrightarrow a = u \wedge b = v$

例: 点的直角坐标  $(3, 4) \neq (4, 3)$

# Cartesian product

## 笛卡尔积

- If  $A$  and  $B$  are two nonempty sets, we define the *product set* or *Cartesian product*  $A \times B$  as the set of all ordered pairs  $(a, b)$  with  $a \in A$  and  $b \in B$ 
  - $A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}$

E.g.,  $A = \{a, b, c, d\}$ ,  $B = \{1, 2, 3\}$

$$A \times B = \{(a, 1), (a, 2), (a, 3), \\ (b, 1), (b, 2), (b, 3), \\ (c, 1), (c, 2), (c, 3), \\ (d, 1), (d, 2), (d, 3)\}$$

$$B \times A = \{(1, a), (1, b), (1, c), (1, d), \\ (2, a), (2, b), (2, c), (2, d), \\ (3, a), (3, b), (3, c), (3, d)\}$$



# Definition of relations

- Let  $A$  and  $B$  be two sets. A **binary relation  $R$  from  $A$  to  $B$**  is a subset of  $A \times B$ .
- If  $R \subseteq A \times B$  and  $(a, b) \in R$ , we say that  $a$  is related to  $b$  by  $R$
- We use the notation  $a R b$  to denote that  $(a, b) \in R$  and  $a \not R b$  to denote that  $(a, b) \notin R$ , when  $(a, b)$  belongs to  $R$ ,  $a$  is said to be related to  $b$  by  $R$ .

例, 设  $A$  是城市的集合,  $B$  是省份的集合,  $A = \{\text{杭州, 石家庄, 南京}\}$ ,  $B = \{\text{河北, 浙江}\}$ ;

定义关系  $R$ :  $a$  是  $b$  的省会, 则  $(a, b) \in R$ , 或  $a R b$

- $A \times B = \{(\text{杭州, 河北}), (\text{杭州, 浙江}), (\text{石家庄, 河北}), (\text{石家庄, 浙江}), (\text{南京, 河北}), (\text{南京, 浙江})\}$
- $R = \{$  ,  $\}$

# Definition of relations

---

- More generally, let  $A_1, A_2, \dots, A_n$  be  $n$  sets. An  $n$ -ary relation  $R$  on these sets is a subset of  $A_1 \times A_2 \times \dots \times A_n$ .
  - The sets  $A_i$  are known as the *domains* of the relation, and  $n$  as its *degree*.

# Inverse Relations

## 逆关系

- Any binary relation  $R$  from  $A$  to  $B$  has an *inverse* relation  $R^{-1}$ , defined by

$$R^{-1} = \{(b, a) \mid (a, b) \in R \wedge a \in A \wedge b \in B\}.$$

E.g.,  $R = \{(a, b) \mid a < b\} \Rightarrow R^{-1} = \{(b, a) \mid a < b\}$

$$R^{-1} = \{(a, b) \mid a > b\}.$$

- E.g., if  $R: \text{People} \rightarrow \text{Foods}$  is defined by  $a R b \Leftrightarrow a \text{ eats } b$ , then:
- $b R^{-1} a \Leftrightarrow b \text{ is eaten by } a$ . (Passive voice.)



# Functions as Relations

## 函数作为关系

- a function  $f: A \rightarrow B$  is a special case of a relation from  $A$  to  $B$ 
  - It satisfies the property: for each  $x \in A$ , there is exactly one pair  $(x, f(x))$  in the relation
- Relations are a generalization of functions
  - They allow unmapped elements from  $A$
  - They allow one-to-many mappings where an element from  $A$  maps to multiple elements from  $B$
  - They also allow many-to-one and many-to-many mappings

*E.g.* :  $R = \{(a, 1), (a, 2)\}$ .



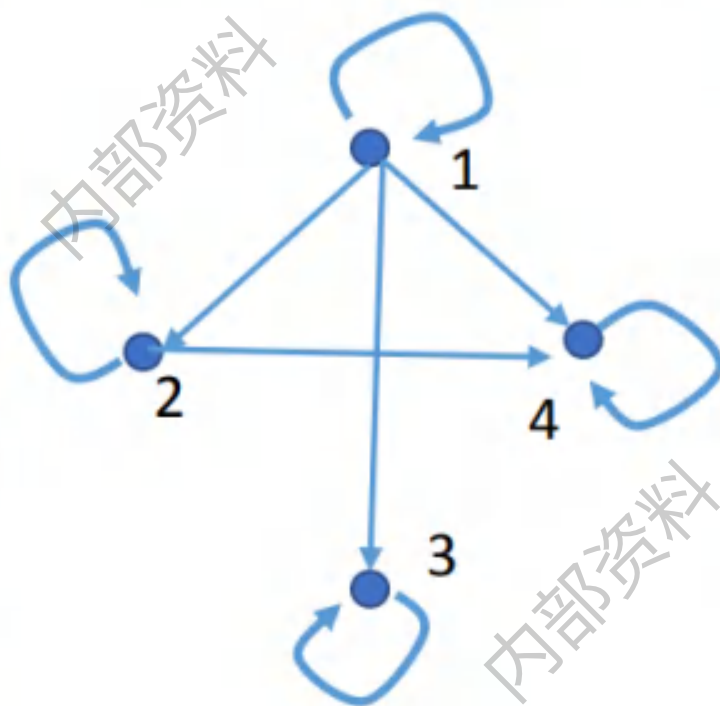
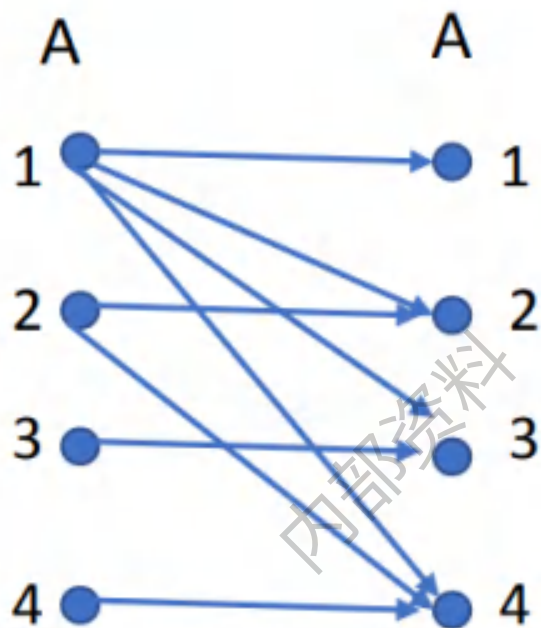
# Relations on a set 集合的关系

- **DEFINITION 2:** *A relation  $R$  on a set  $A$  is a relation from  $A$  to  $A$ , that is, a subset of  $A^2 = A \times A$ .*

# Example 4

Let  $A$  be the set  $\{1,2,3,4\}$ , Which ordered pairs are in the relation  $R = \{(a,b) \mid a \text{ divides } b\}$ ?

- Solution:  $R = \{(1,1), (2,2), (3,3), (4,4), (1,2), (1,3), (1,4), (2,4)\}$



| R | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | x | x | x | x |
| 2 |   | x |   | x |
| 3 |   |   | x |   |
| 4 |   |   |   | x |

## Example 5

Consider these relations on the set of integers :

$$R_1 = \{(a, b) \mid a \leq b\}$$

$$R_2 = \{(a, b) \mid a > b\}$$

$$R_3 = \{(a, b) \mid a = b \text{ 或 } a = -b\}$$

$$R_4 = \{(a, b) \mid a = b\}$$

$$R_5 = \{(a, b) \mid a = b + 1\}$$

$$R_6 = \{(a, b) \mid a + b \leq 3\}$$

Which of these relations contain each of the pairs  
(1, 2)、(2, 2)、(1, 1)、(2, 1) and (1,-1)?

Solution: (1, 2) ---  $R_1, R_6$ ; (2, 1) ---  $R_2, R_5, R_6$ ;

(2, 2) ---  $R_1, R_3, R_4$ ; (1, -1) ---  $R_2, R_3, R_6$ 。

(1, 1) ---  $R_1, R_3, R_4, R_6$ ;



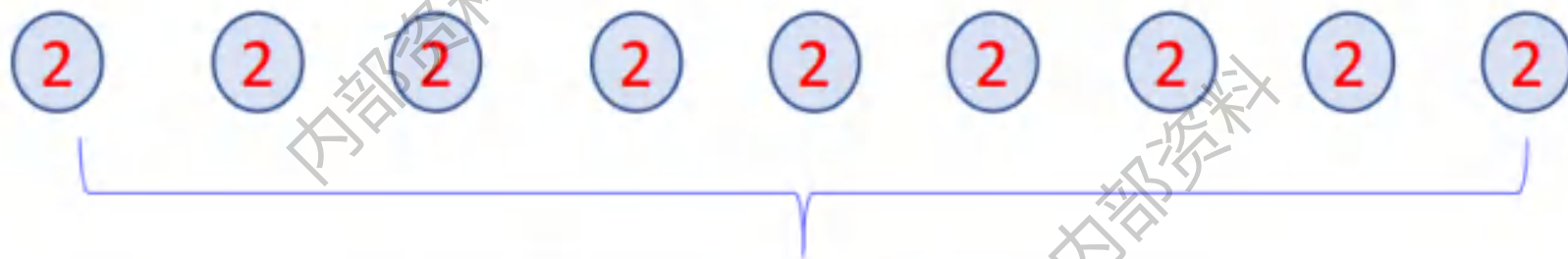
# How many relations

example:  $A = \{a, b, c\}$ , How many relations?

$3^2 \uparrow$   $P(A \times A)$  ?

$3 \times 3$

•  $A \times A = \{(a,a), (a,b), (a,c), (b,a), (b,b), (b,c), (c,a), (c,b), (c,c)\}$



$$P(A \times A): 2^{3^2} = 512$$

## Example 6

How many relations are there on a set with  $n$  elements ?

- $A^2$  has  $n^2$  elements. We know that the number of subsets of a set with  $m$  elements is  $2^m$ , so the desired number is  $2^{n^2}$ .

# 几种特殊的关系

- 假设任意非空集合 $A$ , 可以定义集合 $A$ 上的:
  - 空关系:  $\emptyset$
  - 恒等关系:  $I_A = \{(a, a) \mid \forall a \in A\}$
  - 全域关系:  $U_A = A \times A = \{(a, b) \mid \forall a \forall b (a \in A \wedge b \in A)\}$



# Properties of Relations

## 关系的性质

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- *Reflexive* (自反性)
- *Irreflexive* (反自反性)
- *Symmetric* (对称性)
- *Antisymmetric* (反对称性)
- *Transitive* (传递性)

# reflexive 自反性

- DEFINITION 3 : A relation  $R$  on  $A$  is *reflexive* if  $\forall a \in A, aRa$ .
- Example 8, Which of the relations from Example 5 are reflexive?

$$R_1 = \{(a, b) \mid a \leq b\}$$

$$R_2 = \{(a, b) \mid a = b \text{ 或 } a = -b\}$$

$$R_3 = \{(a, b) \mid a = b\}$$

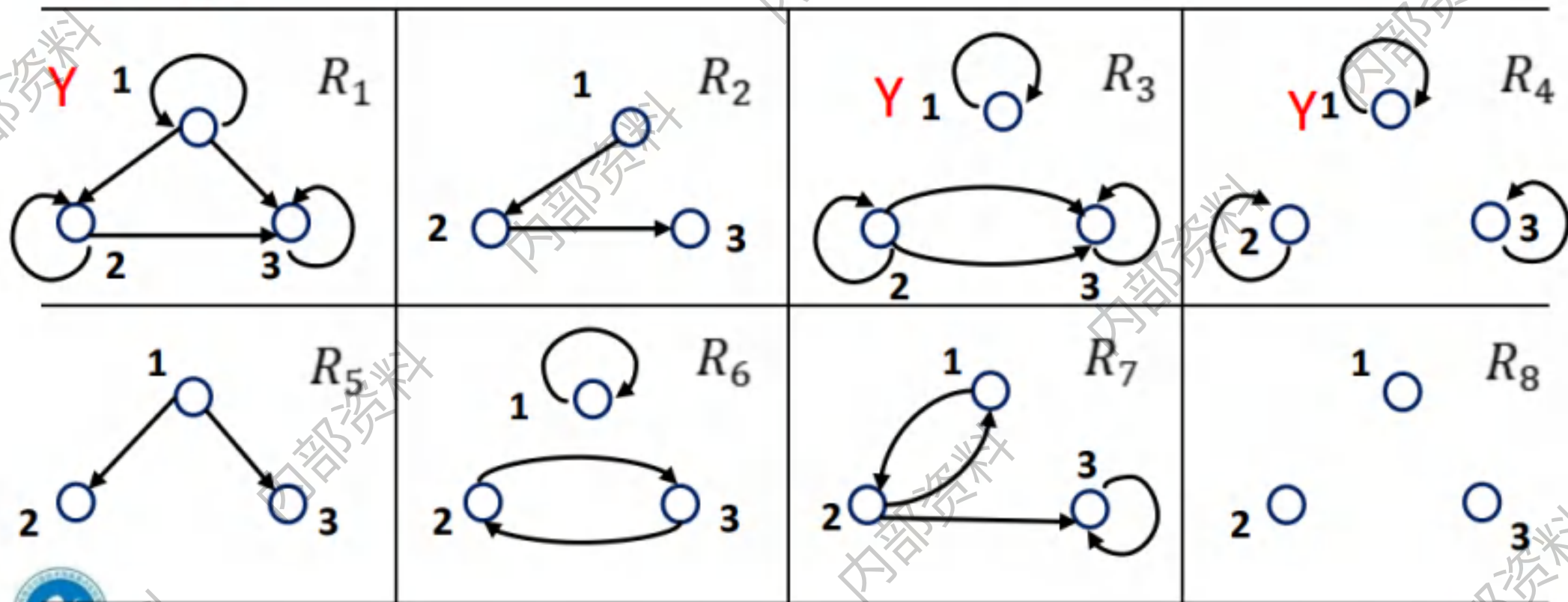
$$R_4 = \{(a, b) \mid a > b\}$$

$$R_5 = \{(a, b) \mid a = b + 1\}$$

$$R_6 = \{(a, b) \mid a + b \leq 3\}$$

# 实例

例:  $A = \{ 1, 2, 3 \}$ , 下面8个关系哪些是自反的  
 $R$  是  $A$  上的关系, 则  $R$  的有向图中,  **$A$  每个元素都有指向自己的环**



# irreflexive 反自反性

- **DEFINITION 3 :** A relation  $R$  on  $A$  is irreflexive if  $\forall a \in A, (a, a) \notin R$ .
- **Example:** Which of the relations are irreflexive?

$$R_1 = \{(a, b) \mid a < b\}$$

$$R_2 = \{(a, b) \mid a \neq b\}$$

\*一个不是自反的关系，不一定是反自反的



# 实例

例,  $A=\{1,2,3\}$ ,  $R_1, R_2, R_3$ 是 $A$ 上的关系, 其中

$$R_1=\{(1,1),(2,2)\}$$

——非自反, 非反自反

$$R_2=\{(1,1),(2,2),(3,3),(1,2)\}$$

——自反

$$R_3=\{(1,3)\}$$

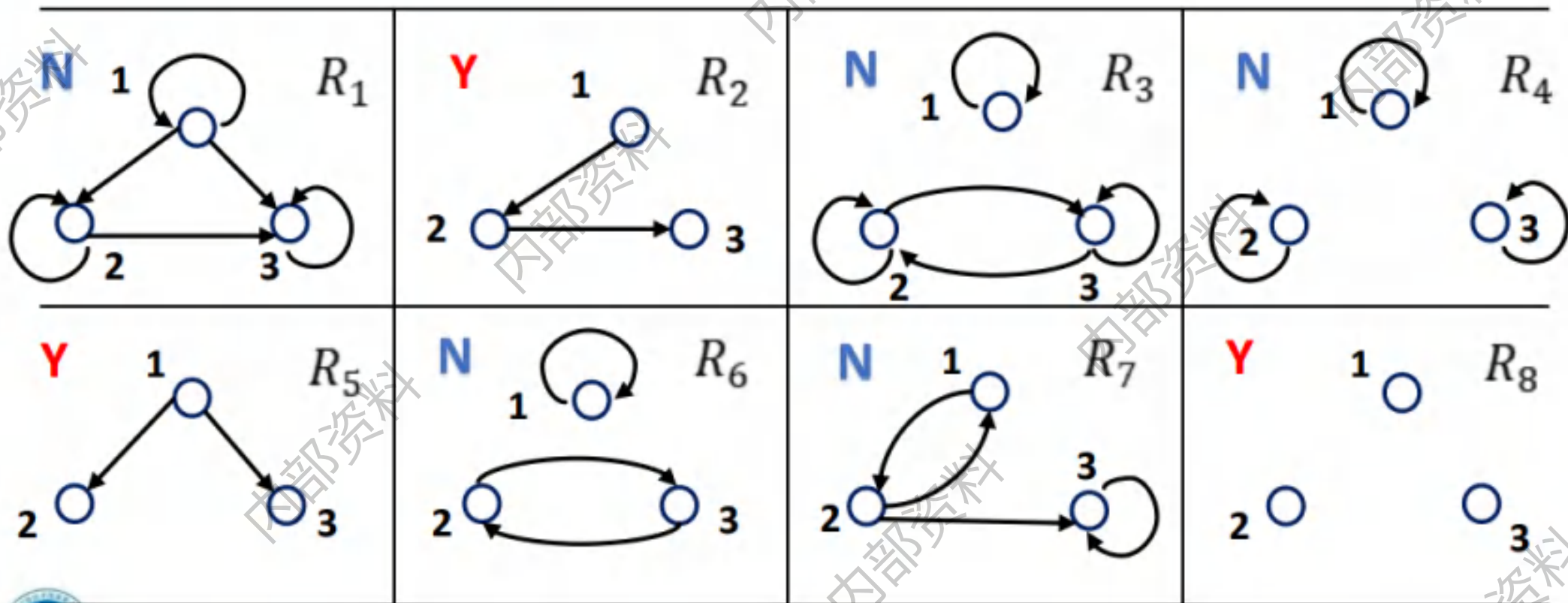
——反自反

哪些自反? 哪些反自反?

多少个反自反?

# 实例

例:  $A=\{1, 2, 3\}$ , 下面8个关系哪些是反自反的  
 $A$ 上的每个元素都无环



# Symmetric对称性

- A binary relation  $R$  on  $A$  is *symmetric* iff  $R = R^{-1}$ , that is, if  $(a, b) \in R \iff (b, a) \in R$ .
- **Example:** Which of the relations are symmetric?

*Symmetric ? :*

$$R_1 = \{(a, b) \mid a + b \leq 3\}$$

$$R_2 = \{(a, b) \mid a = b \text{ 或 } a = -b\}$$

$$R_3 = \{(a, b) \mid a = b\}$$

$$R_4 = \{(a, b) \mid a > b\}$$

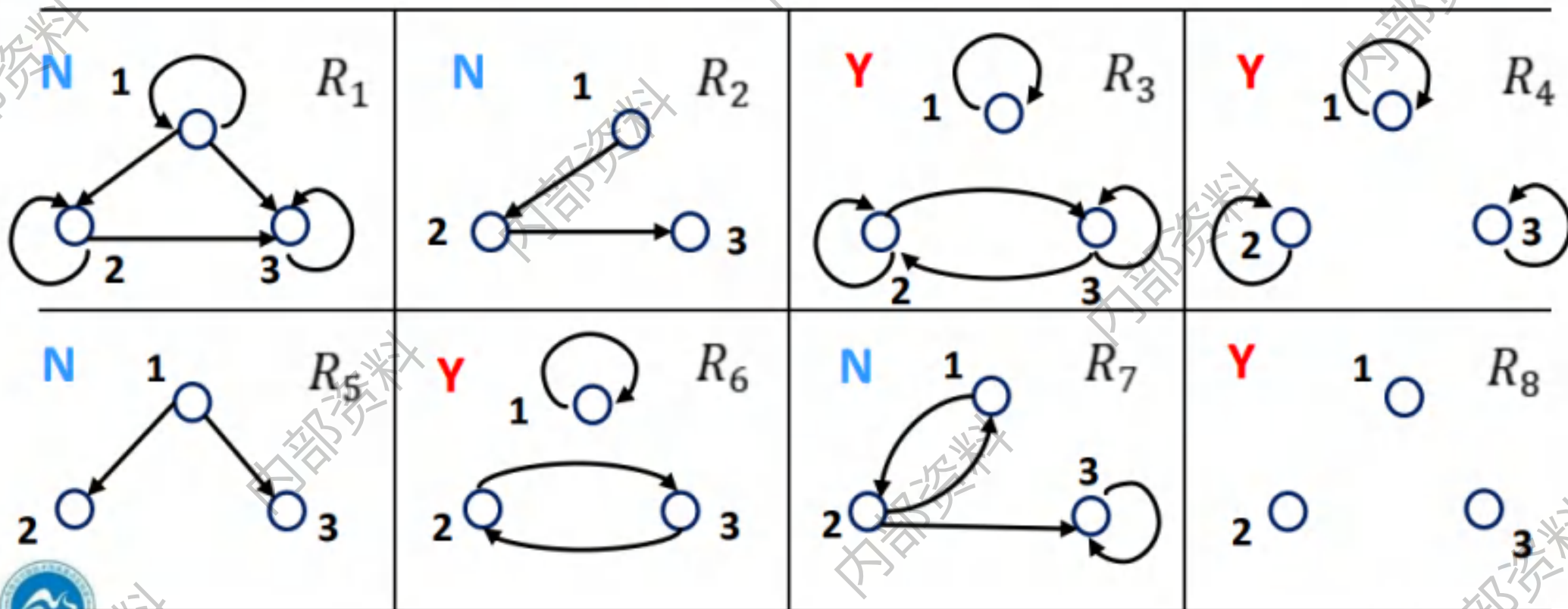
$$R_5 = \{(a, b) \mid a \leq b\}$$

$$R_6 = \{(a, b) \mid a = b + 1\}$$

关系 $R$ 与逆关系 $R^{-1}$  的并?

# Example

对称性，例：  $A=\{1, 2, 3\}$ ，下面8个关系哪些是对称的  
 $R$ 是 $A$ 上的关系，两个不同的节点间，可以无边，如果有边，必须成对出现，且方向相反





# Antisymmetric 反对称性

- **Definition 4:** A relation  $R$  on a set  $A$  such that for all  $a, b \in A$ , if  $(a, b) \in R$  and  $(b, a) \in R$ , then  $a = b$  is called antisymmetric.
- **Example:** Which of the relations are antisymmetric?

antisymmetric

$$R_1 = \{(a, b) \mid a \leq b\}$$

$$R_2 = \{(a, b) \mid a > b\}$$

$$R_3 = \{(a, b) \mid a = b\}$$

$$R_4 = \{(a, b) \mid a = b + 1\}$$

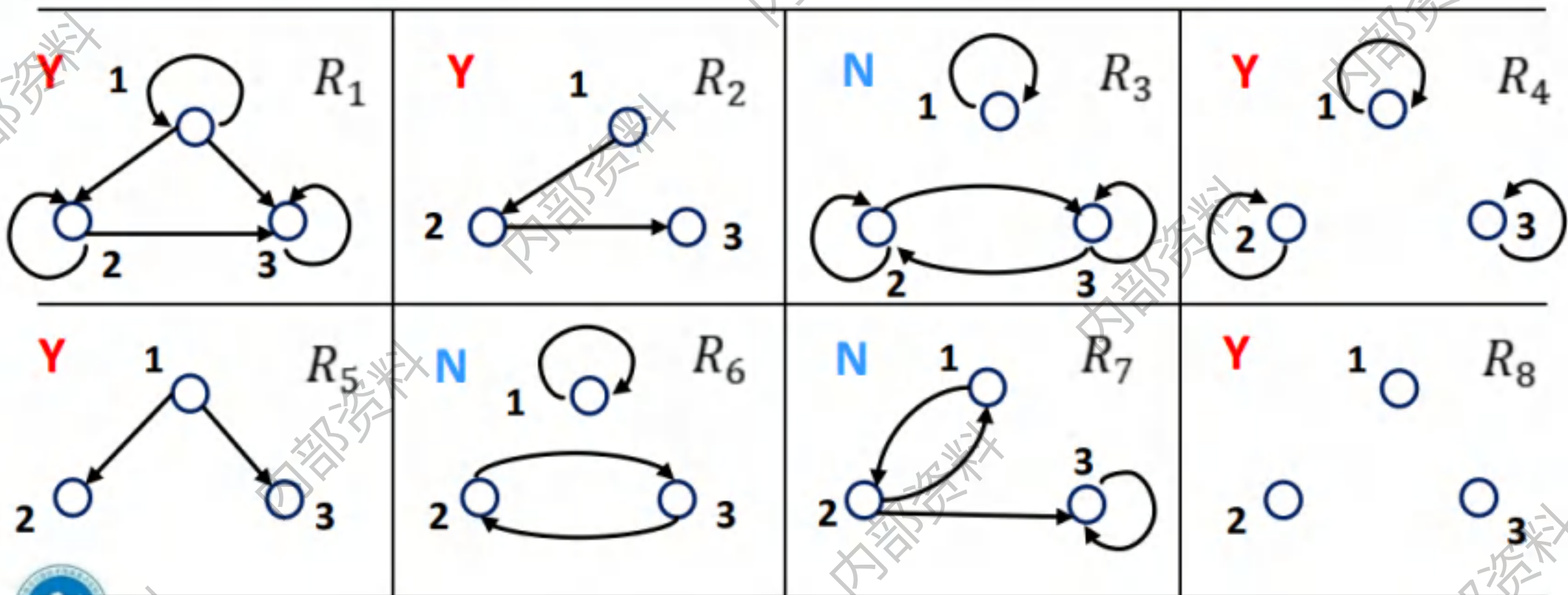
$$R_5 = \{(a, b) \mid a = b \text{ 或 } a = -b\}$$

$$R_6 = \{(a, b) \mid a + b \leq 3\}$$

\*可能有某种关系既是对称的，又是反对称的 ( $I_A$ )

# Example

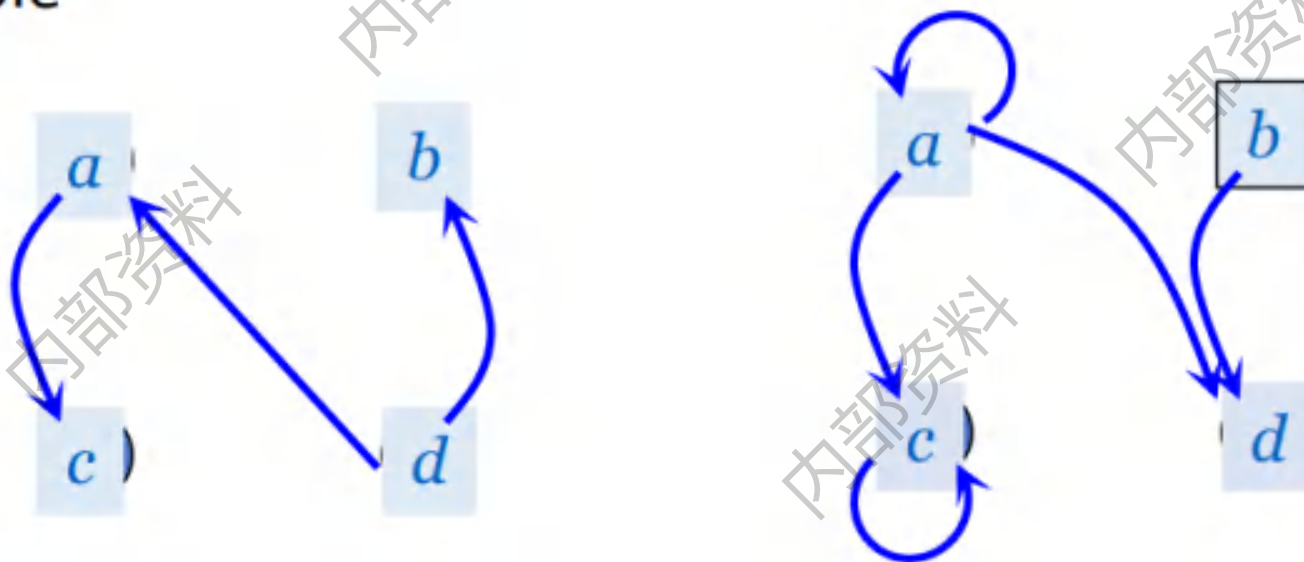
集合  $A = \{1, 2, 3\}$ , 下面8个关系哪些是反对称的  
 $R$  是  $A$  上的关系, 两个不同的节点间, 可以无边, 如果有边, 只能一条边



# Asymmetric (非对称)

- R is *Asymmetric* iff
  - $\forall x \forall y [(x, y) \in R \rightarrow (y, x) \notin R]$
- Note:
  - If there is an arc  $(x, y)$  there must not be an arc  $(y, x)$

Example





# Example

Let  $A = \{1, 2, 3\}$ ,  $R_1$ ,  $R_2$ ,  $R_3$  and  $R_4$  Which of the relations are Symmetric 、 Antisymmetric or Asymmetric ?

$$R_1 = \{(1, 1), (2, 2)\}$$

—symmetric、 antisymmetric.

$$R_2 = \{(1, 1), (1, 2), (2, 1)\}$$

—symmetric.

$$R_3 = \{(1, 2), (1, 3)\}$$

—antisymmetric. Asymmetric

$$R_4 = \{(1, 2), (2, 1), (1, 3)\}$$

—not (symmetric、 antisymmetric 、 Asymmetric)



# Transitive 传递性

- Definition 5: A relation  $R$  is *transitive* iff (for all  $a, b, c$ )  $(a, b) \in R \wedge (b, c) \in R \rightarrow (a, c) \in R$ .
- Example: Which of the relations are transitive?

$$R_1 = \{(a, b) \mid a \leq b\}$$

$$R_2 = \{(a, b) \mid a > b\}$$

$$R_3 = \{(a, b) \mid a = b\}$$

$$R_4 = \{(a, b) \mid a = b \text{ 或 } a = -b\}$$

$$R_5 = \{(a, b) \mid a = b + 1\}$$

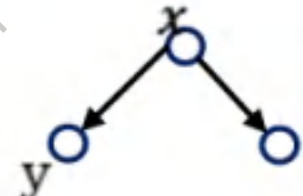
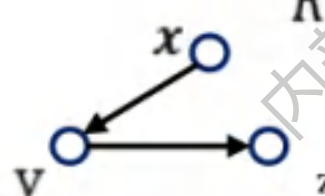
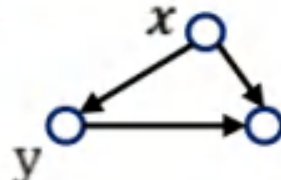
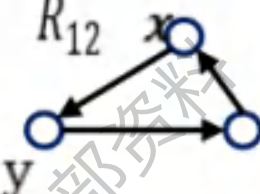
$$R_6 = \{(a, b) \mid a + b \leq 3\}$$

\*前提  $((a, b) \in R \wedge (b, c) \in R)$  不成立,  $R$  也是传递的

# 传递性，例：

传递性  $\Leftrightarrow \forall x \forall y \forall z ((x \in A \wedge y \in A \wedge z \in A \wedge xRy \wedge yRz) \rightarrow xRz)$

## 判断传递性的典型图例

|                                                                                                       |                                                                                                          |                                                                                                        |                                                                                                        |
|-------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| <b>Y</b><br> $R_1$   | <b>Y</b><br> $R_2$      | <b>Y</b><br> $R_3$  | <b>Y</b><br> $R_4$  |
| <b>Y</b><br> $R_5$ |  $R_6$                 | <b>Y</b><br> $R_7$ | <b>Y</b><br> $R_8$ |
|  $R_9$             | <b>Y</b><br> $R_{11}$ |  $R_{12}$         |                                                                                                        |

注：独立节点，空关系，恒等关系都是传递的

# Example

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$A = \{1, 2, 3\}$ , Which of these relations are transitive?

$$R_1 = \{(1, 1), (2, 2)\}$$

——transitive

$$R_2 = \{(1, 2), (2, 3)\}$$

——not transitive

$$R_3 = \{(1, 3)\}$$

——transitive

# Properties of relations (总结)

- A relation  $R$  on a set  $A$  is
- *reflexive*, if
  - $\forall x \in A, (x, x) \in R$
- *irreflexive*, if
  - $\forall x \in A, (x, x) \notin R$
- *symmetric*, if
  - $\forall x \forall y, (x, y) \in R \Rightarrow (y, x) \in R$
- *antisymmetric*, if
  - $\forall x \forall y, (x, y) \in R \text{ and } (y, x) \in R \Rightarrow x = y$
- *transitive*, if
  - $\forall x \forall y \forall z, (x, y) \in R \text{ and } (y, z) \in R \Rightarrow (x, z) \in R$



# Counting the reflexive relations

- How many reflexive relations are there on a set  $A$  with  $n$  elements? (集合  $A$  有  $n$  个元素, 有多少个自反关系)
  - Consider the pairs  $B: \{(x, x) \text{ with } x \in A\}$ . There are  $n$  such pairs. For every configuration of the pairs  $(x, y)$  with  $x \neq y$ , one of the  $2^n$  subsets of  $B$  corresponds to a reflexive relation, and all others do not. So there is one reflexive relation for every  $2^n$  relations, or  $2^{n^2} / 2^n = 2^{n^2-n} = 2^{n(n-1)}$  = ? total reflexive relations.
  - $n^2$  个序偶, 假设:  $(x, x)$  共  $n$  个, 但只能 1 个选择才是自反的, 其他的  $n^2-n$  个序偶随意选。

## 关系的运算

1. 关系作为集合，适用集合的所有运算
2. 关系的合成运算
3. 关系的幂——一种特殊的合成运算
4. 关系的逆运算
5. 关系的补运算

# Combining relations (关系的组合)

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- Since relations are by definition sets, they can be combined in any way that sets can be combined, including union, intersection, and set difference.



## Example 17

- Let  $A=\{1,2,3\}$  and  $B=\{1,2,3,4\}$ .
  - $R1=\{(1,1),(2,2),(3,3)\}$  and  $R2=\{(1,1),(1,2),(1,3),(1,4)\}$
- combine:
- $R1 \cap R2 = \{(1,1)\}$
  - $R1 \cup R2 = \{(1,1),(1,2),(1,3),(1,4),(2,2),(3,3)\}$
  - $R1 - R2 = \{(2,2),(3,3)\}$
  - $R2 - R1 = \{(1,2),(1,3),(1,4)\}$



# Example 18

---

- R1 consists of all ordered pair (a,b), where a is a student who has taken course b
- R2 consists of all ordered pair (a,b), where a is a student who requires course b to graduate.
- What are the relations :  $R1 \cup R2$  ,  $R1 \cap R2$  ,  $R1 - R2$  ,  $R2 - R1$  ,  $R1 \oplus R2$

# Example 19

$R1$  : the less than relation on the set of real number ,  
 $R1 = \{(x, y) \mid x < y\}$  ;

---

$R2$  the greater than relation on the set of real number ;  
 $R2 = \{(x, y) \mid x > y\}$

What are  $R1 \cup R2$  ,  $R1 \cap R2$      $R1 - R2$     $R2 - R1$     $R1 \oplus R2$

solution :

$$R1 \cup R2 = \{(x, y) \mid x \neq y\};$$

$$R1 \cap R2 = \phi;$$

$$R1 - R2 = R1;$$

$$R2 - R1 = R2$$

$$R1 \oplus R2 = R1 \cup R2 - R1 \cap R2 = \{(x, y) \mid x \neq y\};$$

# Composition (关系合成/复合)

- The composition of two relations  $R: A \rightarrow B$  and  $S: B \rightarrow C$ , denoted  $S \circ R$ , is the relation from  $A$  to  $C$  containing all pairs  $(a, c)$  such that there is one  $b \in B$  with  $(a, b) \in R$  and  $(b, c) \in S$ .
- $R: A \rightarrow B$ ,  $S: B \rightarrow C$ ,
- $S \circ R$  表示  $R$  与  $S$  的合成 (composition)
- $S \circ R: A \rightarrow C$ ,  $(a, c) \in S \circ R$ ,
- $\forall a \in A$ ,  $\forall c \in C$ ,  $\exists b \in B$ , 使得  $(a, b) \in R$  且  $(b, c) \in S$ , 则  $(a, c) \in S \circ R$

# Composition

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**Definition:** Suppose

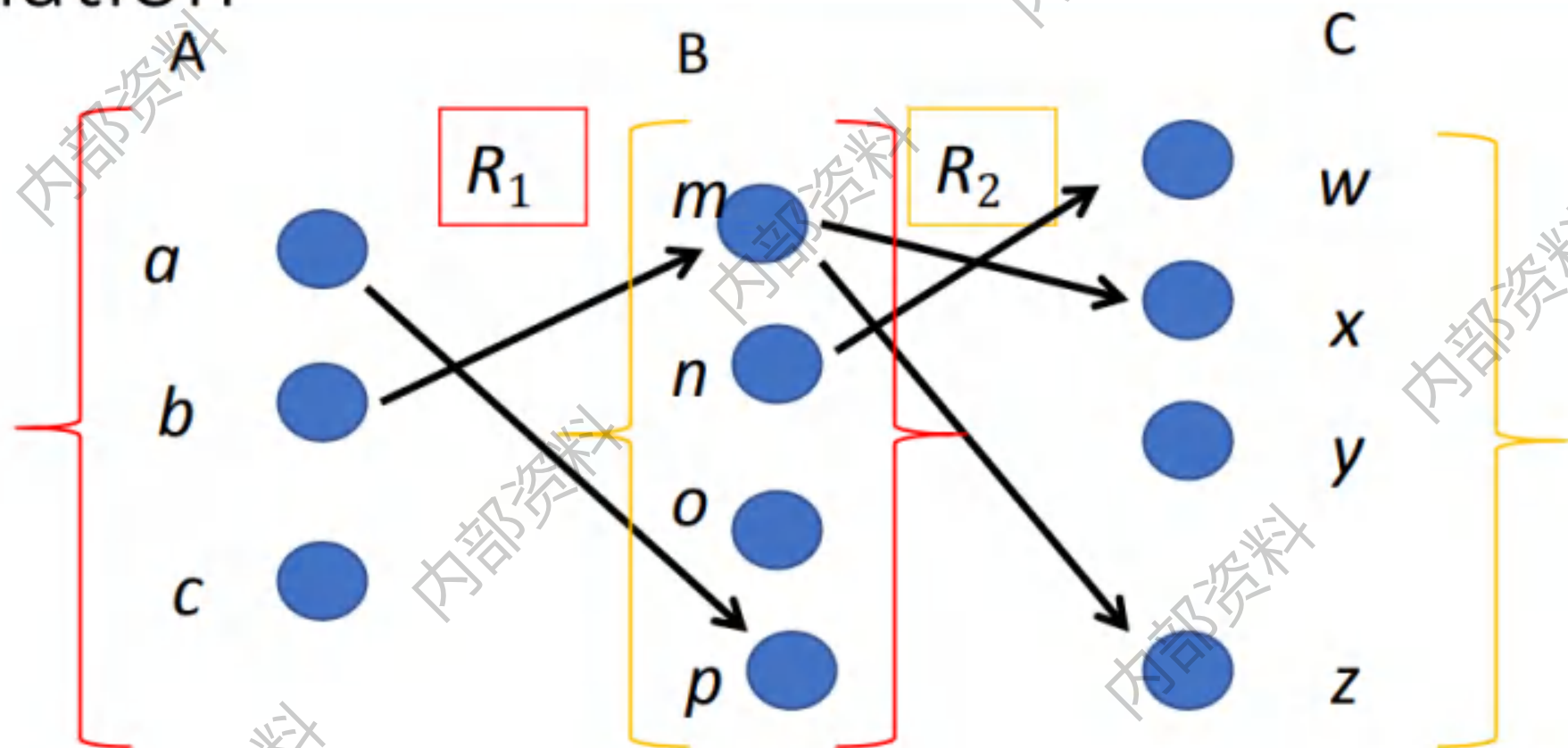
- $R_1$  is a relation from a set  $A$  to a set  $B$ .
- $R_2$  is a relation from  $B$  to a set  $C$ .

Then the *composition* (or *composite*) of  $R_2$  with  $R_1$ , is a relation from  $A$  to  $C$  where

- if  $(x,y)$  is a member of  $R_1$  and  $(y,z)$  is a member of  $R_2$ , then  $(x,z)$  is a member of  $R_2 \circ R_1$ .



# Representing the Composition of a Relation



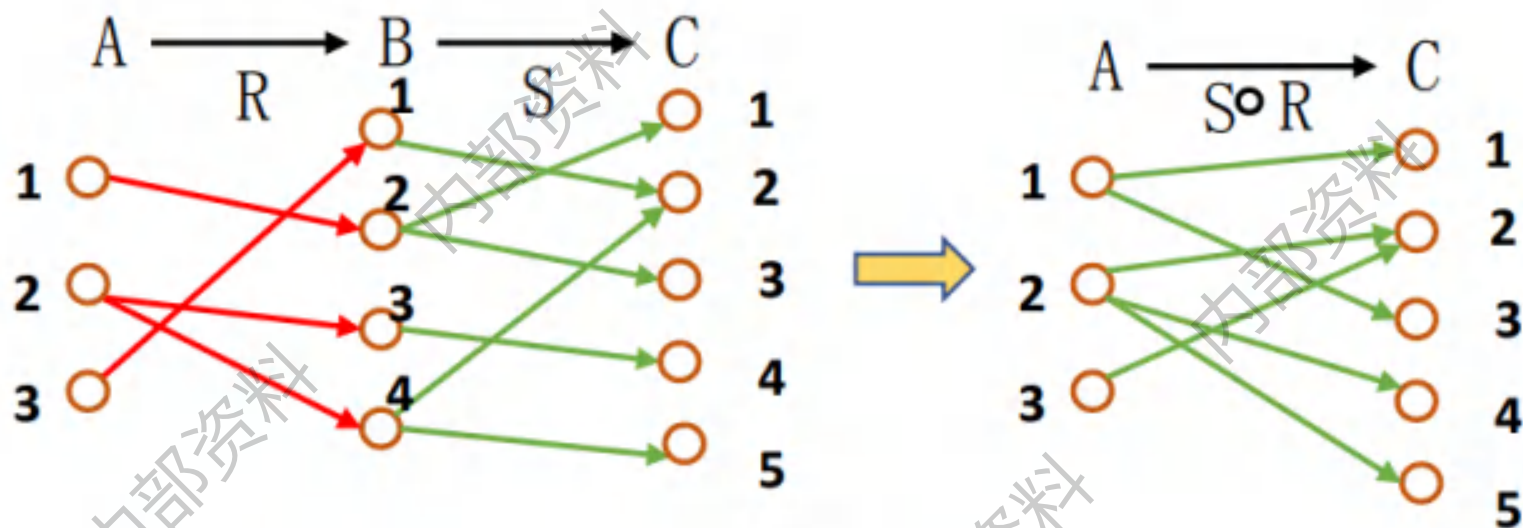
$$R_2 \circ R_1 = \{(b, x), (b, z)\}$$

过河拆桥

# 关系合成的方法

例:  $S \circ R$

1. 有向图  
法



2. 枚举法

$$R = \{(1,2), (2,3), (2,4), (3,1)\}$$

$$S = \{(1,2), (2,1), (2,3), (3,4), (4,2), (4,5)\}$$

$$\text{则 } S \circ R = \{(1,1), (1,3), (2,4), (2,2), (2,5), (3,2)\}$$

## Example 20

---

- What is the composite of the relations  $R$  and  $S$ ,
  - $R$  is the relation from  $\{1,2,3\}$  to  $\{1,2,3,4\}$
  - $R = \{(1,1), (1,4), (2,3), (3,1), (3,4)\}$
  - $S$  is the relation from  $\{1,2,3,4\}$  to  $\{0,1,2\}$
  - $S = \{(1,0), (2,0), (3,1), (3,2), (4,1)\}$
- $$S \circ R = \{(1,0), (1,1), (2,1), (2,2), (3,0), (3,1)\}$$



# Powers of a Relation

## 关系的幂

**Definition:** Let  $R$  be a binary relation on  $A$ . Then the powers  $R^n$  of the relation  $R$  can be defined inductively by:

- Basis Step:  $R^1 = R$
- Inductive Step:  $R^{n+1} = R^n \circ R$

*(see the slides for Section 9.3 for further insights)*

- Negative powers of  $R$  can also be defined if desired, by

$$R^{-n} = (R^{-1})^n$$

$$R^0 = R^1 \circ R^{-1} = I_A$$



# Power

- The  $n^{\text{th}}$  power  $R^n$  of a relation  $R$  on a set  $A$  can be defined recursively by:  
 $R^0 := I_A$ ;  $R^{n+1} := R^n \circ R$  for all  $n \geq 0$ .
  - Negative powers of  $R$  can also be defined if desired, by  
 $R^{-n} := (R^{-1})^n$ .

# Powers of a Relation

## 关系的幂一定理1

---

The powers of a transitive relation are subsets of the relation. This is established by the following theorem:

**Theorem 1:** The relation  $R$  on a set  $A$  is transitive iff  $R^n \subseteq R$  for  $n = 1, 2, 3, \dots$

(the *proof via mathematical induction*)

# Powers of a Relation

## 关系的幂

证明：数学归纳法

1) 证充分性：由  $R^n \subseteq R \Rightarrow R$  传递

- 设  $(a,b) \in R$ ,  $(b,c) \in R$ , 由关系合成知  $(a,c) \in R^2$ , 又  $R^2 \subseteq R$ , 所以  $(a,c) \in R \Rightarrow R$  传递

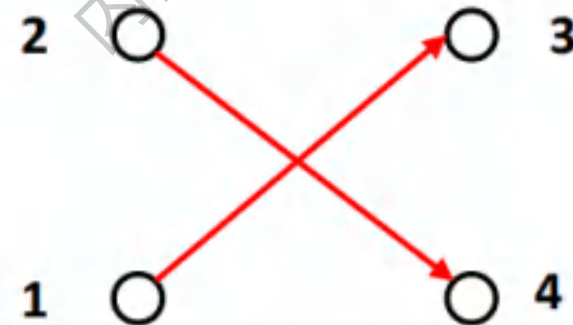
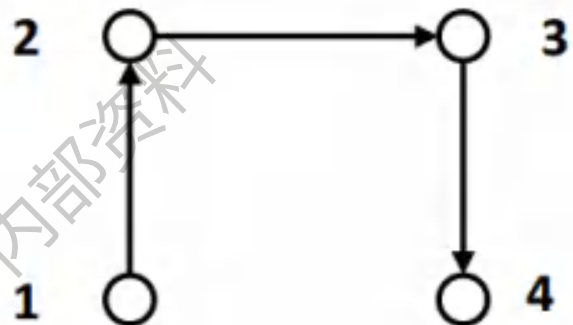
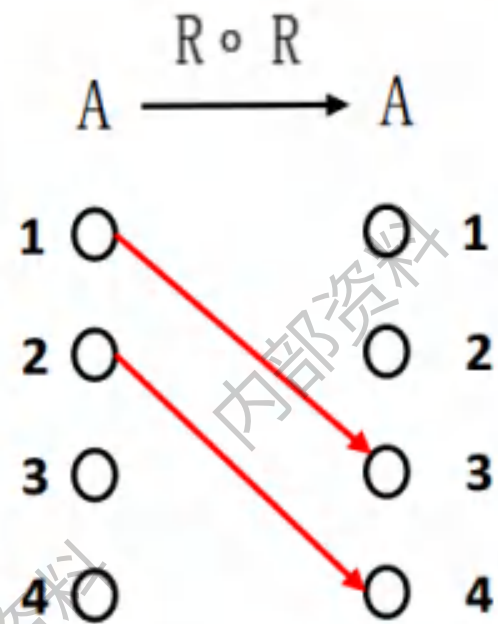
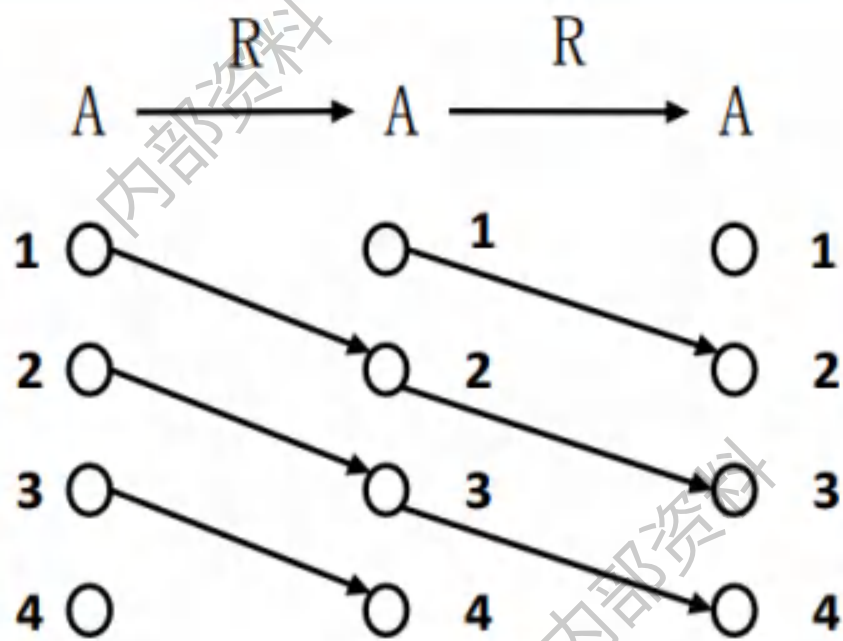
- 同理归纳可证  $R^n \subseteq R \Rightarrow R$  传递

2) 证必要性：由  $R$  传递  $\Rightarrow R^n \subseteq R$

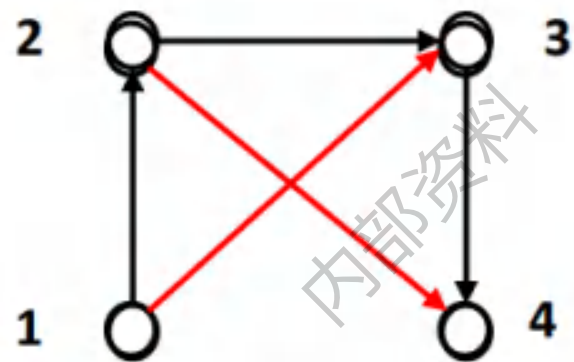
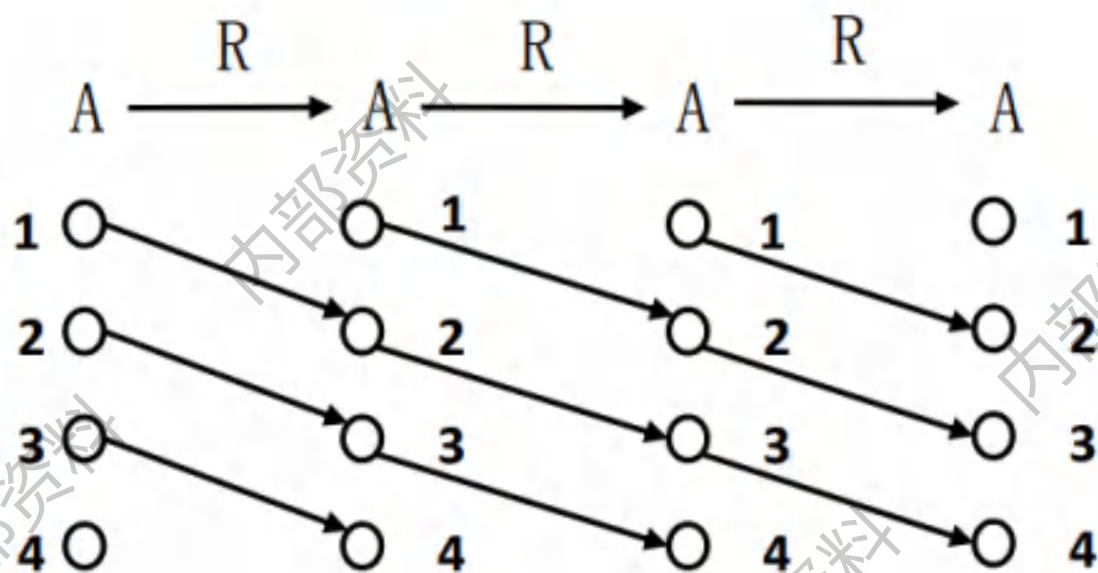
- $n=1$  时, 结果成立

- 设  $R^n \subseteq R$  成立, 希望由  $R$  传递证明  $R^{n+1} \subseteq R$ , 设  $(a,b) \in R^{n+1} = R^n \circ R$ , 一定存在  $x \in A$ , 使得  $(a,x) \in R$ ,  $(x,b) \in R^n$ , 由归纳假设  $R^n \subseteq R$  知,  $(x,b) \in R$ , 因  $R$  传递, 所以  $(a,b) \in R$ , 所以证明了  $R^{n+1} \subseteq R$

# R ∘ R







$R^3$

# Powers of relations

---

- The  $n$ th power of a relation  $R$  for  $n \geq 1$  is defined recursively as

- $R^1 = R$

- $R^{n+1} = R^n \circ R$

## Example 22

---

- Let  $R=\{(1,1),(2,1),(3,2),(4,3)\}$ . Find the power  $R^n, n=2,3,4,\dots$
- 请用关系图表示?

# 关系合成满足结合律

定理：R1，R2，R3为关系，R1是集合Z到集合W的关系，R2是集合Y到集合Z的关系，R3是集合X到集合Y的关系。则：

$$(R1 \circ R2) \circ R3 = R1 \circ (R2 \circ R3)$$



# 关系求逆运算的性质

令 $R$ 、 $S$ 都是从 $X$ 到 $Y$ 的关系, 设 $R \subseteq X \times Y$ ,  $S \subseteq X \times Y$

1.  $(R^{-1})^{-1} = R$

2.  $(R \cup S)^{-1} = R^{-1} \cup S^{-1}$

3.  $(R \cap S)^{-1} = R^{-1} \cap S^{-1}$

4.  $(R - S)^{-1} = R^{-1} - S^{-1}$

5.  $R \subseteq S \Leftrightarrow R^{-1} \subseteq S^{-1}$

6.  $(R \circ S)^{-1} = S^{-1} \circ R^{-1}$

7. 如果 $R$ 是 $A$ 上关系, 则 $R$ 是对称的, 当且仅当 $R^{-1} = R$

作业:

9.1@ 42, 49, 50题 (只做a) b)

## § 9.2: $n$ -ary Relations $n$ 元关系

- An  $n$ -ary relation  $R$  on sets  $A_1, \dots, A_n$  is simply a subset
$$R \subseteq A_1 \times \dots \times A_n.$$
- The sets  $A_i$  are called the domains of  $R$ .
- The **degree of  $R$  is  $n$** .

例：A是航空公司的集合，N是航班号集合，S出发地集合，D目的地集合，T飞行时间集合，五元组  $(A, N, S, D, T)$  的集合构成一个关系  
A, N, S, D, T为关系的域，5阶

| A  | N      | S  | D  | T     |
|----|--------|----|----|-------|
| 国航 | CA1236 | 北京 | 上海 | 9:00  |
| 国航 | CA1238 | 北京 | 成都 | 10:00 |
| 南航 | CZ2355 | 广州 | 济南 | 15:00 |
| 东航 | MU3001 | 广州 | 杭州 | 11:00 |
| 深圳 | ZH1222 | 深圳 | 厦门 | 18:00 |



# Relational Databases

## 关系数据库

- A *relational database* is essentially just an  $n$ -ary relation  $R$ .
- A domain  $A_i$  is a *primary key* for the database if the relation  $R$  is functional in  $A_i$ .
- A *composite key* for the database is a set of domains  $\{A_i, A_j, \dots\}$  such that  $R$  contains at most 1  $n$ -tuple  $(\dots, a_i, \dots, a_j, \dots)$  for each composite value  $(a_i, a_j, \dots) \in A_i \times A_j \times \dots$

# Relational Databases

## 关系数据库

- 一种基于关系概念构建数据库的方法，称为关系数据模型。
- $n$ 元关系 $R$ 是 $A_1 \times A_2 \times \dots \times A_n$ ，笛卡尔积的子集，每个元素是一个 $n$ 元组 $(a_1, a_2, \dots, a_n) \in A_1 \times A_2 \times \dots \times A_n$

# Relational databases

---

- A model for databases, based on relations
- A database is viewed as a collection of *records*, each being a tuple of the form  $(value_1, value_2, \dots, value_n)$

# Database example

• 学生 学号 专业 绩点

| <i>Student name</i> | <i>ID</i> | <i>Major</i>     | <i>GPA</i> |
|---------------------|-----------|------------------|------------|
| John                | 1022      | Computer Science | 3.07       |
| Ellen               | 2773      | Mathematics      | 3.65       |
| Mark                | 4975      | Biology          | 2.88       |
| Ellen               | 5327      | Biology          | 3.50       |

4-元组: (student, ID, Major, GPA)

一条记录: (John, 1022, computer science, 3.07)



# Database keys

- *primary key* (主键) : 能唯一确定 $n$ 元组的其他域的属性作为主键.
- *Composite key* (复合主键) : 当无主键时, 可以选多个域的组合为复合主键
- Care must be taken to maintain the key property when inserting records
- Some databases may have no primary key; instead they use a *composite* key made up from multiple fields
  - A flight database: key = airline + flight number

例,

主键

复合主键

**TABLE 1** Students

| <i>Student_name</i> | <i>ID_number</i> | <i>Major</i>     | <i>GPA</i> |
|---------------------|------------------|------------------|------------|
| Ackermann           | 231455           | Computer Science | 3.88       |
| Adams               | 888323           | Physics          | 3.45       |
| Chou                | 102147           | Computer Science | 3.49       |
| Goodfriend          | 453876           | Mathematics      | 3.45       |
| Rao                 | 678543           | Mathematics      | 3.90       |
| Stevens             | 786576           | Psychology       | 2.99       |

复合主键= major+GPA

# Operations on n-ary Relations

## n元关系的运算

---

- Selection operator (选择)
- Projection operator (投影)
- Join operator (连接)

# Selection Operators

## n元关系的选择

---

- Let  $R$  be an  $n$ -ary relation and  $C$  a condition that elements in  $R$  may satisfy. Then the **selection operator**  $s_C$  maps the  $n$ -ary relation  $R$  to the  $n$ -ary relation of all  $n$ -tuples from  $R$  that satisfy the condition  $C$ .



# Selection Operator Example

例：

| <b>TABLE 1</b> Students. |                  |                  |            |
|--------------------------|------------------|------------------|------------|
| <i>Student_name</i>      | <i>ID_number</i> | <i>Major</i>     | <i>GPA</i> |
| Ackermann                | 231455           | Computer Science | 3.88       |
| Adams                    | 888323           | Physics          | 3.45       |
| Chou                     | 102147           | Computer Science | 3.49       |
| Goodfriend               | 453876           | Mathematics      | 3.45       |
| Rao                      | 678543           | Mathematics      | 3.90       |
| Stevens                  | 786576           | Psychology       | 2.99       |

C1, Major= "coumputer science" ----选出2条记录

C2, ID-number= "678543", ----选出1条记录

# Projection Operators 投影

- Let  $A = A_1 \times \dots \times A_n$  be any  $n$ -ary domain, and let  $\{i_k\} = (i_1, \dots, i_m)$  be a sequence of indices all falling in the range 1 to  $n$ ,
  - That is, where  $1 \leq i_k \leq n$  for all  $1 \leq k \leq m$ .
- Then the *projection operator* on  $n$ -tuples is defined by:

# Projection Example

例  $P_{1, 2}$

**TABLE 3** Enrollments.

| <i>Student</i> | <i>Major</i>     | <i>Course</i> |
|----------------|------------------|---------------|
| Glauser        | Biology          | BI 290        |
| Glauser        | Biology          | MS 475        |
| Glauser        | Biology          | PY 410        |
| Marcus         | Mathematics      | MS 511        |
| Marcus         | Mathematics      | MS 603        |
| Marcus         | Mathematics      | CS 322        |
| Miller         | Computer Science | MS 575        |
| Miller         | Computer Science | CS 455        |

**TABLE 4** Majors.

| <i>Student</i> | <i>Major</i>     |
|----------------|------------------|
| Glauser        | Biology          |
| Marcus         | Mathematics      |
| Miller         | Computer Science |



# Join Operators

## 连接操作

- Puts two relations together to form a sort of combined relation.
- If the tuple  $(A,B)$  appears in  $R_1$ , and the tuple  $(B,C)$  appears in  $R_2$ , then the tuple
- $(A,B,C)$  appears in the join  $J(R_1,R_2)$ .
  - $A$ ,  $B$ , and  $C$  here can also be sequences of elements (across multiple fields), not just single elements.



# Join Example

---

- Suppose  $R_1$  is a teaching assignment table, relating *Professors* to *Courses*.
- Suppose  $R_2$  is a room assignment table relating *Courses* to *Rooms, Times*.
- Then  $J(R_1, R_2)$  is like your class schedule, listing *(professor, course, room, time)*.

例:  $J_2$

**TABLE 5** Teaching\_assignments.

| <i>Professor</i> | <i>Department</i> | <i>Course_number</i> |
|------------------|-------------------|----------------------|
| Cruz             | Zoology           | 335                  |
| Cruz             | Zoology           | 412                  |
| Farber           | Psychology        | 501                  |
| Farber           | Psychology        | 617                  |
| Grammer          | Physics           | 544                  |
| Grammer          | Physics           | 551                  |
| Rosen            | Computer Science  | 518                  |
| Rosen            | Mathematics       | 575                  |

**TABLE 6** Class\_schedule.

| <i>Department</i> | <i>Course_number</i> | <i>Room</i> | <i>Time</i> |
|-------------------|----------------------|-------------|-------------|
| Computer Science  | 518                  | N521        | 2:00 P.M.   |
| Mathematics       | 575                  | N502        | 3:00 P.M.   |
| Mathematics       | 611                  | N521        | 4:00 P.M.   |
| Physics           | 544                  | B505        | 4:00 P.M.   |
| Psychology        | 501                  | A100        | 3:00 P.M.   |
| Psychology        | 617                  | A110        | 11:00 A.M.  |
| Zoology           | 335                  | A100        | 9:00 A.M.   |
| Zoology           | 412                  | A100        | 8:00 A.M.   |



**TABLE 7** Teaching\_schedule.

| <i>Professor</i> | <i>Department</i> | <i>Course_number</i> | <i>Room</i> | <i>Time</i> |
|------------------|-------------------|----------------------|-------------|-------------|
| Cruz             | Zoology           | 335                  | A100        | 9:00 A.M.   |
| Cruz             | Zoology           | 412                  | A100        | 8:00 A.M.   |
| Farber           | Psychology        | 501                  | A100        | 3:00 P.M.   |
| Farber           | Psychology        | 617                  | A110        | 11:00 A.M.  |
| Grammer          | Physics           | 544                  | B505        | 4:00 P.M.   |
| Rosen            | Computer Science  | 518                  | N521        | 2:00 P.M.   |
| Rosen            | Mathematics       | 575                  | N502        | 3:00 P.M.   |

## Association Rules from Data Mining 数据挖掘的关联规则

- 关联规则 ( Association Rules ) : 从大量数据中挖掘潜在的有价值的关联关系
- 事务 ( Transaction ) : 每一次购买的活  
动是一个事务
- 项 ( Item ) : 每一个商品是一个项,
- 项集 ( Item set ) : 项的集合是项集
- 支持度 ( *support* ) : 某一个或多个项  
同时在事务中出现的次数占比



# Association Rules from Data Mining

## 数据挖掘的关联规则

支持度：  $support(I) = \frac{\sigma(I)}{|T|}$        $\sigma(I)$  --项集的计数  
Frequent itemset—频繁项集

| 事务号 | 项 (item)           | 事务号 | 项 (item)       |
|-----|--------------------|-----|----------------|
| 1   | {苹果, 梨, 芒果}        | 5   | {苹果, 苹果酒, 甜甜圈} |
| 2   | {苹果酒, 梨}           | 6   | {梨, 苹果酒, 甜甜圈}  |
| 3   | {苹果, 苹果酒, 甜甜圈, 芒果} | 7   | {梨, 甜甜圈}       |
| 4   | {苹果, 梨, 苹果酒, 甜甜圈}  | 8   | {苹果, 梨, 苹果酒}   |



## Association Rules from Data Mining 数据挖掘的关联规则

支持度： $support(I) = \frac{\sigma(I)}{|T|}$

$$\sigma(\text{苹果}) = 5$$

$$support(\text{苹果}) = \frac{5}{8}$$

商品很多时，  
计算工作量大！

$$\sigma(\text{苹果}, \text{苹果酒}) = 4$$

$$support(\text{苹果}, \text{苹果酒}) = \frac{4}{8}$$

$$\text{置信度 } Confidence(I \rightarrow J) = \frac{\sigma(I \cup J)}{\sigma(I)} = \frac{\sigma(I \cup J) / |T|}{\frac{\sigma(I)}{|T|}}$$

$$Confidence(\text{苹果} \rightarrow \text{苹果酒}) = \frac{\sigma(\text{苹果}, \text{苹果酒})}{\sigma(\text{苹果})}$$

作业:  
\$9.2@ 8, 20

## § 9.3: Representing Relations 关系的表示

- Some ways to represent  $n$ -ary relations:
  - With an explicit list or table of its tuples.
  - With a function from the domain to  $\{T, F\}$ .
    - Or with an algorithm for computing this function.
- Some special ways to represent binary relations:
  - With a zero-one matrix.
  - With a directed graph.
  - $n$ 元关系可以用表，函数形式表示
  - 二元关系还可以用0-1矩阵，有向图表示

# Using Zero-One Matrices

- To represent a binary relation  $R:A \times B$  by an  $|A| \times |B|$  0-1 matrix  $\mathbf{M}_R = [m_{ij}]$ , let  $m_{ij} = 1$  iff  $(a_i, b_j) \in R$ .
- E.g., Suppose Joe likes Susan and Mary, Fred likes Mary, and Mark likes Sally.
- Then the 0-1 matrix representation of the relation **Likes:Boys  $\times$  Girls** relation is:



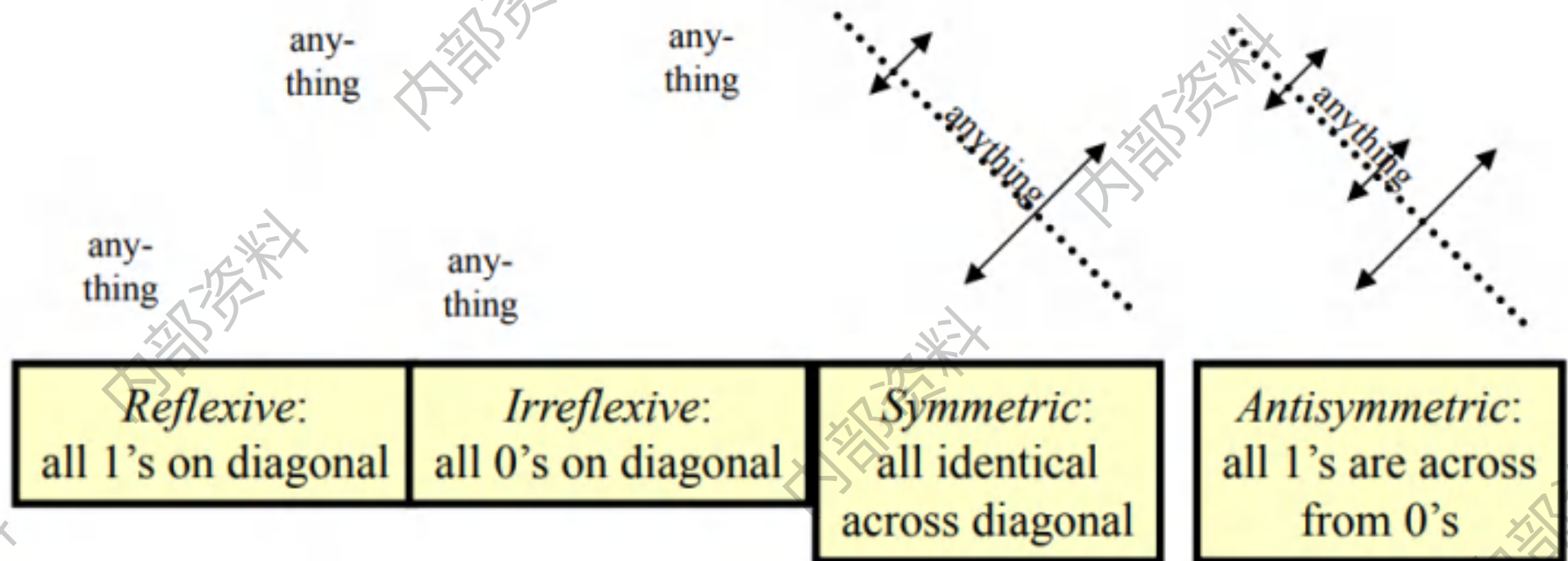
# Example

---

- Consider the matrix
- Since  $M$  is  $3 \times 4$ , let
  - $A = \{a_1, a_2, a_3\}$  and  $B = \{b_1, b_2, b_3, b_4\}$ .
- Then  $(a_i, b_j) \in R$  if and only if  $m_{ij} = 1$ .
  - $R = \{(a_1, b_1), (a_1, b_4), (a_2, b_2), (a_2, b_3), (a_3, b_1), (a_3, b_3)\}$

# Zero-One Reflexive, Symmetric

- Terms: *Reflexive, non-reflexive, irreflexive, symmetric, asymmetric, and antisymmetric.*
  - These relation characteristics are very easy to recognize by inspection of the zero-one matrix.



# Combining Connection Matrices

---

- **Definition:** the *join* of two matrices  $M_1, M_2$ , denoted  $M_1 \vee M_2$ , is the component wise *boolean 'or'* of the two matrices.
- **Fact:** If  $M_1$  is the connection matrix for  $R_1$  and  $M_2$  is the connection matrix for  $R_2$  then the join of  $M_1$  and  $M_2$ ,  $M_1 \vee M_2$  is the connection matrix for  $R_1 \cup R_2$ .



# Combining Connection Matrices

- **Definition:** the *meet* of two matrices  $M_1, M_2$ , denoted  $M_1 \wedge M_2$  is the component wise *boolean 'and'* of the two matrices.
- **Fact:** If  $M_1$  is the connection matrix for  $R_1$  and  $M_2$  is the connection matrix for  $R_2$  then the meet of  $M_1$  and  $M_2$ ,  $M_1 \wedge M_2$  is the connection matrix for  $R_1 \cap R_2$ .



## Example 4

$$\mathbf{M}_{R_1} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \quad \text{and} \quad \mathbf{M}_{R_2} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}.$$

$$\mathbf{M}_{R_1 \cup R_2} = \mathbf{M}_{R_1} \vee \mathbf{M}_{R_2} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix},$$

$$\mathbf{M}_{R_1 \cap R_2} = \mathbf{M}_{R_1} \wedge \mathbf{M}_{R_2} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

# Composite of relations

- Example 5

- Find the matrix representing the relations  $S \circ R$ , where the matrices representing  $R$  and  $S$  are

$$\mathbf{M}_R = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{and} \quad \mathbf{M}_S = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}.$$

$$\mathbf{M}_{S \circ R} = \mathbf{M}_R \odot \mathbf{M}_S = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

# Example 6

- Find the matrix representing the relation  $R^2$ , where the matrices representing  $R$  is

R是否传递?

$$\mathbf{M}_{R^2} = \mathbf{M}_R^{[2]} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

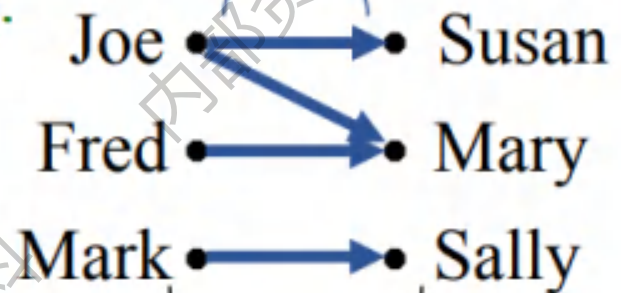
# Using Directed Graphs

- A directed graph or digraph  $G=(V_G, E_G)$  is a set  $V_G$  of vertices (nodes) with a set  $E_G \subseteq V_G \times V_G$  of edges (arcs, links). Visually represented using dots for nodes, and arrows for edges. Notice that a relation  $R:A \times B$  can be represented as a graph  $G_R=(V_G=A \cup B, E_G=R)$ .

Matrix representation  $M_R$ :

Graph rep.  $G_R$ :

Edge set  $E_G$   
(blue arrows)



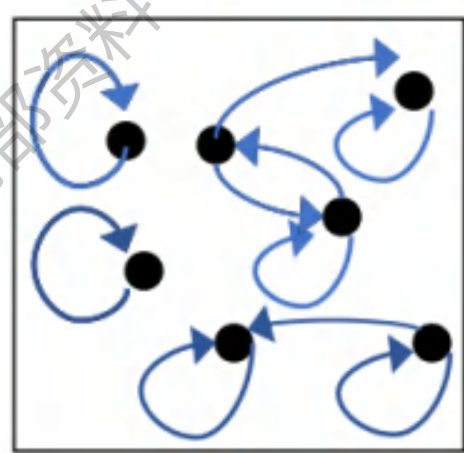
Node set  $V_G$   
(black dots)



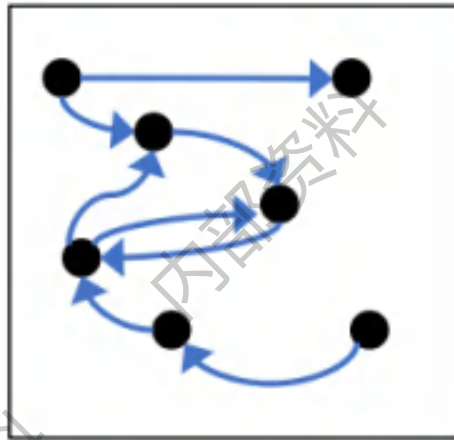
# Digraph Reflexive, Symmetric

## --集合 $A \times A$ 的关系图

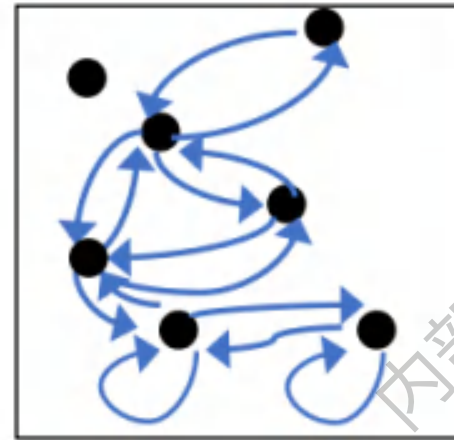
It is extremely easy to recognize the reflexive/irreflexive/ symmetric/antisymmetric properties by graph inspection.



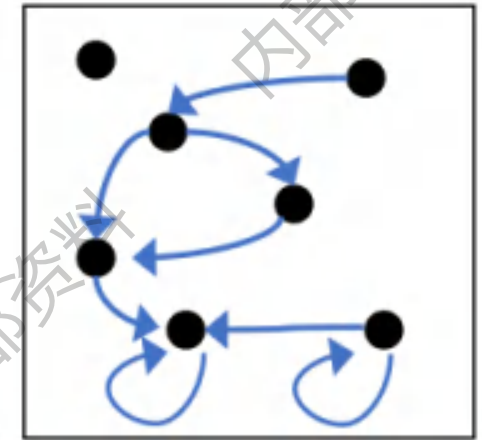
Reflexive:  
Every node  
has a self-loop



Irreflexive:  
No node  
links to itself



Symmetric:  
Every link is  
bidirectional



Antisymmetric:  
No link is  
bidirectional

These are asymmetric & non-antisymmetric

These are non-reflexive & non-irreflexive

# 关系特征、关系图、关系矩阵

| 关系特征 | 关系图特征                                                   | 关系矩阵特征                                   |
|------|---------------------------------------------------------|------------------------------------------|
| 自反   | 每一结点处有一环                                                | 对角线元素均为1                                 |
| 反自反  | 每一结点处无环                                                 | 对角线元素均为0                                 |
| 对称   | 两结点间有相反的两边同时出现                                          | 矩阵为对称矩阵                                  |
| 反对称  | 两结点间没有相反的边成对出现                                          | 当分量 $C_{ij}=1(i \neq j)$ 时<br>$C_{ji}=0$ |
| 传递   | 如果结点 $v_1, v_2$ 间有边, $v_2, v_3$ 间有边, 则 $v_1, v_3$ 间必有边。 | $R^n \subseteq R$                        |

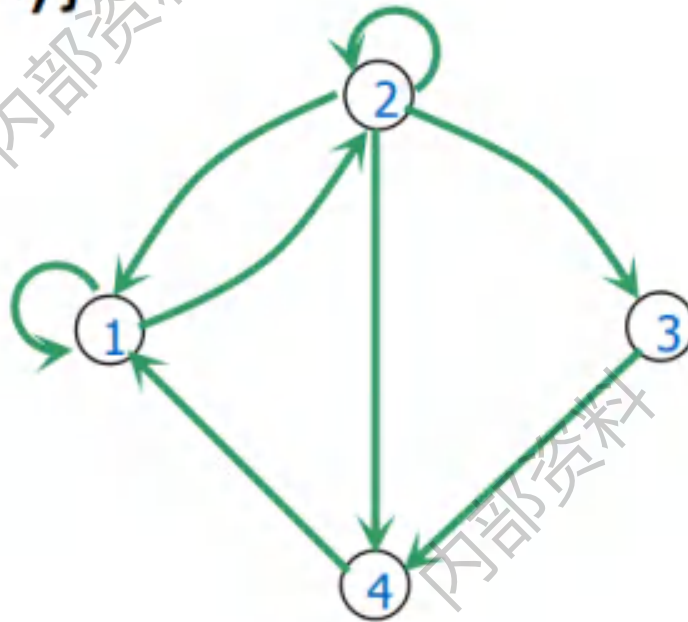
# Example

- Let

- $A = \{1, 2, 3, 4\}$

- $R = \{(1, 1), (1, 2), (2, 1), (2, 2), (2, 3), (2, 4), (3, 4), (4, 1)\}$

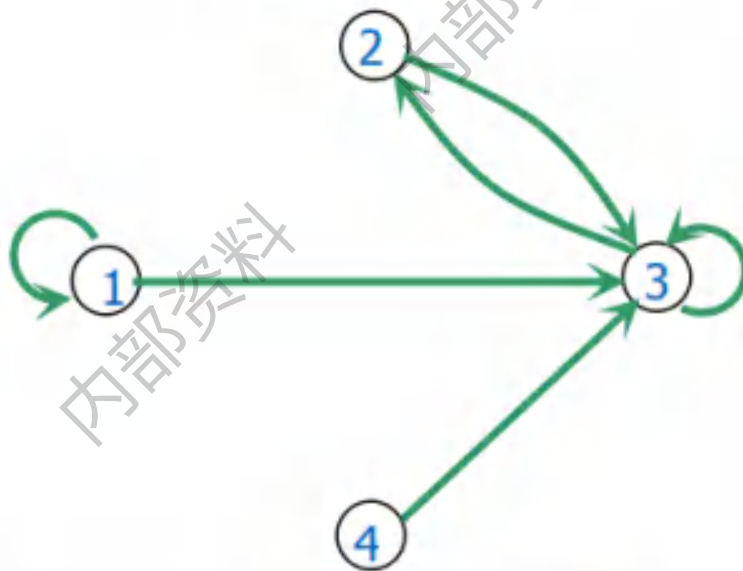
- Then





# Example

- Find the relation determined by



$$-R = \{(1, 1), (1, 3), (2, 3), (3, 2), (3, 3), (4, 3)\}$$



# Example

---

通过矩阵判断自反、反自反、对称、反对称

# 小结：二元关系的运算

---

## 二元关系

作为集合，存在并，交，差，补和对称差的运算；

作为函数 也存在复合运算和逆运算；

作为特殊的集合和函数的推广 还存在闭包运算。

# 作业

---

- § 9.1 – 42, 49, 50 (c)不做)
- § 9.2 – 8, 20
- § 9.3 – 14, 32
- § 9.4 – 20, 22, 28

## § 9.4: Closures of Relations 关系闭包

---

- $R$  is a relation on a set  $A$ . For any property  $P$ , the “ $P$  closure” of  $R$  is defined as the “smallest” subset of  $A \times A$  containing  $R$  with property  $P$ .

- 1) 闭包包含  $R$
- 2) 闭包具有性质  $P$
- 3) 闭包是具有上述2点的所有关系中最小的



## § 9.4: Closures of Relations

- The **reflexive closure** (自反闭包) of a relation  $R$  on  $A$  is obtained by adding  $(a,a)$  to  $R$  for each  $a \in A$ .  
I.e., it is  $R \cup I_A$
- The **symmetric closure** (对称闭包) of  $R$  is obtained by adding  $(b,a)$  to  $R$  for each  $(a,b)$  in  $R$ .  
I.e., it is  $R \cup R^{-1}$
- The **transitive closure** (传递闭包) or **connectivity relation** of  $R$  is obtained by repeatedly adding  $(a,c)$  to  $R$  for each  $(a,b), (b,c)$  in  $R$ .
  - I.e., it is

# Reflexive closure 自反闭包

[定义1]: 称  $R' = R \cup I_A$  是  $R$  的 **自反闭包**, 记为  $r(R)$ 。

[定理1]: 设  $R$  是  $A$  上的二元关系, 则  $r(R)$  是包含  $R$ , 具有自反性的最小关系。

证明: 对  $\forall x \in A$ ,  $(x, x) \in I_A \subseteq R \cup I_A$ , 且  $R \subseteq R \cup I_A$ 。 (**包含  $R$ , 且自反**)  
若  $R'$  为包含  $R$  且具有自反性, 则  $I_A \subseteq R'$ ,  $R \subseteq R'$ ,  
 $I_A \cup R \subseteq R'$ 。即  $I_A \cup R$  为最小。 ( **$R$  最小**)

[推论1]:  $R$  是自反闭包当且仅当  $R$  是自反的。

证明:

(1)  $R$  是自反闭包  $\Rightarrow R$  有自反性:  $R = I_A \cup R \Rightarrow I_A \subseteq R$ ;

(2)  $R$  具有自反性  $\Rightarrow R$  是自反闭包:  $I_A \subseteq R$ ,  $R \cup I_A = R$ ;

# 生成自反闭包的方法

## Method:

- Add loops to all vertices on the digraph representation of  $R$ .
- Put 1's on the diagonal of the connection matrix of  $R$ .

- $R$ 的有向关系图中，节点加自环
- 关系矩阵中，对角线对应位置  $0 \rightarrow 1$



# Symmetric closure 对称闭包

**[定义2]:** 称  $R \cup R^{-1}$  是  $R$  的**对称闭包**, 记为  $s(R)$ 。

**[定理2]:** 设  $R$  是  $A$  上的二元关系, 则  $R \cup R^{-1}$  是对称的, 包含  $R$  的最小关系, 其中  $R^{-1}$  是  $R$  的逆关系)。



# Symmetric closure

## 对称闭包

### 证明:

1) 证对称性: 若  $(a, b) \in R \cup R^{-1}$ , 则  $(a, b) \in R$  或  $(a, b) \in R^{-1}$ ,  
 $\Rightarrow (b, a) \in R^{-1}$  或  $(b, a) \in R$   
故  $(b, a) \in R \cup R^{-1}$  (对称性),

2) 证  $s(R) \supseteq R$  (根据定义, 是显然的)

3) 证  $s(R)$  是最小的:

设另外存在一个对称的二元关系  $R' \supseteq R$ ,

对  $\forall (a, b) \in R \cup R^{-1}$ ,

若  $(a, b) \in R \Rightarrow (a, b) \in R' \Rightarrow R \subseteq R'$

若  $(a, b) \in R^{-1} \Rightarrow (b, a) \in R \Rightarrow (b, a) \in R'$

又  $R'$  具有对称性,  $(a, b) \in R' \Rightarrow R^{-1} \subseteq R'$

故  $R \cup R^{-1} \subseteq R'$ .

**[推论2]：R 是对称闭包时，当且仅当 R 是对称的。**

**证明：**(1)R具有对称性 $\Rightarrow$  闭包，

$$(i) \text{ 因对称, } (a, b) \in R \Rightarrow (b, a) \in R, \Rightarrow (a, b) \in R^{-1} \\ \Rightarrow R \subseteq R^{-1}$$

$$(ii) \text{ 又 } (b, a) \in R^{-1}$$

$$\text{即 } \forall (b, a) \in R^{-1}, (a, b) \in R$$

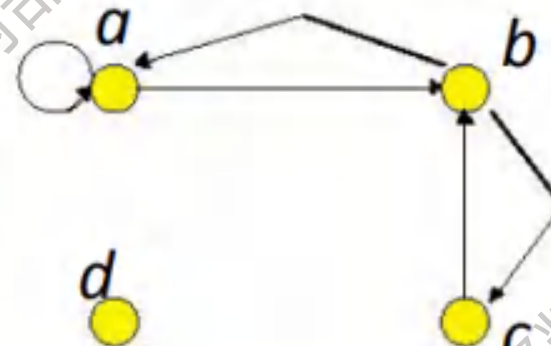
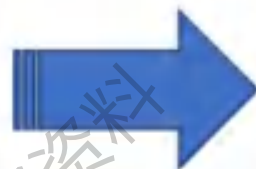
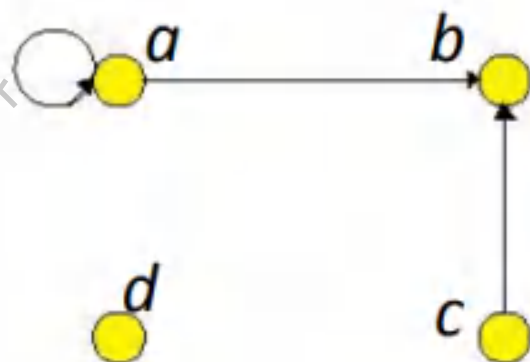
$$\Rightarrow (b, a) \in R$$

$$\Rightarrow R^{-1} \subseteq R$$

$$(i) \text{ 和 } (ii) \Rightarrow R = R^{-1} \Rightarrow R \cup R^{-1} = R ;$$

(2)R是对称闭包时， $R = s(R) = R \cup R^{-1} \Rightarrow R$ 具有对称性。

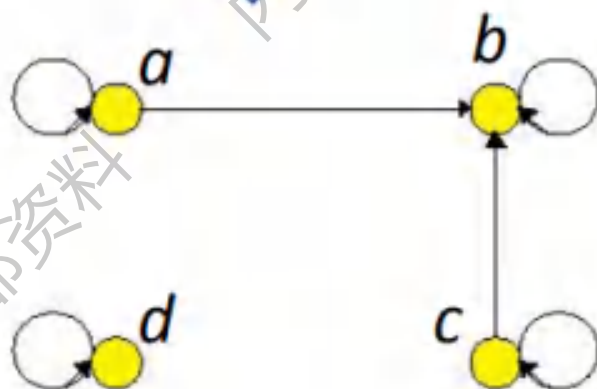
# Example



$s(R)$



$R$



$r(R)$

# Examples

1. 整数集上的关系  $R = \{(a, b) \mid a < b\}$  的自反闭包是什么?

解:  $R \cup I_R = \{(a, b) \mid a < b\} \cup \{(a, a) \mid a \in \mathbb{Z}\}$

2. 正整数集上的关系  $R = \{(a, b) \mid a > b\}$  的对称闭包是什么?

解:  $R \cup R^{-1} = \{(a, b) \mid a > b\} \cup \{(b, a) \mid a > b\} = \{(a, b) \mid a \neq b\}.$



# Transitive closure

## 传递闭包

---

- The transitive closure of a relation  $R$  is the smallest transitive relation containing  $R$ .
- 所有的传递关系中包含 $R$ 的最小的一个

# Paths in Directed Graphs

## 有向图路径

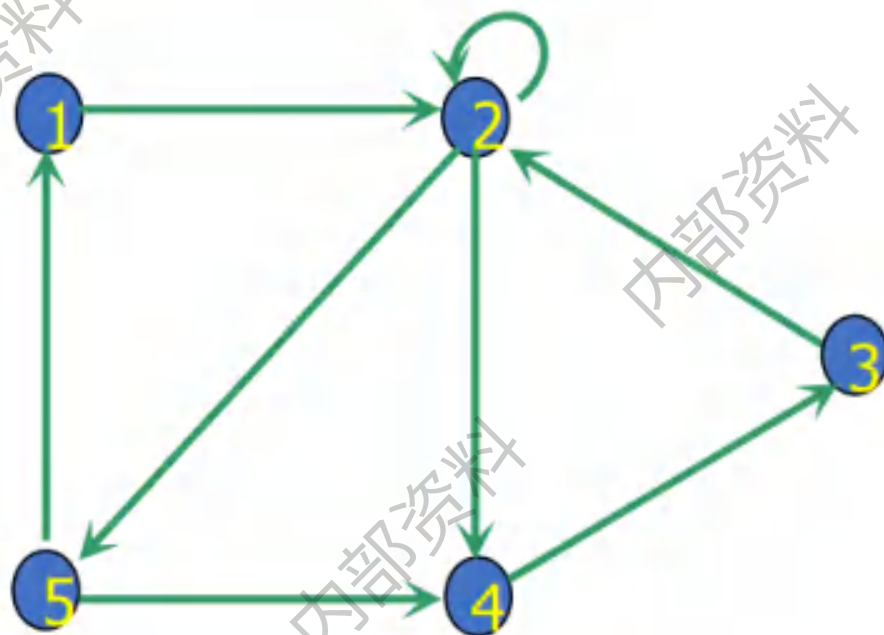
- A *path* of length  $n$  from node  $a$  to  $b$  in the directed graph  $G$  (or the binary relation  $R$ ) is a sequence  $(a, x_1), (x_1, x_2), \dots, (x_{n-1}, b)$  of  $n$  ordered pairs in  $E_G$  (or  $R$ ).
  - If any path from  $a$  to  $b$  exists, then we say that  $a$  is *connected to*  $b$ . (“You can get there from here.”)
  - An empty sequence of edges is considered a path of length 0 from  $a$  to  $a$ .
- A path of length  $n \geq 1$  from  $a$  to itself is called a *circuit* or a *cycle*.

# 关系中的路径

- 有向图中的路径:  $a, x_1, x_2, x_3, \dots, x_{n-1}, b$
  - $(a, x_1) \in R, (x_1, x_2) \in R, (x_2, x_3) \in R, \dots,$   
 $(x_{n-1}, b) \in R$
- $\Leftrightarrow R$ 中存在一条 $a$ 到 $b$ 的路径

# Example-路径

- $\pi_1 : 1, 2, 5, 4, 3$  长度4
- $\pi_2 : 1, 2, 5, 1$  长度3的回路
- $\pi_3 : 2, 2$  长度1的环



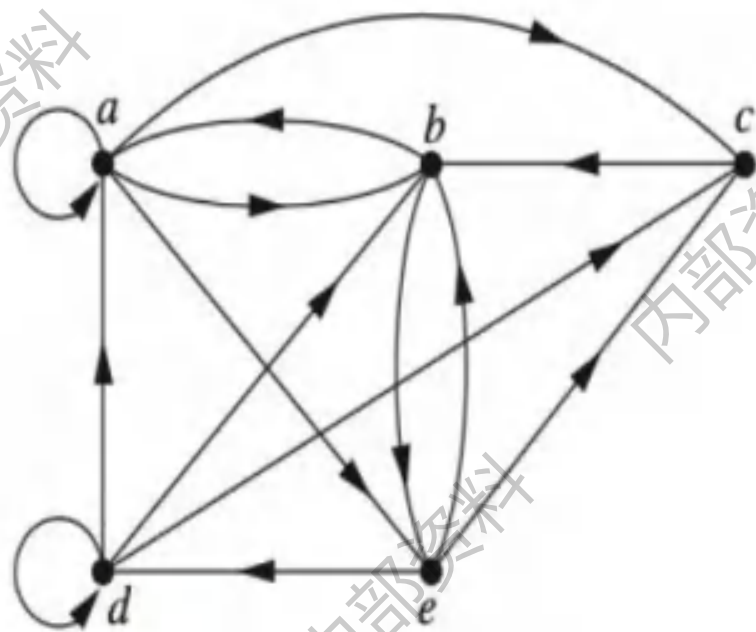


# 有向图路径

例：图中，

- $a, b, e, d$ —长度3
- $a, e, c, d, b$ —长度4
- $b, a, c, b, a, a, b$ —长度6
- $e, b, a, b, a, b, e$ —长度6

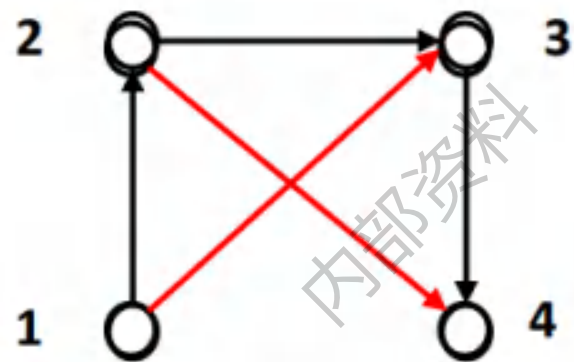
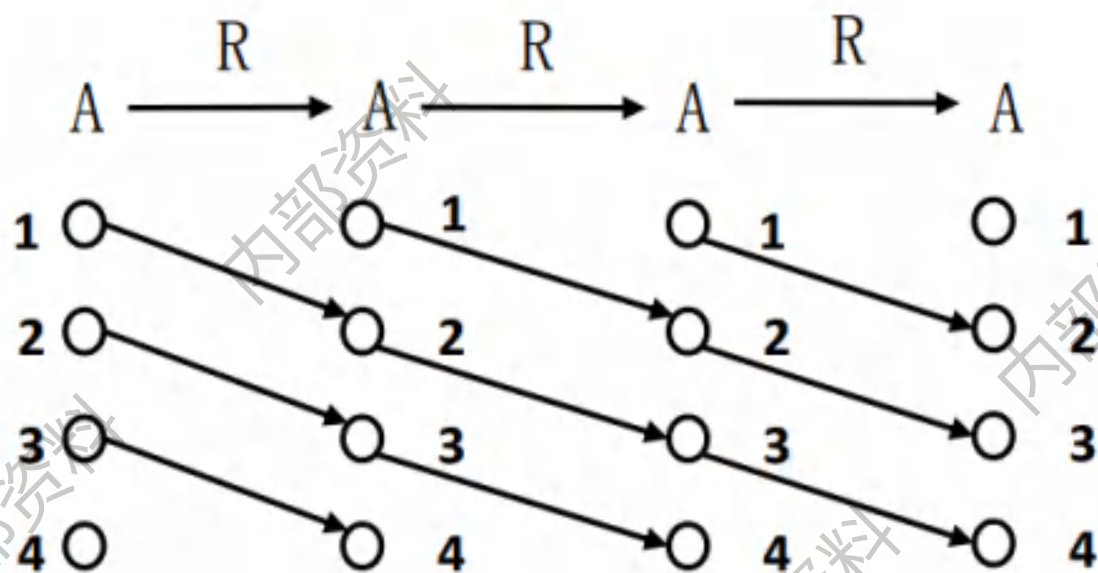
路径或回路？



**FIGURE 1** A Directed Graph.

# Theorem 1

- Let  $R$  be a Relation on a set  $A$ . This is a path of length  $n$ , where  $n$  is a positive integer, from  $a$  to  $b$  if and only if  $(a, b) \in R^n$ .
- 设  $a$  到  $b$  有长度为  $n$  的路径,  $n$  为正整数, 当且仅当  $(a, b) \in R^n$
- 数学归纳法证明 (略)



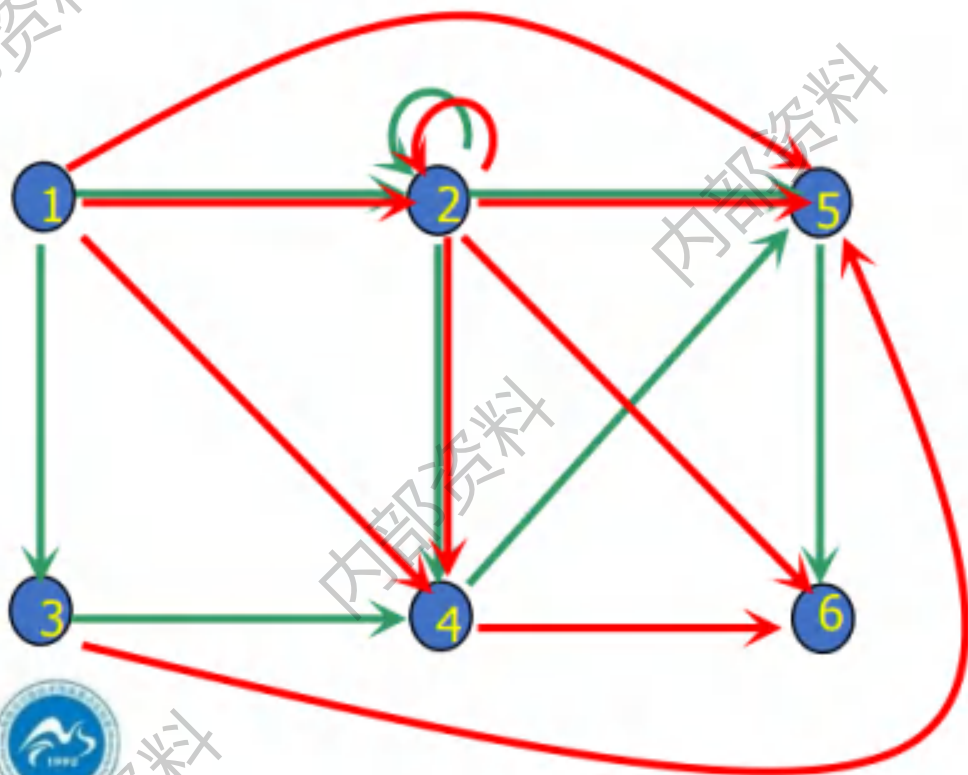
$$R \cup R^2 \cup R^3 \cup R^3$$



$$R^3$$

# Example

- Let  $A = \{1, 2, 3, 4, 5, 6\}$
- $R$  is shown as in figure
- $R^2 = ?$



$1 R^2 2$  since  $1 R 2$  and  $2 R 2$   
 $1 R^2 4$  since  $1 R 2$  and  $2 R 4$   
 $1 R^2 5$  since  $1 R 2$  and  $2 R 5$   
 $2 R^2 2$  since  $2 R 2$  and  $2 R 2$   
 $2 R^2 4$  since  $2 R 2$  and  $2 R 4$   
 $2 R^2 5$  since  $2 R 2$  and  $2 R 5$   
 $2 R^2 6$  since  $2 R 5$  and  $5 R 6$   
 $3 R^2 5$  since  $3 R 4$  and  $4 R 5$   
 $4 R^2 6$  since  $4 R 5$  and  $5 R 6$ .



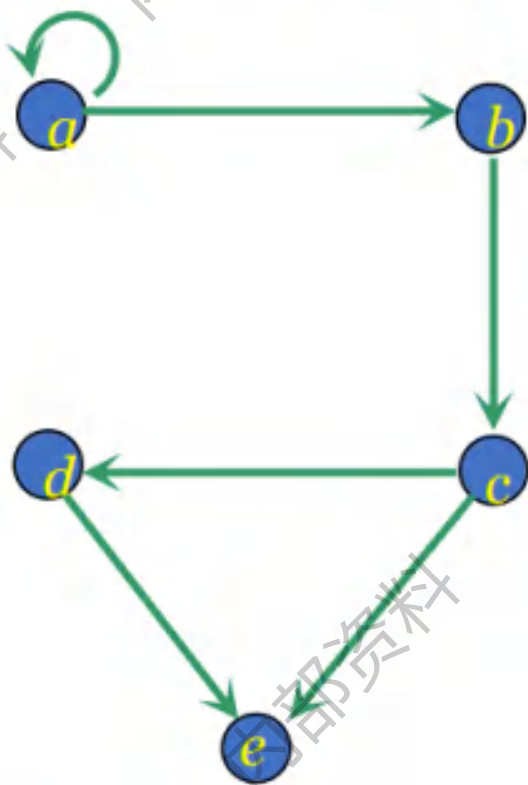
# Example

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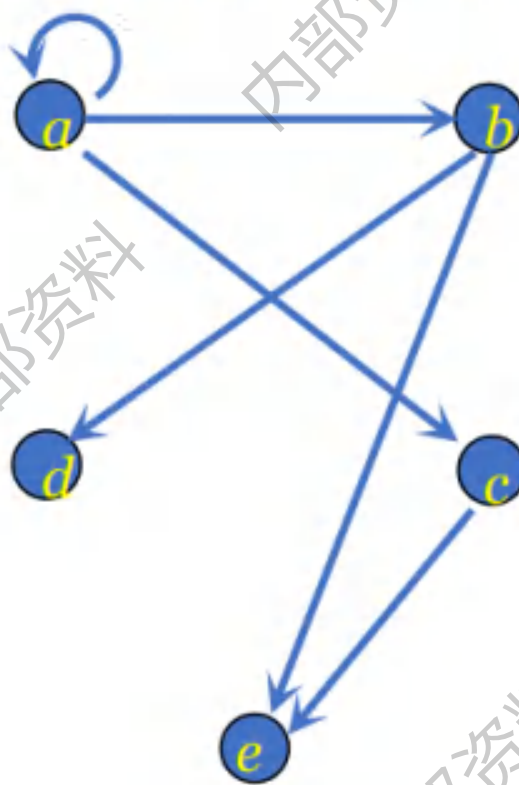
- Let  $A = \{a, b, c, d, e\}$ 
  - $R = \{(a, a), (a, b), (b, c), (c, e), (c, d), (d, e)\}$ .
- Compute (a)  $R^2$ ; (b)  $R^\infty$

# Solution

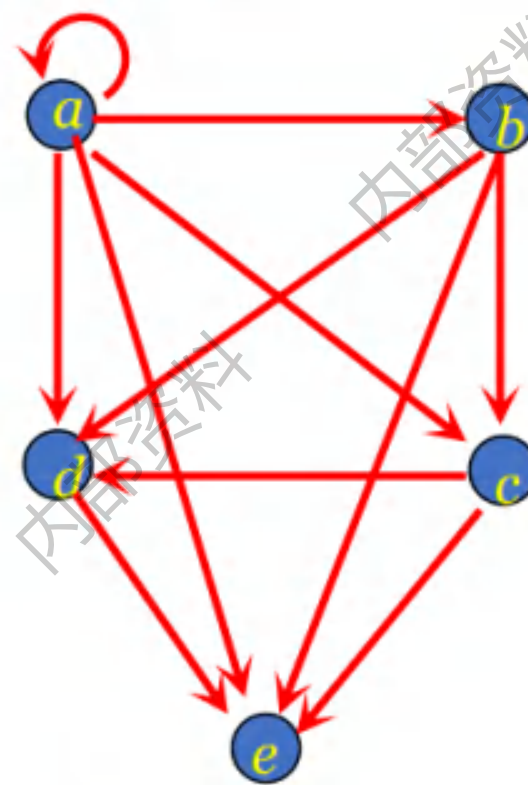
- $R = \{(a, a), (a, b), (b, c), (c, e), (c, d), (d, e)\}$ .



$R$



$R^2$



$R^\infty$

$$R \cup R^2 \cup R^3 \cup \dots \cup R^n = ?$$

# Transitive Closure

## 传递闭包

---

- The transitive closure of a relation  $R$  is the smallest transitive relation containing  $R$ .
- $R$ 的传递闭包用 $t(R)$ 表示



## Theorem 2

- Let  $R$  be a relation on a set  $A$ , then  $R^*$  is the transitive closure of  $R$ .

- 传递关系闭包:

$$R^* = t(R) = R \cup R^2 \cup R^3 \cup \dots = \bigcup_{i=1}^{\infty} R^i$$

- Proof: we must show that  $R^*$ 
  - 1) is a transitive relation (传递)
  - 2) contains  $R$  (包含  $R$ )
  - 3) is the smallest transitive relation which contains  $R$  (证明包含  $R$  的最小传递关系)

# Proof of Part 1

- 证明传递

- Suppose  $(x,y)$  and  $(y,z)$  are in  $R^*$ . Show  $(x,z)$  is in  $R^*$ . ( $= R \cup R^1 \cup R^m \dots \cup R^n \dots$ )

- By definition of  $R^*$ ,  $(x,y)$  is in  $R^m$  for some  $m$  and  $(y,z)$  is in  $R^n$  for some  $n$ .

- Then  $(x,z)$  is in  $R^n \circ R^m = R^{m+n} \subseteq R^*$ .

- Hence,  $R^*$  must be transitive.

- 设  $(x,y)$  和  $(y,z) \in R^*$ . 则一定存在  $m, n$ , 使得  $(x,y) \in R^m, (y,z) \in R^n$   
 $\Rightarrow (x,z) \in R^n \circ R^m = R^{m+n} \subseteq R^*$

# Proof of Part 2

- --- 证明包含  $R$
- Easy from the definition of  $R^*$
- 显然:  $R \subseteq R^*$

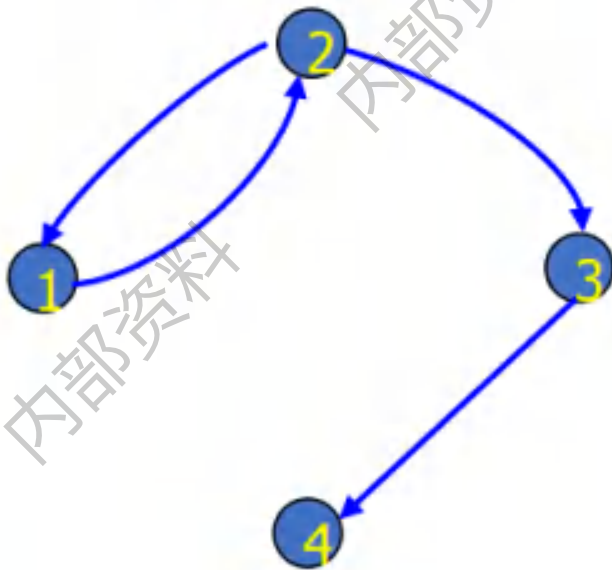
## Proof of Part 3—证 $R^*$ 最小

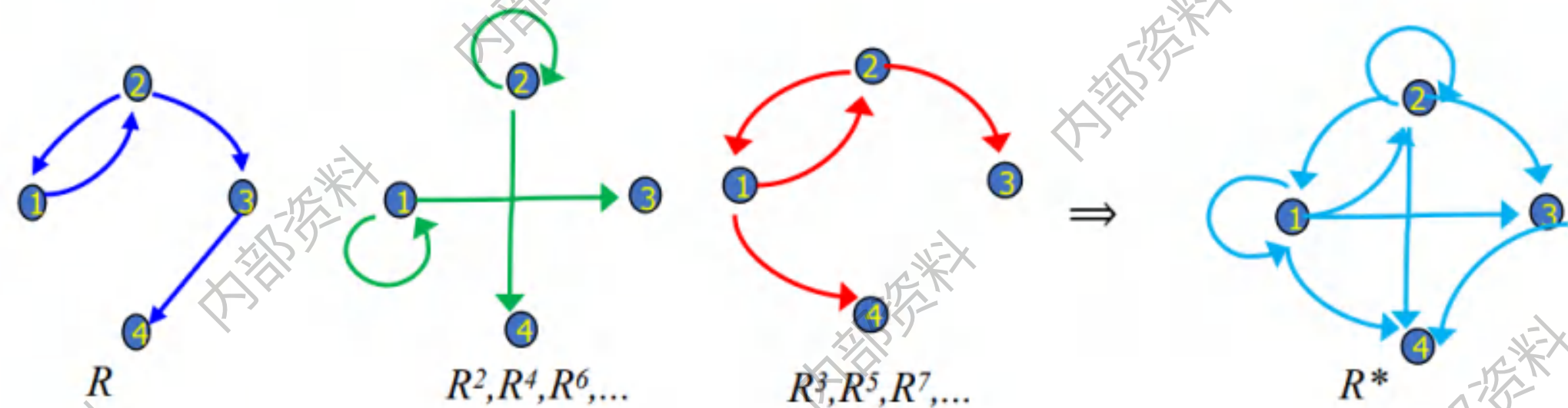
- Now Suppose  $S$  is any transitive relation that contains  $R$ , show  $S$  contains  $R^*$  (that is  $R^*$  is the smallest such relation).
- 设 $S$ 是包含 $R$ 的任意一个传递关系,  $R \subseteq S$ , 且 $S$ 传递, 只要证明 $R^* \subseteq S$ 即可。归纳法:
- $n=1$ 时, 显然 $R \subseteq S$ , 假设 $n=k$ 时,  $R^k \subseteq S$ 成立, 任意取 $(x,y) \in R^{k+1}$ , 则 $\exists z$ 使得  $(x,z) \in R^k, (z,y) \in R$ , 根据假设 $R^k \subseteq S$ , 所以  $(x,z) \in S$ , 显然  $(z,y) \in S$ , 根据 $S$ 是传递的, 所以 $(x,y) \in S$ , 于是 $R^{k+1} \subseteq S$
- 所以 $R^* (= R \cup R^2 \cup \dots \cup R^m \cup \dots \cup R^n \dots) \subseteq S$
- Q.E.D.



# Example

- Let
  - $A = \{1, 2, 3, 4\}$
  - $R = \{(1, 2), (2, 3), (3, 4), (2, 1)\}$
- Find the transitive closure of  $R$ .





$$(M_R)_{\odot}^2 = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} = (M_R)_{\odot}^4 = (M_R)_{\odot}^6 = \dots$$

$$(M_R)_{\odot}^3 = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} = (M_R)_{\odot}^5 = (M_R)_{\odot}^7 = \dots$$

$$M_{R^*} = M_R \vee (M_R)_{\odot}^2 \vee (M_R)_{\odot}^3 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

计算有限集合A的传递闭包是否求无穷次并运算？

$$t(R) = R \cup R^2 \cup R^3 \cup \dots = \bigcup_{i=1}^{\infty} R^i$$

# Theorem 3

- Let  $A$  be a set with  $|A|=n$ , and let  $R$  be a relation on  $A$ . Then
- $t(R) = R \cup R^2 \cup R^3 \dots \cup R^n$

证明：记  $t(R) = R^*$ ，假设  $a, b \in A$  且  $(a, b) \in R^*$ ，则存在一个正整数  $m$ ，使得  $(a, b) \in R^m$  成立。即存在如下一系列序偶  $aRx_1, x_1Rx_2, x_2Rx_3, \dots, x_{m-1}Rb$ 。用反证法，假设满足此序列的最小的  $m > n$ ，那么在这个序列中一定有  $0 \leq i < j \leq m$ ，使得  $x_i = x_j$ 。故该序列可以重写为  $aRx_1, x_1Rx_2, \dots, x_{i-1}Rx_i, x_iRx_{j+1}, \dots, x_{m-1}Rb$ ，其中红色部分有  $i$  个序偶，绿色部分有  $m-j$  个序偶，存在一个更短的路径，这表明  $aR^k b$  存在，其中  $k = m - j + i = m - (j - i)$ ，易知存在  $k < m$ 。这与  $m$  是最小的假设矛盾，因此  $k \leq n$ 。



# Simple Transitive Closure Alg.

## 定理3-算法思路

A procedure to compute  $R^*$  with 0-1 matrices.

**procedure** *transClosure*( $M_R$ :rank- $n$  0-1 mat.)

**A** := **B** :=  $M_R$ ;

**for**  $i := 2$  to  $n$  **begin**

**A** :=  $A \odot M_R$ ; **B** :=  $B \vee A$  {join}

**end** {note **A** represents  $R^i$ , **B** represents}

**return B** {Alg. takes  $\Theta(n^4)$  time}

# 闭包几个有用的结论

**定理** 设  $R \subseteq A \times A$  且  $A \neq \emptyset$ , 则

- (1)  $R$  自反  $\Leftrightarrow r(R) = R$ ;
- (2)  $R$  对称  $\Leftrightarrow s(R) = R$ ;
- (3)  $R$  传递  $\Leftrightarrow t(R) = R$ . #

**定理** 设  $R_1 \subseteq R_2 \subseteq A \times A$  且  $A \neq \emptyset$ , 则

- (1)  $r(R_1) \subseteq r(R_2)$ ;
- (2)  $s(R_1) \subseteq s(R_2)$ ;
- (3)  $t(R_1) \subseteq t(R_2)$ . #

## 小结

$r(R) = R \cup I_A$  ----自反闭包

$s(R) = R \cup R^{-1}$  -----对称闭包

$t(R) = R \cup R^2 \cup R^3 \cup \dots \cup R^n$  ——传递闭包

$R$ 是传递的，充要条件  $R^n \subseteq R$

要求：求闭包的几种方法

1.枚举法

2.有向图方法

3.关系矩阵法

## Example

---

#1. Let  $R$  be the relation on  $\{1, 2, 3, 4\}$  such that

$$R = \{(1, 1), (1, 4), (2, 3), (3, 1), (3, 3), (4, 4)\}.$$

Find:

- (a) the reflexive closure of  $R$ .
- (b) the symmetric closure of  $R$ .
- (c) the transitive closure of  $R$ .

**Solution:**

- (a)  $\{(1, 1), (1, 4), (2, 2), (2, 3), (3, 1), (3, 3), (4, 4)\}.$
- (b)  $\{(1, 1), (1, 3), (1, 4), (2, 3), (3, 1), (3, 2), (3, 3), (4, 1), (4, 4)\}.$
- (c)  $\{(1, 1), (1, 4), (2, 1), (2, 3), (2, 4), (3, 1), (3, 3), (3, 4), (4, 4)\}.$



# Warshall 算法

# Some definitions

- Let

- $A = \{a_1, a_2, \dots, a_n\}$

- $R$  be a relation on  $A$

- *Interior vertices* (内点)

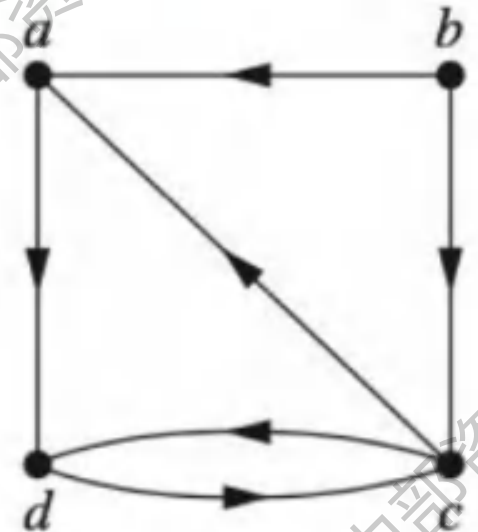
- $a, x_1, x_2, \dots, x_i, b$  (路径)

- $a, x_1, x_2, \dots, x_i, b, b$

- $a, x_1, a, x_2, \dots, x_i, b$

# Some definitions

- $W_k$ : a Boolean matrix, for  $1 \leq k \leq n$ 
  - $W_k$  has a 1 in position  $i, j$ 
    - If and only if
      - there is a path from  $a_i$  to  $a_j$  in  $R$  whose interior vertices, if any, come from the set  $\{a_1, a_2, \dots, a_k\}$
- What about  $W_0 W_n$ ?
  - Let  $W_0 = M_R$
  - $W_n = W_\infty$
  - $W_0, W_1, W_2, \dots, W_n$



# Warshall's Algorithm

---

- Procedure

- begin with the matrix of  $R$ , and
- compute each matrix,  $W_k$  from the previous matrix  $W_{k-1}$  and,
- reach  $W^\infty$  in  $n$  steps,

计算 $W_k$ 基于前一个矩阵 $W_{k-1}$ ，第 $n$ 步结束。



# Warshall's Algorithm

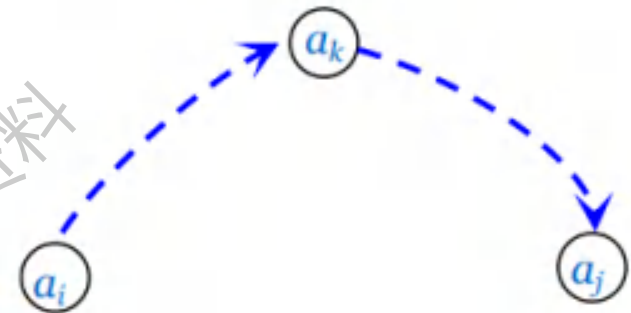
- Suppose
  - $W_k = [t_{ij}]$
  - $W_{k-1} = [s_{ij}]$
- If  $t_{ij} = 1$ , then there must be a path from  $a_i$  to  $a_j$  whose interior vertices come from the set  $\{a_1, a_2, \dots, a_k\}$ .
  - Whether  $a_k$  is an interior vertex ?
  - Two cases



$t_{ij} = 1$ 的条件:  $a_i$  到  $a_j$  的路径, 内点来自前k个节点

# Warshall's Algorithm

- $a_k$  is not an interior vertex
  - then all interior vertices must actually come from the set  $\{a_1, a_2, \dots, a_{k-1}\}$
  - so  $s_{ij} = 1$ .
- $a_k$  is an interior vertex
  - Assume  $a_k$  appears only once (why?)
  - Two subpaths
    - $a_i$  to  $a_k$  and  $a_k$  to  $a_j$
  - $s_{ik} = 1$  and  $s_{kj} = 1$



# Warshall's Algorithm

---

- The basis for Warshall's Algorithm

- $t_{ij} = 1$

- If and only if

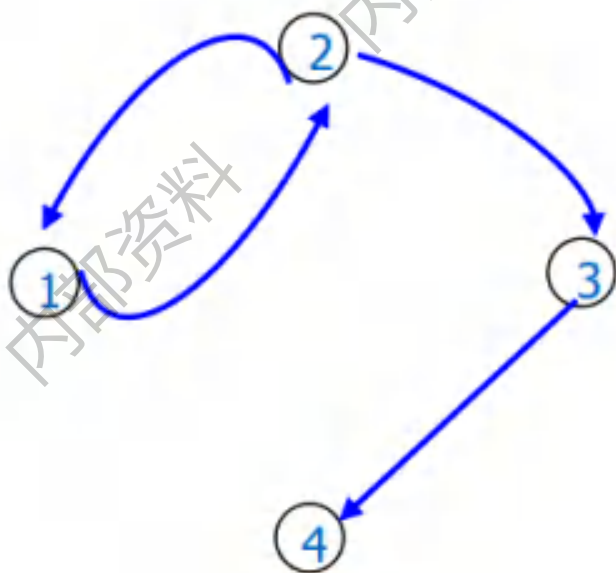
- either

- $s_{ij} = 1$

- $s_{ik} = 1$  and  $s_{kj} = 1$

## Example 2 (1)

- Let
  - $A = \{1, 2, 3, 4\}$
  - $R = \{(1, 2), (2, 3), (3, 4), (2, 1)\}$
- Find the transitive closure of  $R$ .



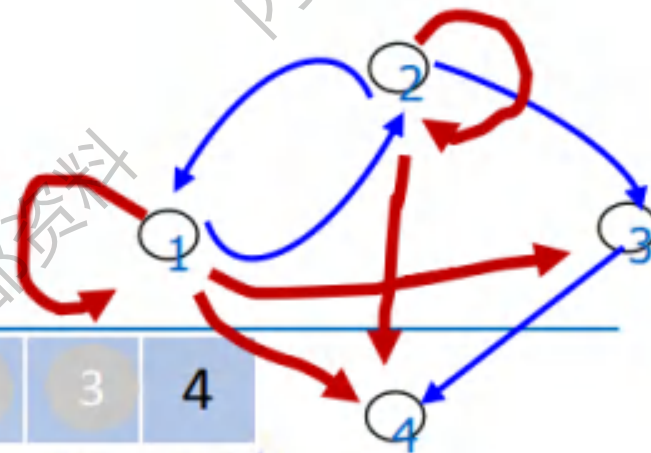


# Warshall's Algorithm

---

- **Step1:**
  - First transfer to  $W_k$  all 1's in  $W_{k-1}$ .
- **Step2:**
  - List the locations  $p_1, p_2, \dots$ , in column  $k$  of  $W_{k-1}$ , where the entry is 1
  - List the locations  $q_1, q_2, \dots$ , in row  $k$  of  $W_{k-1}$ , where the entry is 1
- **Step3:**
  - Put 1's in all the positions  $p_i, q_j$  of  $W_k$  (if they are not already there)

# Example 2



$$W_0 = \begin{bmatrix} & \begin{matrix} 1 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{bmatrix} = M_R,$$

$$W_1 = \begin{bmatrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{bmatrix}$$

$$W_2 = \begin{bmatrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{bmatrix},$$

$$W_3 = \begin{bmatrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{bmatrix} = W_4 = W^\infty$$

# Warshall's Algorithm

---

## Algorithm Warshall

CLOSURE  $\leftarrow$  MAT

For  $k = 1$  thru  $N$

For  $i = 1$  thru  $N$

For  $j = 1$  thru  $N$

CLOSURE[i,j]  $\leftarrow$  CLOSURE[i,j]

$\vee (\text{CLOSURE}[i,k] \wedge \text{CLOSURE}[k,j])$

End of Algorithm Warshall

# Warshall's Algorithm

---

## 另一种方式



## 例：用warshall算法求 $t(R)$

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

(a)

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

(b)

$$\begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

(c)

$$\begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

(d)

# Analysis

- Complexity of Algorithm

- Warshall

- $n^3$

- $M^\infty = M_R \vee (M_R)_{\odot}^2 \vee \dots \vee (M_R)_{\odot}^n$

- $n^4$

# Warshall算法

1) 置新矩阵  $A := M$

2)  $i := 1$

3) 对所有  $j$ , 若  $A[j, i] = 1$ , 则对

$k = 1, 2, \dots, n$

4)  $i + 1$

5) 若  $i \leq n$ , 则转3), 否则停止。

# 作业

---

- § 9.4 – 20, 22, 28



# Closures of Relations

- The *reflexive closure* (自反闭包) -----  $R \cup I_A$
- The *symmetric closure* (对称闭包) --  $R \cup R^{-1}$
- The *transitive closure* (传递闭包)

$$t(R) = R \cup R^2 \cup R^3 \cup \dots \cup R^n$$

Warshall's Algorithm

# 几种重要的关系

- 等价关系
- 偏序关系
- 相容关系

## § 9.5: Equivalence Relations 等价关系

---

- An *equivalence relation* (e.r.) on a set  $A$  is simply any binary relation on  $A$  that is **reflexive, symmetric, and transitive.**

同时具有性质：

- (1) 自反的
- (2) 对称的
- (3) 传递的

# Example

- Let  $A$  be the set of all triangles in the plane and let  $R$  be the relation on  $A$  defined as follows:
  - $R = \{(a, b) \in A \times A \mid a \text{ is congruent to } b\}$ .
- It is easy to see that  $R$  is an equivalence relation.
- $R$ : 全等三角形的关系



# Example

- Let  $A = \mathbb{Z}$  and let
  - $R = \{(a, b) \in A \times A \mid a \text{ and } b \text{ yield the same remainder when divided by } 2\}$ .
- Show
  - congruence mod 2 is an equivalence relation
- note
  - 2 is called the *modulus*
  - $a \equiv b \pmod{2}$ 
    - “ $a$  is congruent to  $b$  mod 2.”(模2同余)

$R$ 为整数集合上的模2同余关系，等价关系？

# Solution, *a typical way*

- 证自反、对称、传递
- First
  - clearly  $a \equiv a \pmod{2}$ .
  - Thus  $R$  is *reflexive*.
- Second
  - if  $a \equiv b \pmod{2}$ , then  $a$  and  $b$  yield the same remainder when divided by 2, so  $b \equiv a \pmod{2}$ .
  - $R$  is *symmetric*.
- Finally
  - suppose that  $a \equiv b \pmod{2}$  and  $b \equiv c \pmod{2}$ . Then  $a$ ,  $b$ , and  $c$  yield the same remainder when divided by 2.
  - Thus,  $a \equiv c \pmod{2}$ .
  - $R$  is *transitive*.
- Hence congruence mod 2 is an equivalence relation.

例：

集合  $A = \{1, 2, 3, 4, 5, 6, 7\}$ ,  $R$  是  $A$  上的模3同余关系，即，

$R = \{ (a, b) \mid a - b \text{ 可被3整除} \}$ 。

解：

$R = \{ (1, 1), (1, 4), (1, 7), (2, 2), (2, 5), (3, 3), (3, 6),$

$(4, 1), (4, 4), (4, 7), (5, 2), (6, 3), (7, 1), (7, 4), (7, 7) \}$

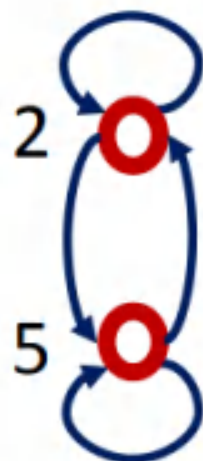
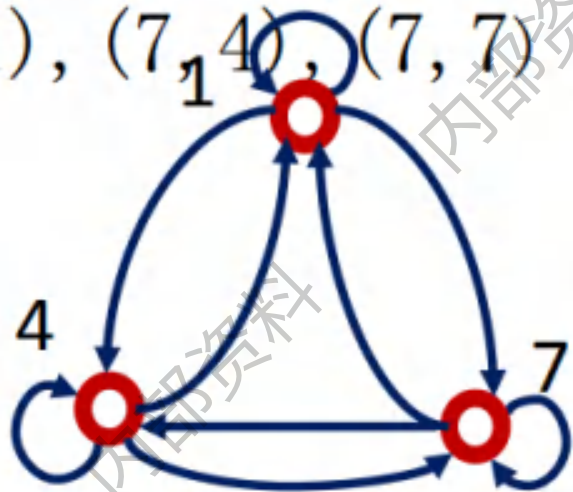


# 等价关系的有向图

画例中模3同余关系R的关系图

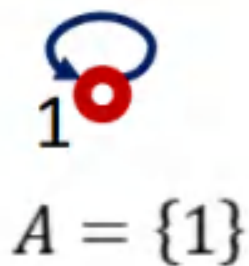
$R = \{$

$(1, 1), (1, 4), (1, 7), (2, 2), (2, 5), (3, 3), (3, 6),$   
 $(4, 1), (4, 4), (4, 7), (5, 2), (5, 5), (6, 3), (6, 6)$   
 $(7, 1), (7, 4), (7, 7)\}$

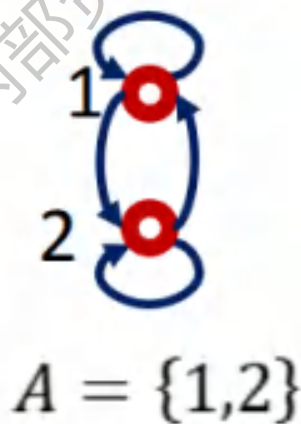




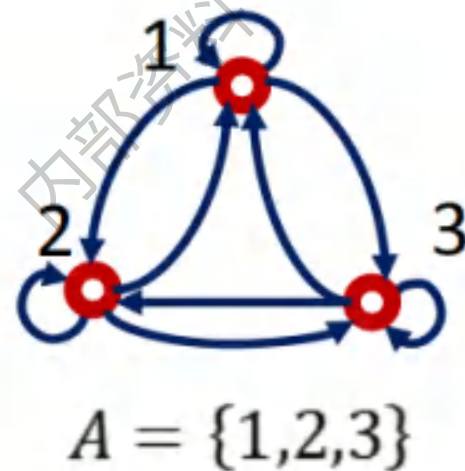
完全关系（全域关系）图：集合A上的全域关系即是 $A \times A$ ，下面是当A中只有1，或2，或3个元素时的全域关系图



(a)



(b)



(c)

思考题： $A=\{1,2,3\}$ , 可构造多少个A中不同的等价关系？

解：1) 三个子图的情况， 1个等价关系  
2) 二个子图的情况， 3个等价关系  
3) 一个子图的情况， 1个等价关系

思考题：A集合为班级同学，设计等价关系？

# Equivalence Classes 等价类

- Let  $R$  be any equiv. rel. on a set  $A$ .
- The equivalence class of  $a$ ,  
 $[a]_R = \{b \mid b \in A \wedge aRb\}$  (optional subscript  $R$ )

-Each such  $b$  (including  $a$  itself) is called  $a$  representative (代表元) of  $[a]_R$ .

例：集合 $A=\{1, 2, 3, 4, 5, 6, 7\}$ ,  $R$ :模3同余

模3同余，每一个子图就是一个等价类

$[1]=\{1,4,7\}=[4]=[7]$  -----余数为1 的等价类

$[2]=\{2,5\}=[5]$  -----余数为2 的等价类

$[3]=\{3,6\}=[6]$  -----余数为0 的等价类

等价类 $[1]$ 代表元为1，也可以选元素4为代表元，则 $[4]$ 表示该等价类。 $[1]$ 和 $[4]$ 代表的是同一个等价类

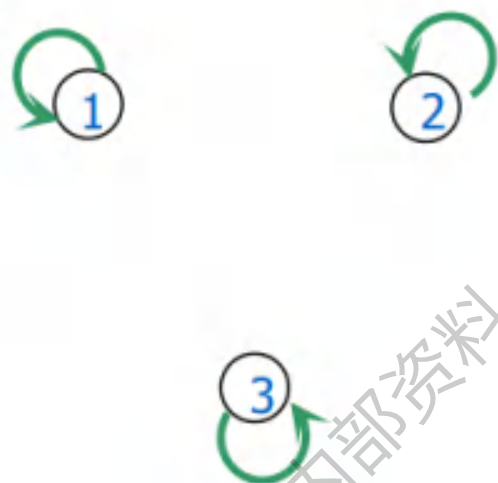


## 如何根据等价关系图求等价类？

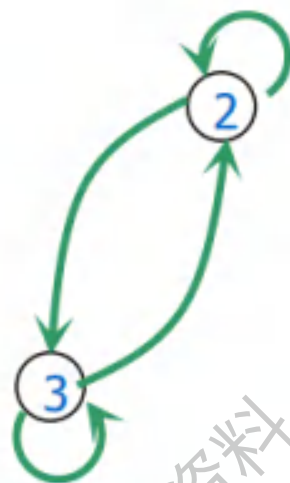
例：R关系图中每个独立子图上的节点，构成一个等价类。

独立子图个数=不同等价类个数

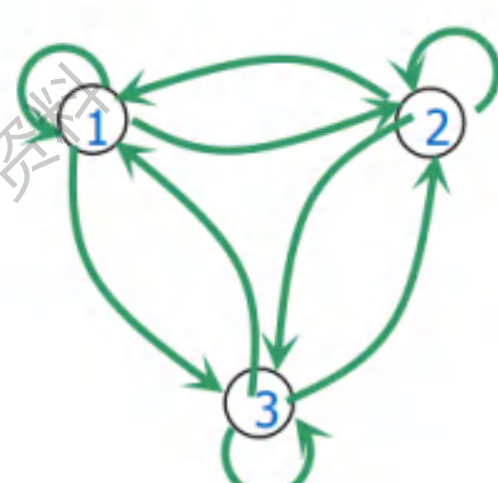
下图中，每个等价关系各几个等价类？说出对应的各个等价类



R1



R2



R3

例:

集合A为全班同学，R表示户籍属于同一个省份的关系。

$R = \{ (a, b) \mid a, b \text{ 属于同省份} \}$

等价类有哪些?

代表元?

### 3. 等价类性质

(1)  $R$  是  $A$  上等价关系, 任意  $a \in A$ , 若  $x, y \in [a]_R$ , 必有  $(x, y) \in R$ 。

含义: 同一个等价类中的元素, 彼此有等价关系。

(2)  $R$  是  $A$  上等价关系, 任意  $a, b \in A$ ,  $[a]_R \cap [b]_R = \emptyset$ , 当且仅当  $(a, b) \notin R$ 。

(3)  $[a]_R = [b]_R$ , 当且仅当  $(a, b) \in R$ 。

(4)  $R$  的所有等价类构成的集合是  $A$  的一个划分。

# Partitions 划分

- A *partition* of a set  $A$  is the set of all the equivalence classes  $\{A_1, A_2, \dots\}$  for some equivalence relation on  $A$ .
  - The  $A_i$ 's are all disjoint and their union =  $A$ .
- They “partition” the set into pieces. Within each piece, all members of that set are equivalent to each other.



# Partitions 划分

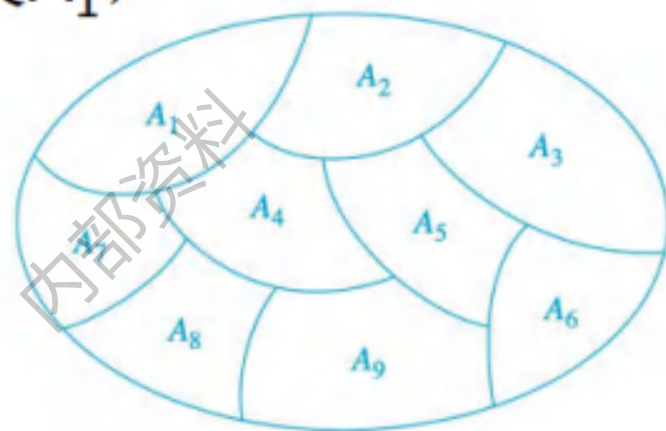
注:

等价关系 $R$ 把 $A$ 的元素分为若干类, 各类之间没有公共元素。 $R$ 是对集合 $A$ 进行的一个划分。

[定义]集合的划分: 把集合 $A$ 分为若干子集 $A_1, A_2, \dots$ , 满足:

- (1)  $A_i$  非空
- (2) 当  $i \neq j$  时  $A_i \cap A_j = \Phi$
- (3)  $\forall a \in A, \exists i,$   
使  $a \in A_i (i=1, 2, \dots)$

则集合  $P_r(A) = \{A_1, A_2, \dots, A_n, \dots\}$  称为 $A$ 的一个划分/partition。



A上最多有几个划分?

例：  $A = \{1, 2, 3, 4\}$

下列是否是划分？

$P_1 = \{\{1\}, \{2, 4\}, \{3\}\}$

$P_2 = \{\{1, 3\}, \{2, 4\}, \{3\}\}$

$P_3 = \{\emptyset, \{1, 2\}, \{3, 4\}\}$

$P_4 = \{\{1, 2, 3, 4\}\}$

$P_5 = \{\{1\}, \{2\}, \{3\}\}$

[定义]平凡划分:

集合A的最粗或最细的划分, 称为平凡划分。

例:  $A=\{1,2,3,4,5\}$

$P1=\{\{1,2,3,4,5\}\}$  ,  $P2=\{\{1\},\{2\},\{3\},\{4\},\{5\}\}$

# Equivalence Relations and Partitions

## 等价关系和划分

---

- If  $P$  is a partition of a set  $A$ , then  $P$  can be used to construct an equivalence relation on  $A$ .
- $P$ 为集合 $A$ 上的划分，通过 $P$ 可以构造 $A$ 上的等价关系？



# Theorem 1

- Let  $P$  be a partition of a set  $A$ . Define the relation  $R$  on  $A$  as follows:
  - $a R b$ 
    - If and only if
    - $a$  and  $b$  are members of the **same block**.
- Then  $R$  is an equivalence relation on  $A$ .

$R$ 定义为:  $a, b$  “同一个块”, 则  $a R b$ ,  $R$ 为: 等价关系?

例: 同学关系? 同一个宿舍?

# Proof

- 关系 $R$ 定义: 为 $a, b$ 同属于同一个块, 则有关系
- If  $a \in A$ , then clearly  $a$  is in the same block as itself; **so  $a R a$ .** (证自反)
- If  **$a R b$** , then  $a$  and  $b$  are in the same block; **so  $b R a$ .** (证对称)
- If  **$a R b$  and  $b R c$** , then  $a, b$ , and  $c$  must all lie in the same block of  **$P$** . Thus  **$a R c$ .** (证传递)

- Since  $R$  is reflexive, symmetric, and transitive,  $R$  is an equivalence relation.
- $R$  will be called the **equivalence relation determined by  $P$ .**



# Example

- ---通过划分 $P$ 确认一个等价关系
- Let  $A = \{1, 2, 3, 4\}$  and consider the partition
- $P = \{\{1, 2, 3\}, \{4\}\}$  of  $A$ . Find the equivalence relation  $R$  on  $A$  determined by  $P$ .
- Solution
  - The blocks of  $P$  are  $\{1, 2, 3\}$  and  $\{4\}$ .
  - Each element in a block is related to every other element in the same block and only to those elements.
  - Thus  $R = \{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3), (4, 4)\}$ .



# Lemma 1(引理)

- Let  $R$  be an equivalence relation on a set  $A$ , and let  $a \in A$  and  $b \in A$ .
- Then
  - $a R b$ 
    - if and only if
  - $R(a) = R(b)$

$R$ 是 $A$ 上的等价关系,如果 $a R b$ ,当且仅当 $R(a) = R(b)$



# Proof of lemma 1

- 1) 证  $R(a) = R(b) \Rightarrow a R b$ 
  - Since  $R$  is reflexive,  $b \in R(b)$ ,  $R(b) = R(a)$ ,  $\Rightarrow b \in R(a)$  so  $a R b$ .
- 2) 证  $a R b \Rightarrow R(a) = R(b)$ 
  - choose  $x \in R(b)$
  - $b R x$
  - $a R b$
  - $a R x$  (transitive)
  - $x \in R(a)$ .
  - Thus  $R(b) \subseteq R(a)$
  - A completely similar argument shows that,  $R(a) \subseteq R(b)$
  - $R(b) \subseteq R(a)$  and  $R(a) \subseteq R(b)$
- so  $R(a) = R(b)$

• QED

# Theorem 2

- Let  $R$  be an equivalence relation on  $A$ , and let  $\mathbf{P}$  be the collection of all distinct relative sets  $R(a)$  for  $a$  in  $A$ .
  - Then  $\mathbf{P}$  is a partition of  $A$ , and  $R$  is the equivalence relation determined by  $\mathbf{P}$ .
- 设 $R$ 是 $A$ 上的等价关系， $P$ 是所有不同关系集 $R(a)$ 的集合， $a \in A$ ，那么 $P$ 是 $A$ 的一个划分，并且 $R$ 是由 $P$ 确定的等价关系。

# Proof

## • 1) 证由等价关系R, 确定P :

- 首先证明: 任何一个元素a一定属于P的某个元素R(a)
- 其次证明: 如果 $R(a) \neq R(b)$ , 则 $R(a) \cap R(b) = \emptyset$

•  $P = \{R(a) \mid a \in A\}$

—因为R是自反的, 所以  $a R a, \Rightarrow a \in R(a)$

—所以  $\forall a$  一定属于划分P中的某个块

—证:  $R(a) \neq R(b)$ , 则 $R(a) \cap R(b) = \emptyset$ ,

—转换为证  $R(a) \cap R(b) \neq \emptyset$ , 则  $R(a) = R(b)$ .

- Contrapositive(换质位命题、对换式、逆反式)



Proof :

证明如果  $R(a) \cap R(b) \neq \emptyset$  , 那么  $R(a) = R(b)$  ?

- Let  $c \in R(a) \cap R(b)$  , then  $c \in R(a)$  and  $c \in R(b)$ 
  - $\Rightarrow a R c, b R c$
  - $c R b$ , since  $R$  is symmetric
  - $a R c, c R b \Rightarrow a R b$ , since  $R$  is transitive
  - $a R b \Rightarrow R(a) = R(b)$  by Lemma 1
- So,  $P$  is a partition
- $a R b$  iff  $a$  and  $b$  belong to the same block of  $P$ , Thus  $P$  determines  $R$ .
- 2) 当且仅当  $a, b$  同属于  $P$  中的一个块, 则  $P$  可确定等价关系  $R$ , (定理1)

• QED



例：集合 $A=\{1, 2, 3, 4\}$ ，等价关系如下：

$$R = \{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3), (4, 4)\}$$

求划分 $P$

解：  $R(1)=R(2)=R(3)=\{1, 2, 3\}$  ,  $R(4)=\{4\}$

$$P = \{R(1), R(4)\}$$

例：集合 $A=\{1, 2, 3, 4\}$ , 恒等关系如下：

$$I_R = \{(1, 1), (2, 2), (3, 3), (4, 4)\}$$

求划分 $P$

解：  $R(1)=\{1\}; R(2)=\{2\}; R(3)=\{3\}, R(4)=\{4\}$

$$P = \{R(1), R(2), R(3), R(4)\}$$

例：上例的全域关系的划分？

# Equivalence classes and quotient set

## 等价类和商集

- If  $R$  is an equivalence relation on  $A$ , then the sets  $R(a)$  are traditionally called *equivalence classes* of  $R$ , denoted by  $[a]$
- The partition  $\mathbf{P}$  consists of all equivalence classes of  $R$  is denoted by  $A/R$  and called
  - the *quotient set* (商集), or
  - the *partition of  $A$  induced by  $R$* , or,
  - *$A$  modulo  $R$* .

等价关系 $R$ 确定 $A$ 上的等价类，等价类的集合，称为 $A$ 上关于 $R$ 的商集

# Example 7

- Let

- $A = \{1, 2, 3, 4\}$

- $R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 3), (3, 3), (4, 4)\}.$

- Determine  $A/R$ .

- Solution

- $R(1) = \{1, 2\} = R(2)$

- $R(3) = \{3, 4\} = R(4).$

- Hence  $A/R = \{[1,], [3]\} = \{R(1), R(3)\}$



# Determining partitions $A/R$

- **STEP 1:** Choose any element of  $A$  and compute the equivalence class  $R(a)$ .
- **STEP 2:** if  $R(a) \neq A$ , choose an element  $b$ , not included in  $R(a)$ , and compute the equivalence class  $R(b)$ .
- **STEP 3:** If  $A$  is not the union of previously computed equivalence classes, then choose an element  $x$  of  $A$  that is not in any of those equivalence classes and compute  $R(x)$ .
- **STEP 4:** Repeat step 3 until all elements of  $A$  are included in the computed equivalence classes. If  $A$  is countable, this process could continue indefinitely. In that case, continue until a pattern emerges that allows you to describe or give a formula for all equivalence classes.

## Example 14

- What are the sets in the partition of the integers arising from **congruence modulo 4**?

模4同余产生的整数划分是什么？

$\{ R(0), R(1), R(2), R(3) \}$

或者

$\{ [0], [1], [2], [3] \}$

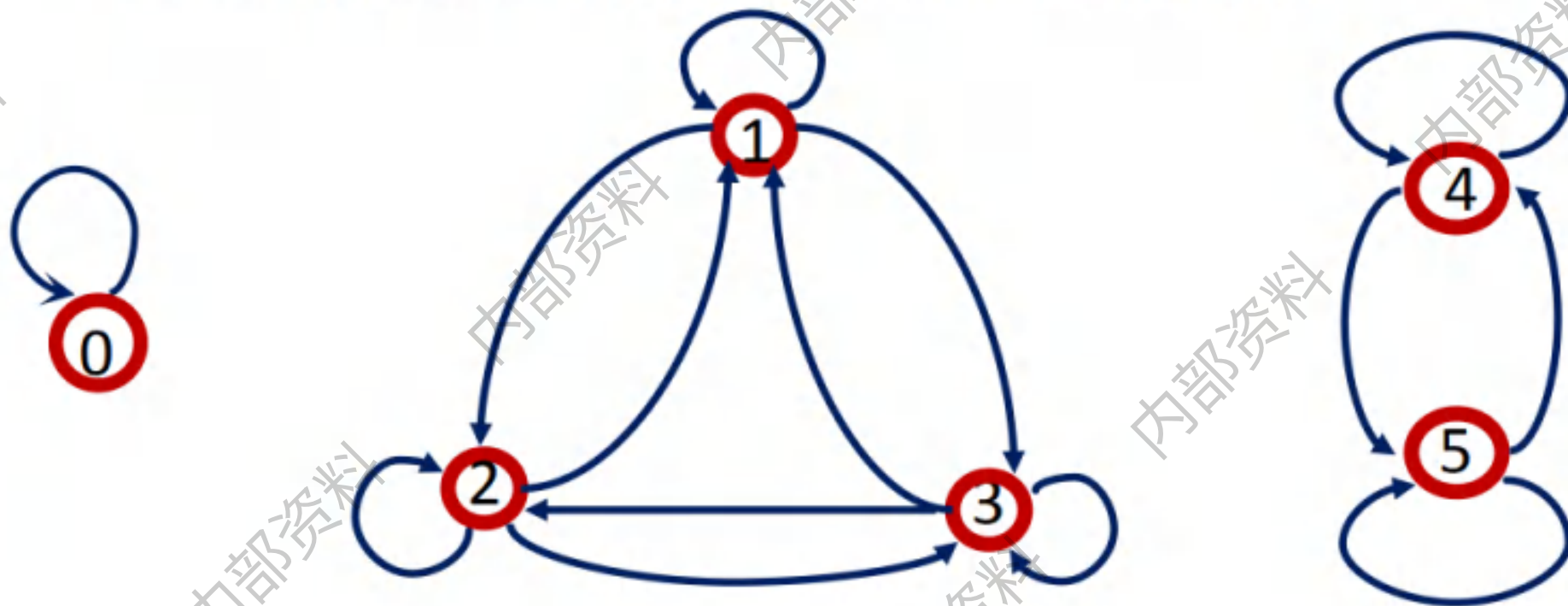
## Example 8

- Let  $R$  be the equivalence relation defined in congruence modulo 2, determine  $A / R$ .
- Solution
  - $R(0) = \{\dots, -6, -4, -2, 0, 2, 4, 6, \dots\}$  — 无穷偶数集合
  - $R(1) = \{\dots, -5, -3, -1, 1, 3, 5, 7, \dots\}$  --- 无穷奇数集合
  - Hence  $A/R$  consists of the set of even integers and the set of odd integers
  - $A/R = \{R(0), R(1)\}$  --- 有限集合



# 有限集合划分的判断：

- (1) 等价关系，用穷举法很难判别
- (2) 用关系图比较直观，明确
- (3) 用集合的划分来研究等价关系



三个不连通的图



# 相容关系的定义

**定义：**给定集合 $X$ 上的关系 $R$ ，若 $R$ 是自反的、对称的，则 $R$ 是 $A$ 上相容关系。

**例：** $X$ 是由一些英文单词构成的集合。

$$X = \{fly, any, able, key, book, pump, fit\}$$

$X$ 上的关系 $R$ :  $R = \{(\alpha, \beta) | \alpha \in X, \beta \in X \text{ 且 } \alpha \text{ 与 } \beta \text{ 含有相同字母}\}$

— $R$ 具有自反性，任意一个 $X$ 上的单词 $\alpha$ ， $\alpha$ 和 $\alpha$ 具有相同字母。

— $R$ 具有对称性，如果 $\alpha$ 与 $\beta$ 具有相同字母，则 $\beta$ 与 $\alpha$ 也具有相同字母。

— $R$ 不具有传递性，例如： $fly$ 与 $any$ 具有相同字母 $y$ ， $fly$ 与 $fit$ 具有相同字母 $f$ ，但 $any$ 和 $fit$ 没有相同字母；

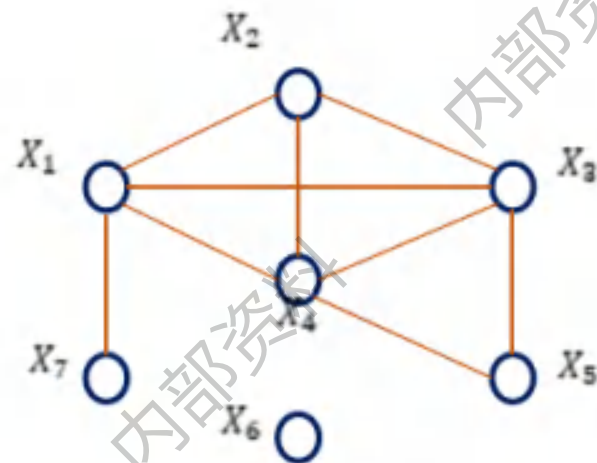
# 相容关系的关系图

相容关系图的简化：（1）不画环；（2）简化双向边：两条对称边用一条无向边代替。

令  $x_1 = fly, x_2 = any, x_3 = able, x_4 = key, x_5 = book, x_6 = pump, x_7 = fit$ ,

**关系  $R$ : 有相同字母**

$$X = \{x_1, x_2, x_3, x_4, x_5, x_6, x_7\}$$



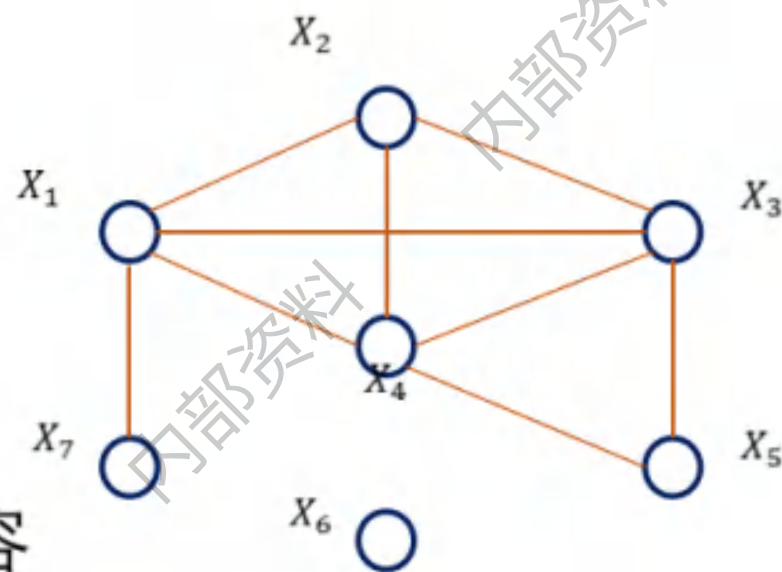
$r$  的简化图

# 相容类及最大相容类

**定义：** 设 $R$ 是集合 $X$ 上的相容关系,  $C \subseteq X$ , 如果对于 $C$ 中任意元素 $x, y$ , 有 $(x, y) \in R$ , 称 $C$ 是 $R$ 的一个相容类。

上例中 $\{x_1, x_2\}$ ,  $\{x_3, x_4\}$ ,  $\{x_1, x_2, x_3\}$ ,  
 $\{x_2, x_3, x_4\}$ ,  $\{x_1, x_2, x_4\}$ ,  $\{x_3, x_4, x_5\}$ ,  
 $\{x_1, x_3, x_4\}$ ,  $\{x_1, x_2, x_3, x_4\}$ ,  $\{x_1, x_7\}$ ,  
 $\{x_6\}$

都是相容类。上述相容类中, 有些相容类间有真包含关系, 例如 $\{x_1, x_2\}$ 和 $\{x_1, x_2, x_3, x_4\}$ 之间。



$R$ 的简化图



# 相容类及最大相容类

**定义：** 设 $R$ 是集合 $X$ 上的相容关系， $C$ 是 $R$ 的一个相容类，如果 $C$ 不能被其它相容类所真包含，则称 $C$ 是一个最大相容类。

也可以说， $C$ 是一个相容类，如果 $C$ 中加入任意一个新元素，就不再是相容类， $C$ 就是一个最大相容类。

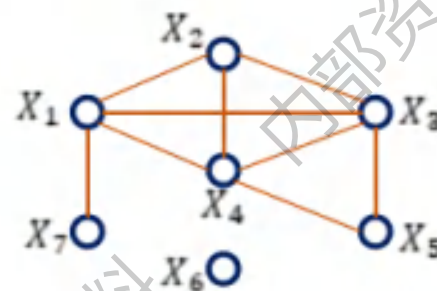
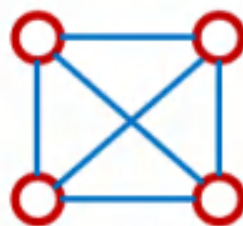
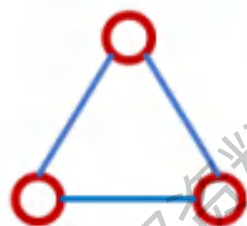
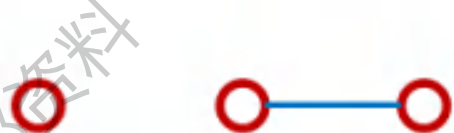
$\{x_3, x_4, x_5\}$ ,  
 $\{x_1, x_2, x_3, x_4\}$ ,  
 $\{x_1, x_7\}$ ,  $\{x_6\}$ 都是最大相容类。

令 $x_1 = fly$ ,  $x_2 = any$ ,  $x_3 = able$ ,  $x_4 = key$ ,  $x_5 = book$ ,  $x_6 = pump$ ,  $x_7 = fit$ ,  $X = \{x_1, x_2, x_3, x_4, x_5, x_6, x_7\}$   $X$ 上的具有相同字母关系 $R$



# 相容类及最大相容类

从简化图找最大相容类：找最大完全多边形，即含有结点最多，每个结点都与其它所有结点相连的多边形。



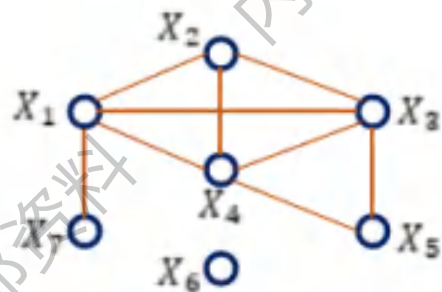
$R$ 的简化图

在相容关系简化图中，每个最大完全多边形的结点集合构成一个最大相容类。

上例中最大相容类 $\{x_1, x_2, x_3, x_4\}$ ,  $\{x_3, x_4, x_5\}$ ,  $\{x_1, x_7\}$ ,  $\{x_6\}$ 分别对应最大完全四、三、一、0边形。

# 最大相容类-完全覆盖

**定义：** $R$ 是 $X$ 中的相容关系，由 $R$ 的所有**最大相容类**为元素构成的集合，称之为 $X$ 的完全覆盖。记作 $C_r(X)$ 。



$\{x_1, x_2, x_3, x_4\}$ ,  $\{x_3, x_4, x_5\}$ ,  $\{x_1, x_7\}$ ,  $\{x_6\}$ 是 $r$ 的最大相容类。

$$C_r(X) = \{\{x_1, x_2, x_3, x_4\}, \{x_3, x_4, x_5\}, \{x_1, x_7\}, \{x_6\}\}$$

## 思考： 等价关系的运算

- $R_1 \cap R_2$ ——等价关系
- $R_1 \cup R_2$ ——相容关系

# 等价关系的运算

[定理2]:

若 $R_1, R_2$ 是 $A$ 上的等价关系, 则 $R_1 \cap R_2$ 也是等价关系。

证明:  $R_1 \cap R_2 = R_3$ ,  $R_3$ 传递。

(1)  $\forall a \in A$ , 由 $R_1, R_2$ 自反

$(a, a) \in R_1, (a, a) \in R_2$ , 于是

$(a, a) \in R_1 \cap R_2 = R_3$ ,  $R_3$ 自反

——自反

(2) 若  $(a, b) \in R_3$ , 则  $(a, b) \in R_1$

且  $(a, b) \in R_2$ , 由 $R_1, R_2$ 对称

$(b, a) \in R_1, (b, a) \in R_2$

于是  $(b, a) \in R_1 \cap R_2 = R_3$ ,  $R_3$ 对称

——对称



(3) 若  $(a, b) \in R_3, (b, c) \in R_3$

于是  $(a, c) \in R_1 \cap R_2 = R_3, R_3$  传递。

——传递

归纳以上三点,  $R_3 = R_1 \cap R_2$  也是等价的二元关系。

[定理3]: 若  $R_1, R_2$  是  $A$  上等价关系, 则  $R_1 \cup R_2$  是相容关系。

葛 注:  $R_1 \cup R_2$  不一定是等价关系。

证明： 设  $R_4 = R_1 \cup R_2$

(1)  $\forall a \in A, (a, a) \in R_1, (a, a) \in R_2$ , 于是  $(a, a) \in R_1 \cup R_2 = R_4$ 。  
——自反性

(2) 若  $(a, b) \in R_4, (a, b) \in R_1$  或  $(a, b) \in R_2$  由  $R_1, R_2$  对称性,  $(b, a) \in R_1$  或  $(b, a) \in R_2$ ,  
于是  $(b, a) \in R_1 \cup R_2 = R_4$  ——对称性

注：  $R_4 = R_1 \cup R_2$  是相容的，即  $R_1 \cup R_2$  不一定是等价关系

若  $(a, b) \in R_4, (b, c) \in R_4$

$(a, b)$  与  $(b, c)$  可能分别属于  $R_1$  与  $R_2$ , 无法保证  $R_1 \cup R_2$  的传递性。

例：

$$A = \{a, b, c\} \text{ 滢}$$

$$R_1 = \{ (a, a), (b, b), (c, c), (a, b), (b, a) \}$$

$$R_2 = \{ (a, a), (b, b), (c, c), (b, c), (c, b) \}$$

$$R_4 = R_1 \cup R_2 = \{ (a, a), (b, b), (c, c), (a, b), (b, a), (b, c), (c, b) \} \text{ 滢}$$

显然,  $(a, b), (b, c) \in R_1 \cup R_2$

但  $(a, c) \notin R_1 \cup R_2$  (不具有传递性)。

# 作业

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- § 9.4 – 20, 22, 28
- § 9.5 – 16, 56, 60, 64



# § 9.6: Partial Orderings

## --偏序

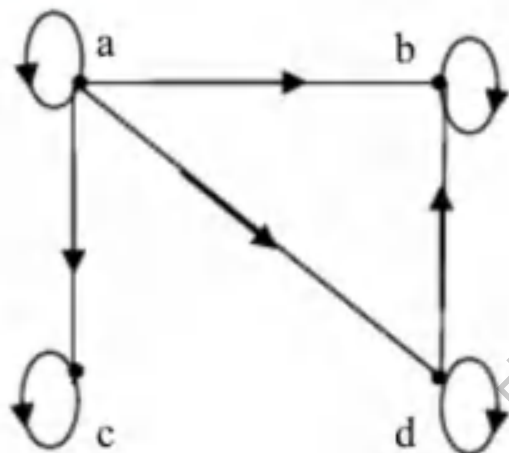
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- Partial Order
  - Linear order (线序)
  - Quasiorder (拟序)
  - Product Partial Order (乘积偏序)
  - Lexicographic Order (词典顺序)
- Hasse Diagram (哈斯图)
- Topological Sorting (拓扑排序)

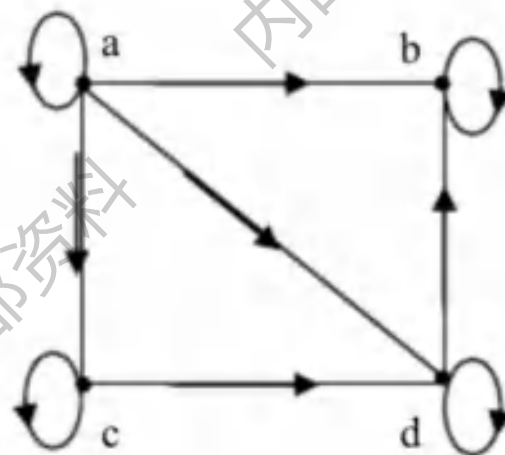
## § 9.6: Partial Orderings 偏序

- **Partial order (偏序关系)**: A relation  $R$  on a set  $A$  is called a *partial order* if  $R$  is reflexive (自反), antisymmetric (反对称), and transitive (传递)
- 用符号 “ $\leq$ ” 表示偏序

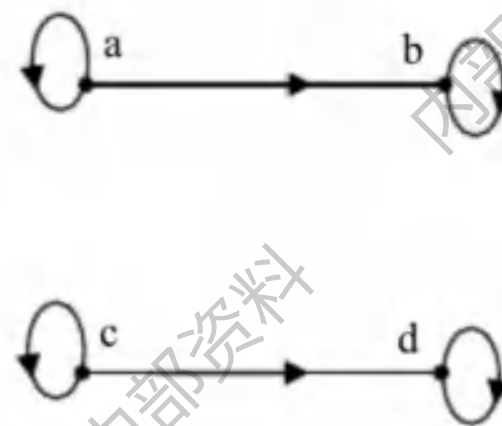
例：下面关系图是否是偏序关系



(a)



(b)



(c)

例：判断关系矩阵是否为偏序关系

$$\begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(a)

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

(b)

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

(c)



# PARTIALLY ORDERED SETS (偏序集)

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- **Partially ordered set:** The set  $A$  together with the partial order  $R$  is called a *partially ordered set* (偏序集), or simply a *poset*, and we will denote this poset by  $(A, R)$ .

# Examples

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- $(\mathbb{Z}^+, \leq)$
- $(\mathbb{Z}^+, \geq)$
- $(\mathbb{Z}^+, |)$
- $(\mathbb{Z}^+, <)$  --- is not a poset

# linear order (线序关系)

- If  $(A, \leq)$  is a poset, the elements  $a$  and  $b$  are said to be
  - *Comparable* (可比的) if  $a \leq b$  or  $b \leq a$ .
  - *incomparable* (不可比的) if neither  $a \leq b$  nor  $b \leq a$ .
- 比如: poset  $(\mathbb{Z}^+, |)$ , 3和9是可比的, 5和7是不可比的

# linear order (线序关系)

- If every pair of elements in a poset  $A$  is comparable, we say that  $A$  is called a *totally ordered or linearly ordered set* (全序集或线序集), and the partial order is called a *totally order or linear order* (全序关系或线序关系). We also say that  $A$  *totally ordered set* is a *chain* (链).
  - $(\mathbb{Z}^+, \leq)$  ---任何两个元素是可比的, 是链
  - $(\mathbb{Z}^+, |)$  ---有些元素间不可比, 比如2不能整除3, 不是链.



# Well-ordered (良序)

- $(S, \leq)$  is a **well-ordered set** if it is a poset such that  $\leq$  is a **total ordering** and every **nonempty** subset of  $S$  has a **least element**.

有限关系集合的全序集一定是良序集

无限关系集合的全序集不一定是良序集

例: 集合  $(\mathbb{Z}_+, \leq)$ , 是良序集, 任何子集总能找到最小元

例: 集合  $(\mathbb{Z}, \leq)$ , 不是良序集, 因为  $\mathbb{Z}$  的负整数子集, 没有最小元

# Quasiorder(拟序关系)

- **Quasiorder**: A relation  $R$  on a set  $A$  is called **quasiorder** if it is **transitive** and **irreflexive**.

**Transitive** (传递)

**Irreflexive** (反自反)

拟序关系是否具有反对称性?

**Example:**

$(P(S), \subset)$ ,  $(Z^+, <)$

如 $R$ 为拟序, 则 $R \cup I_A$ 为偏序。偏序是拟序的扩展, 拟序是偏序的缩减

# Product partial order (乘积偏序)

$((a,b), (a',b'))$

- This ordering is called *product partial order*.
- 乘积偏序要满足 有序对的第一元素和第二元素分别同时满足偏序关系

$$a, a' \in A \text{ and } b, b' \in B$$

$$a \leq a' \wedge b \leq b' \Rightarrow (a,b) \leq (a',b')$$



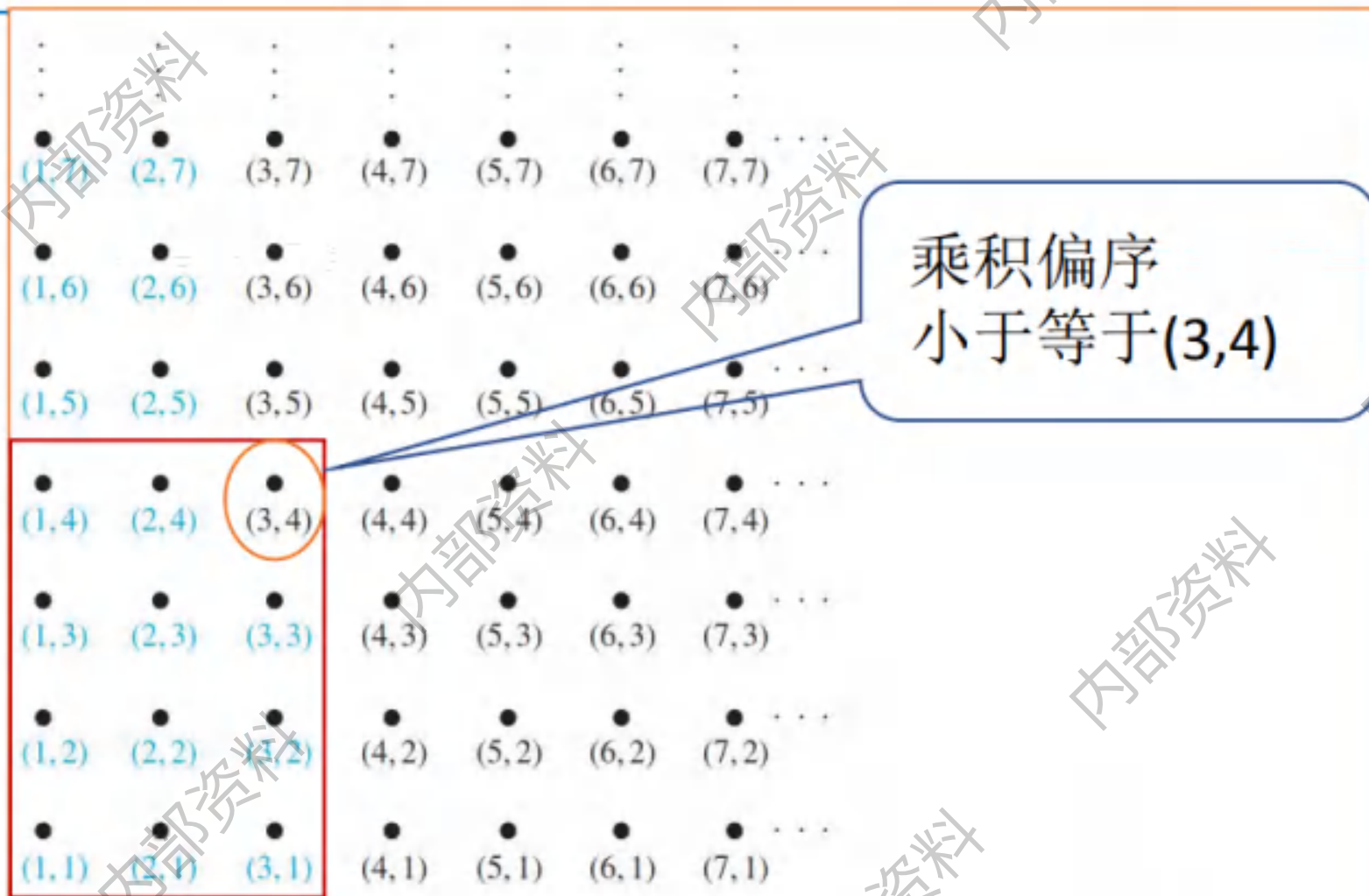
# Lexicographic order (词典顺序)

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- This ordering is called *lexicographic*, or “*dictionary*” *order*.
- Extended to Cartesian products  $A_1 \times A_2 \times \dots \times A_n$
- Ordinary alphabet help helping helper helping
- 词典排序: 首先比较第一元素, 依次比较第二元素.....



例9： $\mathbb{Z}^+ \times \mathbb{Z}^+$ 中的有序对，按词典排序小于(3, 4)的



**FIGURE 1** The Ordered Pairs Less Than (3, 4) in Lexicographic Order.

# Covering Relation 覆盖关系

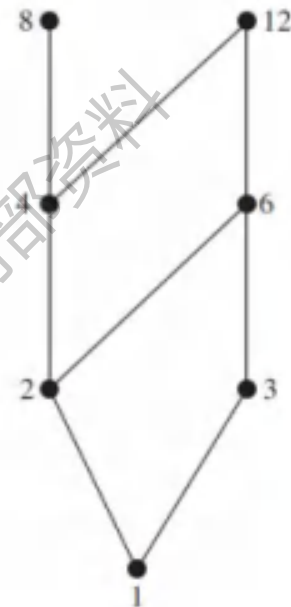
Let  $(S, \preceq)$  be a poset. We say that an element  $y \in S$  covers an element  $x \in S$  if  $x < y$  and there is no element  $z \in S$  such that  $x < z < y$ . The set of pairs  $(x, y)$  such that  $y$  covers  $x$  is called the covering relation  $\prec$  of  $(S, \preceq)$ .

例:  $S = \{1, 2, 3, 4, 6, 8, 12\}$ , 偏序集  
 $(S, |)$  的覆盖关系?  
 $R = \{ (1, 2), (1, 3), \dots \}$

$(2, 4)$  ?

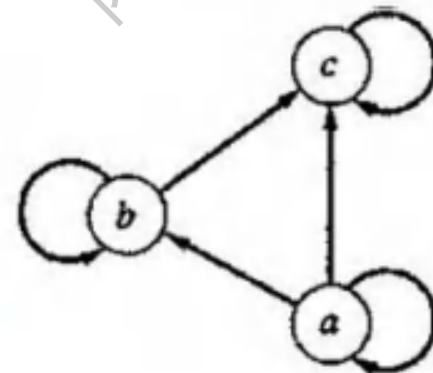
$(2, 8)$  ?

$(2, 6)$  ?



求关系图的传递闭包?

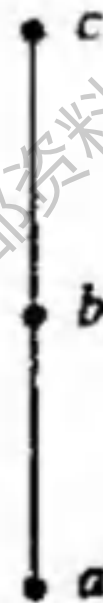
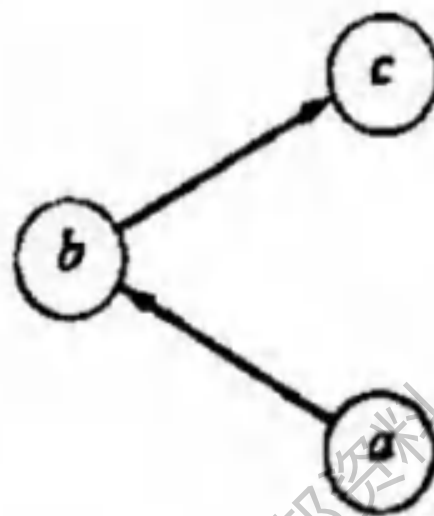
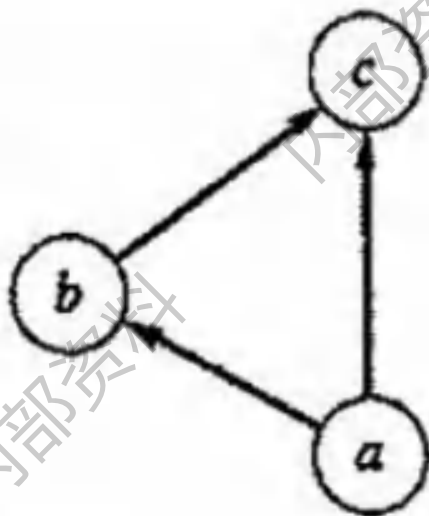
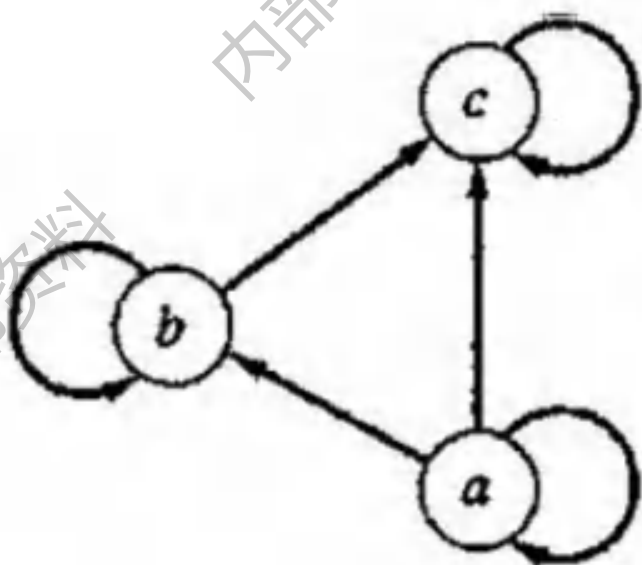
# Hasse Diagrams (哈斯图)



- Hasse Diagram is the digraph of a partial order on a finite set  $A$  that
  - deletes all self-cycles (删除自环)
  - eliminates all edges that are implied by the transitive property (取消传递性边)
  - draws the digraph with all edges pointing upward (边方向向上)

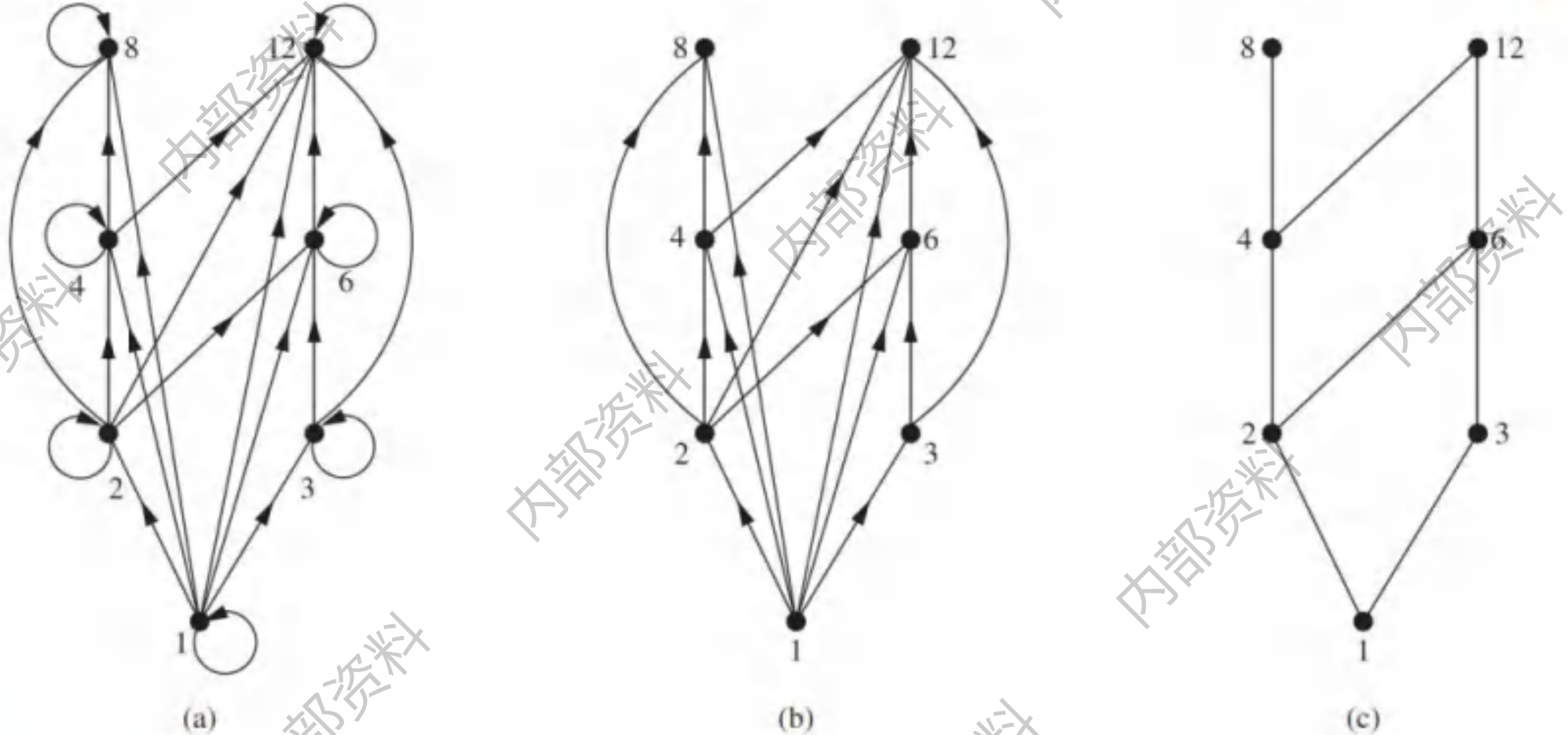


# Hasse Diagram





# Example 13



**FIGURE 3** Constructing the Hasse Diagram of  $(\{1, 2, 3, 4, 6, 8, 12\}, |)$ .

# 例-根据覆盖关系求偏序

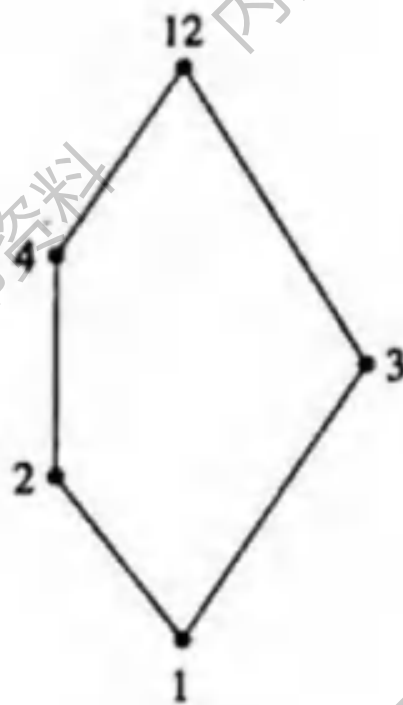
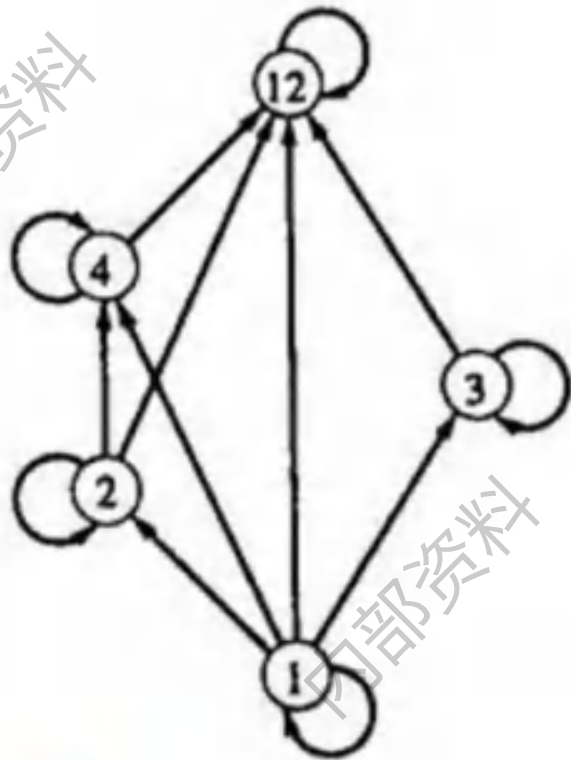
$$\begin{bmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

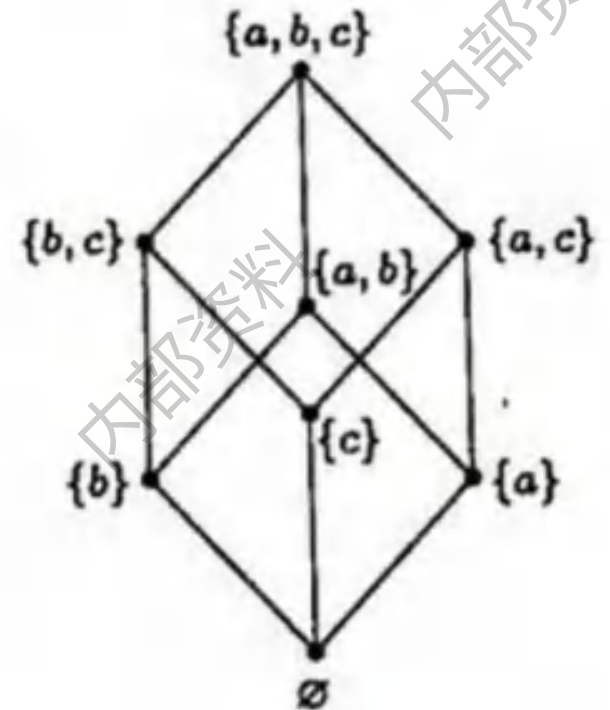
$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

# Hasse Diagram

•  $(\{1, 2, 3, 4, 12\}, |)$



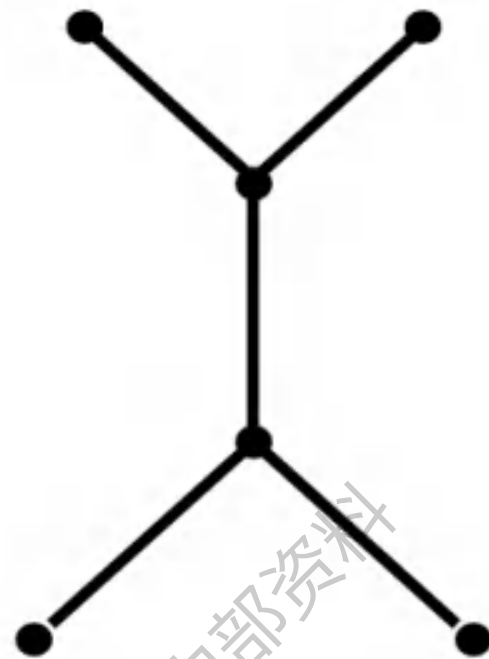
•  $(P(\{a, b, c\}), \subseteq)$



例1:

$A = \{2, 3, 6, 12, 24, 36\}$  中,  $(A, \leq)$

为偏序集整除关系, 求偏序集的  
哈斯图:



例2:

$A = \{1, 2, 3, 4, 5, 6, 8, 10, 12, 15, 16\}$

$(A, \leq)$  为偏序集整除关系



# 偏序集的重要元素

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Extremal elements  
(极值元素)

# Extremal elements

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- *maximal element* (极大元)
- *minimal element* (极小元)
- *greatest element* (最大元)
  - denoted by  $1$  and often called the *unit element* (单位元)
- *least element* (最小元)
  - denoted by  $0$  and often called the *zero element* (零元)

# Maximal(minimal) element

## 极大 (极小) 元

- **极大元**: 设  $(P, \leq)$  是偏序集, 子集  $A \subseteq P$ , 若  $a \in A$ , 且在  $A$  中找不到一个元素  $b$  ( $b \neq a$ ), 使  $a \leq b$ , 则  $a$  为  $A$  的极大元
- **极小元**: 设  $(P, \leq)$  是偏序集, 子集  $A \subseteq P$ , 若  $a \in A$ , 且在  $A$  中找不到一个元素  $b$  ( $b \neq a$ ), 使  $b \leq a$ , 则  $a$  为  $A$  的极小元.
- **注意: 极大元(极小元)只能在子集  $A$  中找, 不能在全集中找**

An element  $a \in A$  is called a maximal element of  $A$  if there is no element  $c$  in  $A$  such that  $a < c$ .

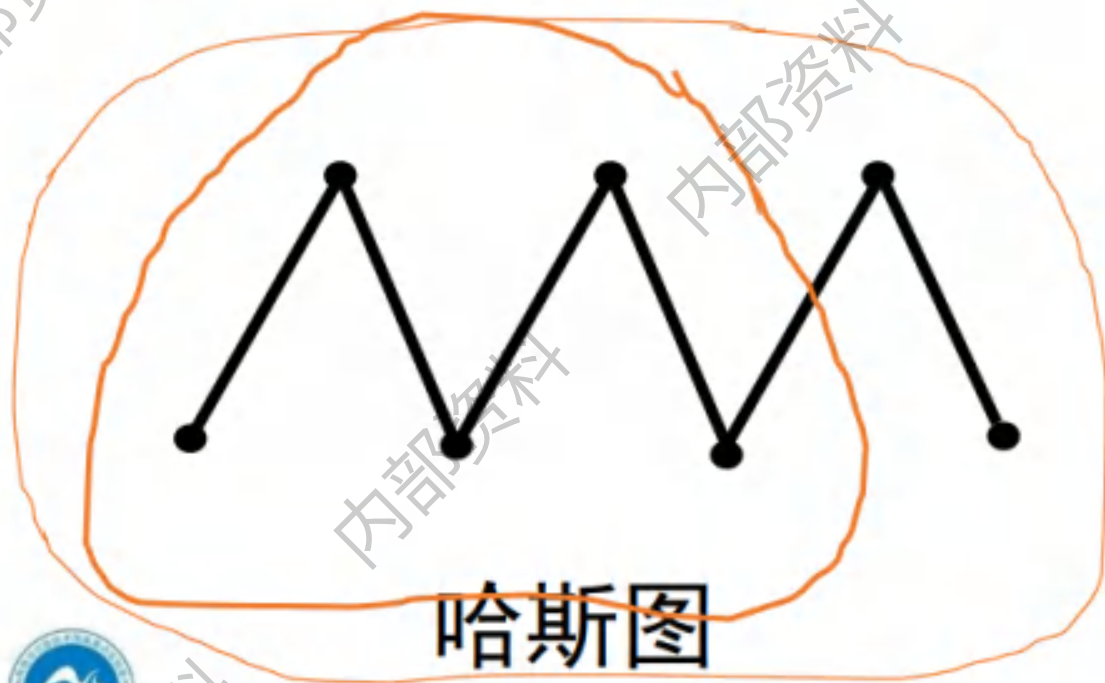
An element  $b \in A$  is called a minimal element of  $A$  if there is no element  $c$  in  $A$  such that  $c < b$ .

例:  $P = \{2, 3, 5, 7, 14, 15, 21\}$ , 整除关系, 子集

$A = \{2, 7, 3, 21, 14\}$

极大元:  $\{14, 21\}$

极小元:  $\{2, 7, 3\}$



当  $A = P$  时

极大元为 14, 15, 21

极小元为 2, 3, 5, 7.

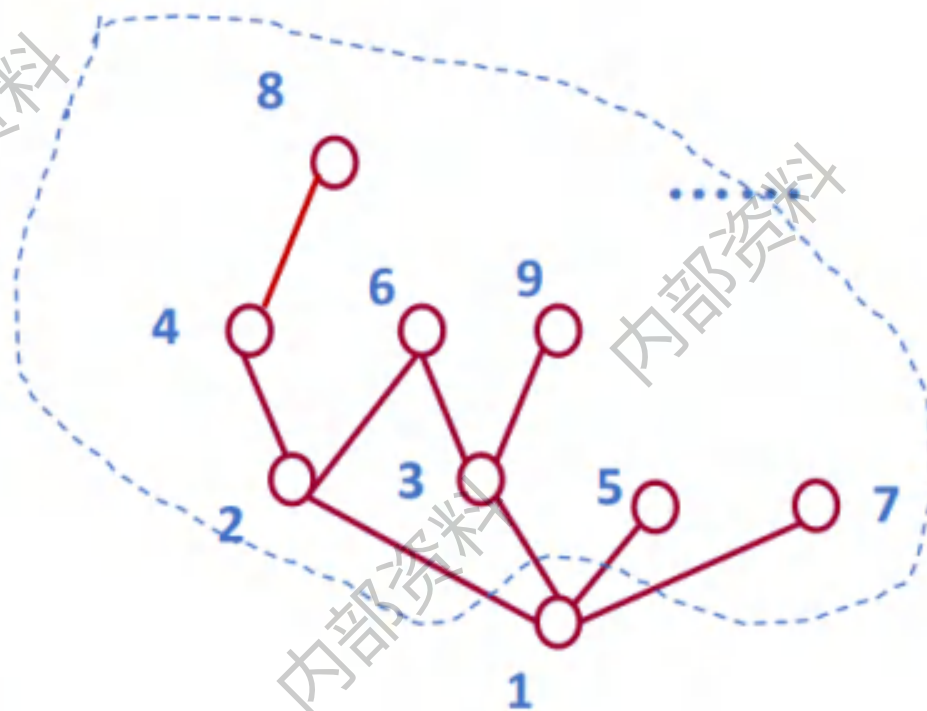


例：自然数集合 $N$ 上的整除关系  
“ $|$ ”构成偏序集  $(N, |)$

$A = \{2, 3, 4, 5, 6, 7, 8, 9\}$

$A$ 中极大元：8, 6, 9, 5, 7

$A$ 中极小元：2, 3, 5, 7



注意：极大元、极小元并不唯一，且同一元素即可能是极大元，又可能是极小元

# Theorem 1

- Let  $A$  be a finite nonempty poset with partial order  $\leq$ , then  $A$  has at least one maximal element and at least one minimal element.

有限非空偏序集，集合 $A$ 至少有一个极大元和一个极小元

# Greatest (least) element

## 最大元 (最小元)

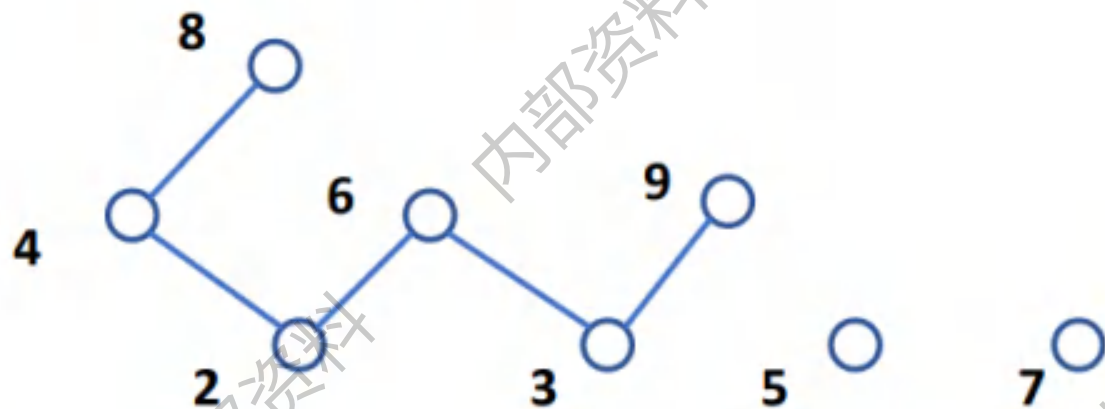
**定义：**偏序集  $(P, \leq)$  中，子集  $A \subseteq P$ ，若  $a \in A$ ，对  $\forall b \in A$ ，都有  $b \leq a$  则  $a$  为  $A$  的最大元；若  $a \in A$ ，对  $\forall b \in A$ ，都有  $a \leq b$  则  $a$  为  $A$  的最小元。

**注意：**最大元和最小元在子集  $A$  中找。子集  $A$  的最大元 (最小元) 如果存在，则与  $A$  中的任意元素必须可比

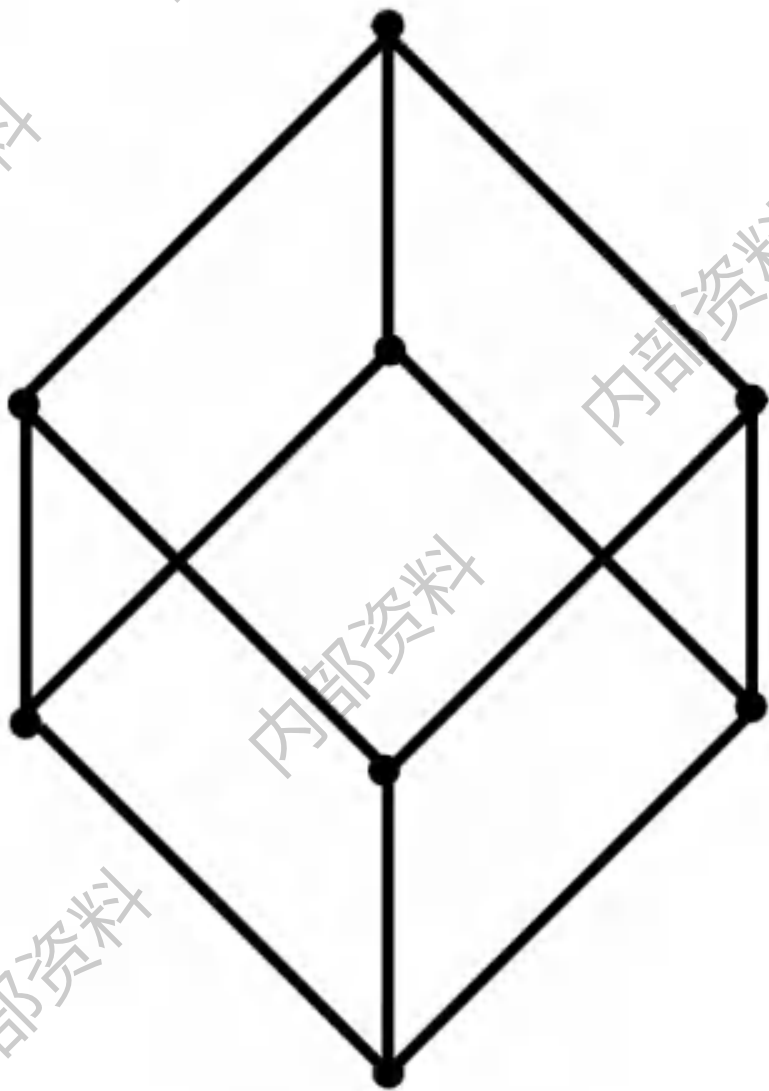
例： $(N, |)$  的哈斯图

$A = \{2, 3, 4, 5, 6, 7, 8, 9\}$

无最大元，无最小元



$(P\{a,b,c\}, R \subseteq )$

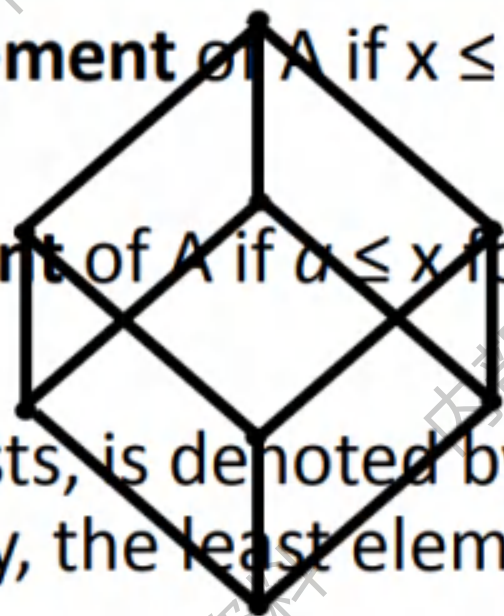




# Greatest (least) element

## 集合A上的最大（最小）元

- An element  $a \in A$  is called a **greatest element** of  $A$  if  $x \leq a$  for all  $x \in A$ .
- An element  $a \in A$  is called a **least element** of  $A$  if  $a \leq x$  for all  $x \in A$ .
- The greatest element of a poset, if it exists, is denoted by  $1$ , and is often called the **unit** element. Similarly, the least element is called **zero** element.



➤ 如果A是全集，则最大元，通常用 **1** 表示--称为单位元

➤ A的最小元，通常用 **0** 表示--称为零元

# Lub (最小上界)

# Glb (最大下界)

- poset  $A$  and a subset  $B \subseteq A$ . (B是A的子集)

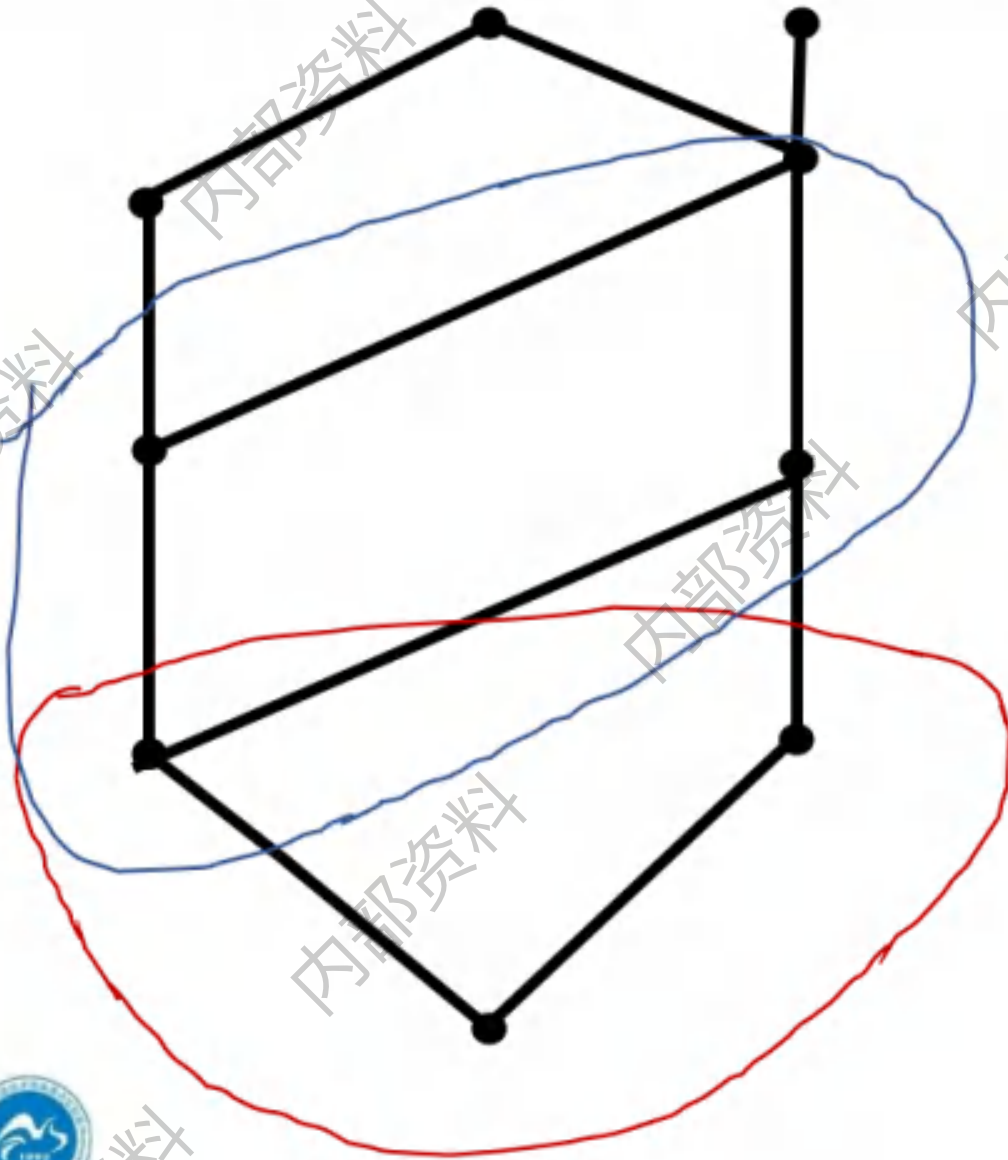
注意:

- An element  $a \in A$  is called **a upper bound (上界)** of  $B$  if  $b \leq a$  for all  $b \in B$ .  
上界和下界都在全集A中找, 不限于子集B
- An element  $a \in A$  is called **a lower bound (下界)** of  $B$  if  $a \leq b$  for all  $b \in B$ .
- 上界和下界可以不唯一, 也可以不存在
- 如果存在最小上界和最大下界, 则一定唯一
- An element  $a \in A$  is called **a least upper bound (LUB)** of  $B$  if  $a$  is an upper bound of  $B$  and  $a \leq a'$ , whenever  $a'$  is an upper bound of  $B$ .
- An element  $a \in A$  is called **a greatest lower bound (GLB)** of  $B$  if  $a$  is an lower bound of  $B$  and  $a' \leq a$ , whenever  $a'$  is an lower bound of  $B$ .



**Example 18**: Find the lower and upper bounds of

$\{a, b, c\}, \{j, h\}, \{b, e, d, f\}$



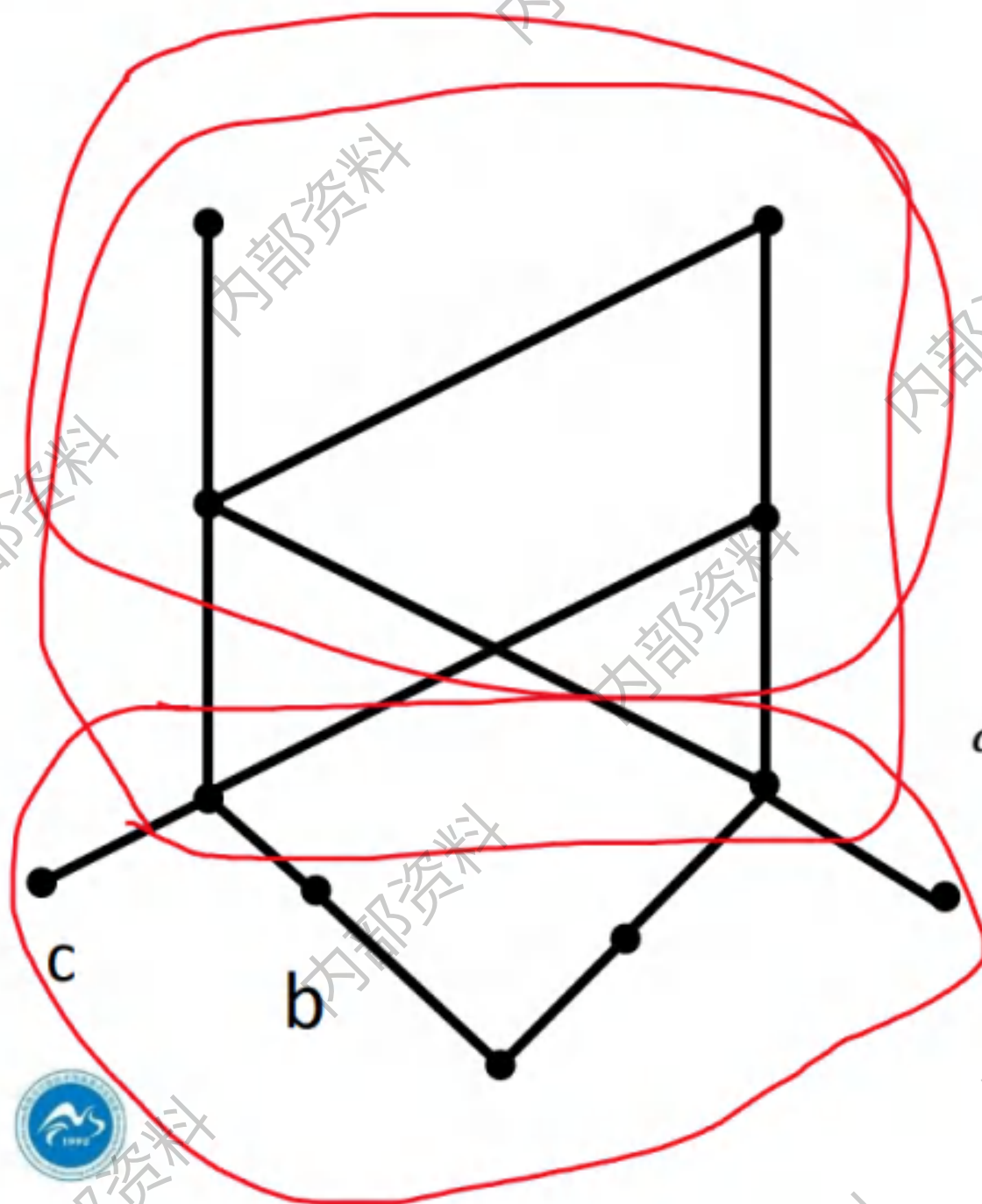
LUB and GLB of  $\{b, d, g\}$  ?

## Example 20

Find the GLB and LUB of  $\{3,9,12\}$  and  $\{1,2,4,5,10\}$ , in the poset  $(\mathbb{Z}^+, |)$

- LUB of  $\{3,9,12\}$ : =LCM(最小公倍数)36, GLB: =GCD(最大公约数) 3
- $\{1,2,4,5,10\}$ , LUB: =LCM ,20, GLB: =GCD, 1





$h, i, j, k$  为  $B$  的上界

$f, g$  为  $B' = \{j, h, k, i\}$  的下界

$a, b, c, d, e$  也为  $B' = \{j, h, k, i\}$  的下界

$a$  是  $\{f, g, h, i, j, k\}$  的下界

$b, c, d, e$  不是  $\{f, g, h, i, j, k\}$  的下界

# Theorem 1,2,3

- 总结

- Let  $(A, \leq)$  be a poset

- If  $A$  is a finite nonempty, then  $A$  has at least **one** maximal element and at least **one** minimal element. (有限非空集合极大元/极小元各至少一个)
- $A$  has at most **one** greatest element and at most **one** least element. (最大元/最小元各最多一个)
- A subset  $B$  of  $A$  has at most **one** LUB and at most **one** GLB. (子集 $B$ 的最小上界/最大下界各最多一个)

思考：什么情况可比，什么情况不可比？

# Topological Sorting (拓扑排序)

---

- If  $A$  is a poset with partial order  $\leq$ , we sometimes need to find *a linear order* for the set  $A$  that will merely be an extension of the given partial order in the sense that
  - if  $a \leq b$ , then  $a < b$ .
- The process of constructing a linear order such as  $<$  is called *topological sorting*



# SORT Algorithm

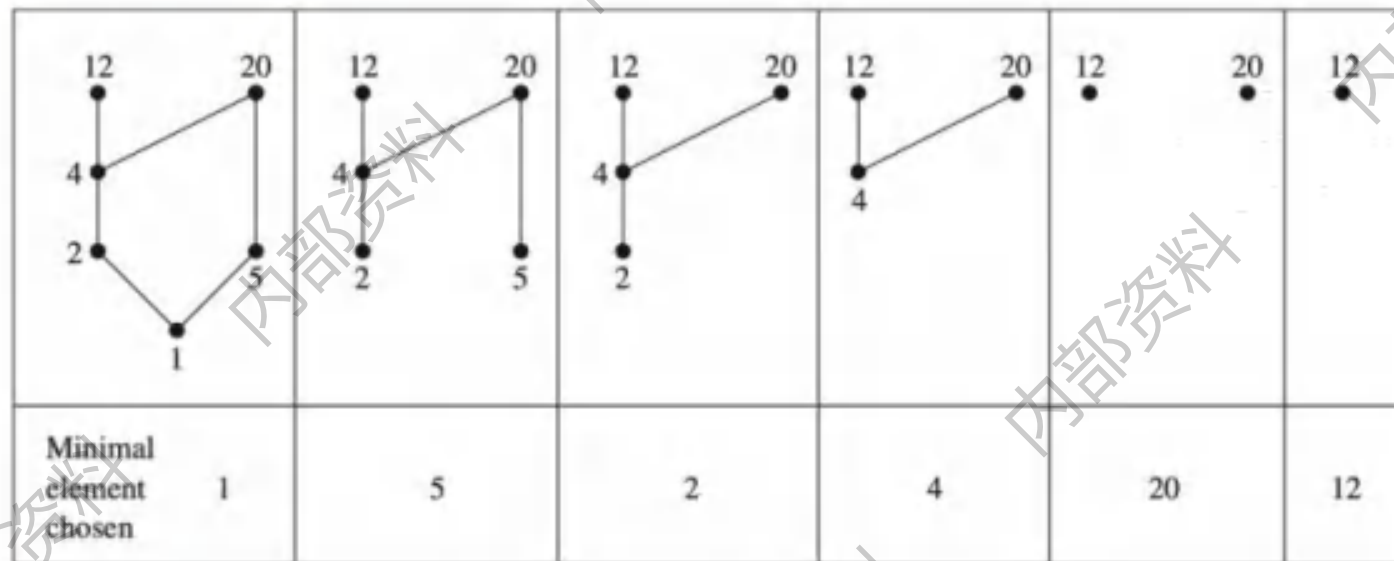
## 拓扑排序算法

- For finding a topological sorting of a finite poset  $(A, \leq)$ .
- Step1 Choose a minimal element  $a$  of  $A$ .
- Step2 Make  $a$  the next entry of SORT and replace  $A$  with  $A - \{a\}$ .
- Step3 Repeat steps 1 and 2 until  $A = \{\}$ .
- End of Algorithm

每一步从 $A$ 中删除1个极小元，直至 $A = \{\}$



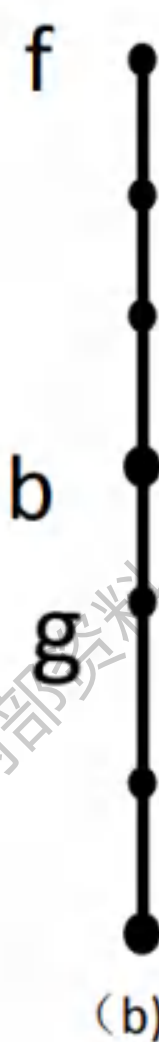
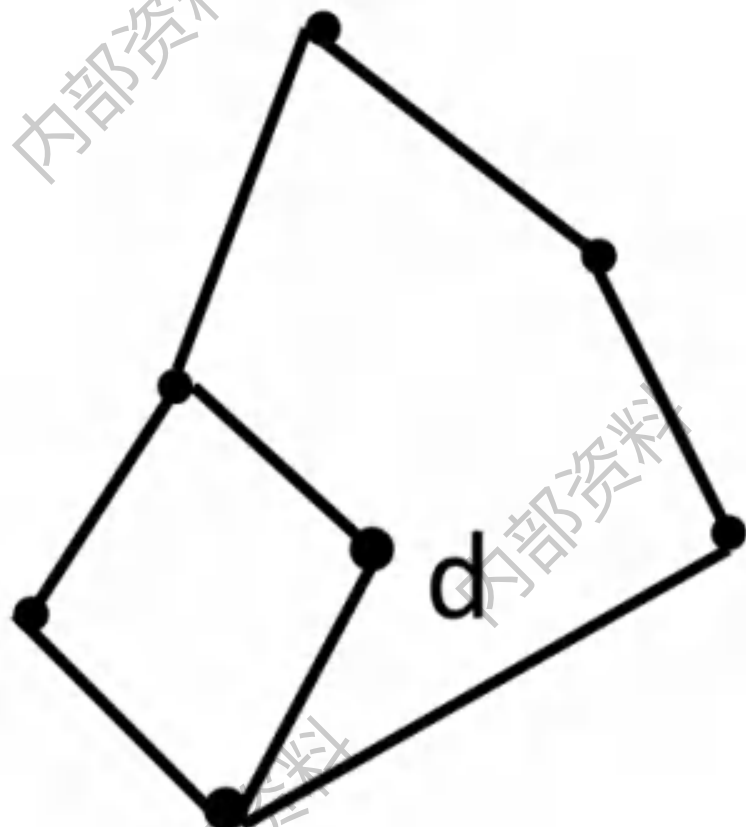
- Example 26
- Find a compatible total ordering for the poset
- $(\{1, 2, 4, 5, 12, 20\}, |)$



**FIGURE 9** A Topological Sort of  $(\{1, 2, 4, 5, 12, 20\}, |)$ .

全序关系:  $1 < 5 < 2 < 4 < 20 < 12$ .

例



# 二元关系

## 概念及表示方法

- 枚举法
- 有向图法
- 矩阵法
- 谓词公式法

## 性质

自反

对称

传递

反对称

反自反

等价关系

相容关系

偏序关系

有向图

等价类

商集

划分

简化图

相容类

最大相容类

完全覆盖

全序

哈斯图

重要元素

## 运算

复合

求逆

闭包

计算方法

运算性质

# Lattices (格)



# lattices

---

- Definition lattices sublattice (格、子格定义)
- Properties of lattices (格性质)
- Special types of lattices (特殊的格)
  - bounded distributive (有界分配格)
  - complemented (有补格)
  - modular (模格)

# Lattice (格)

- A *lattice* is a poset in which every subset  $\{a, b\}$  consisting of two elements has a least upper bound (LUB) and a greatest lower bound (GLB).
  - denote  $\text{LUB}(\{a, b\})$  by  $a \vee b$  and call it the *join* (并) of  $a$  and  $b$ .
  - denote  $\text{GLB}(\{a, b\})$  by  $a \wedge b$  and call it the *meet* (交) of  $a$  and  $b$ .
- $\text{LUB}(\{a, b\}) = a \vee b$ ,  $\text{GLB}(\{a, b\}) = a \wedge b$

特殊的偏序集！

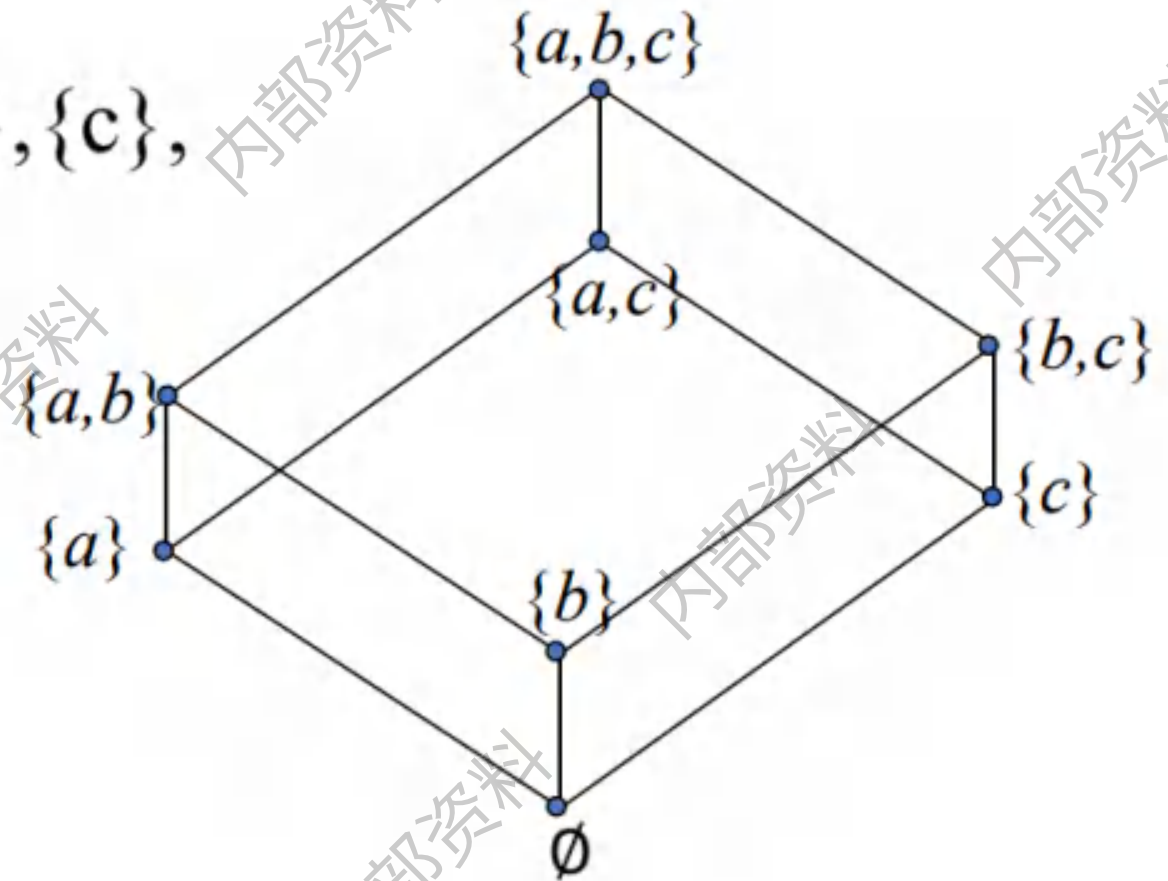
# Example

- Let  $S$  be a set and let  $L = P(S)$ .
  - As we have seen,  $\subseteq$ , containment, is a partial order on  $L$ . Let  $A$  and  $B$  belong to the poset  $(L, \subseteq)$ .
- $A \vee B$  is the set  $A \cup B$ . To see that, note that
  - $A \subseteq A \cup B$ ,  $B \subseteq A \cup B$ ,
  - and if  $A \subseteq C$  and  $B \subseteq C$ ,
  - then it follows that  $A \cup B \subseteq C$ .
- Similarly,  $A \wedge B$  is the set  $A \cap B$ .
- Thus,  $L$  is a lattice.



# Example

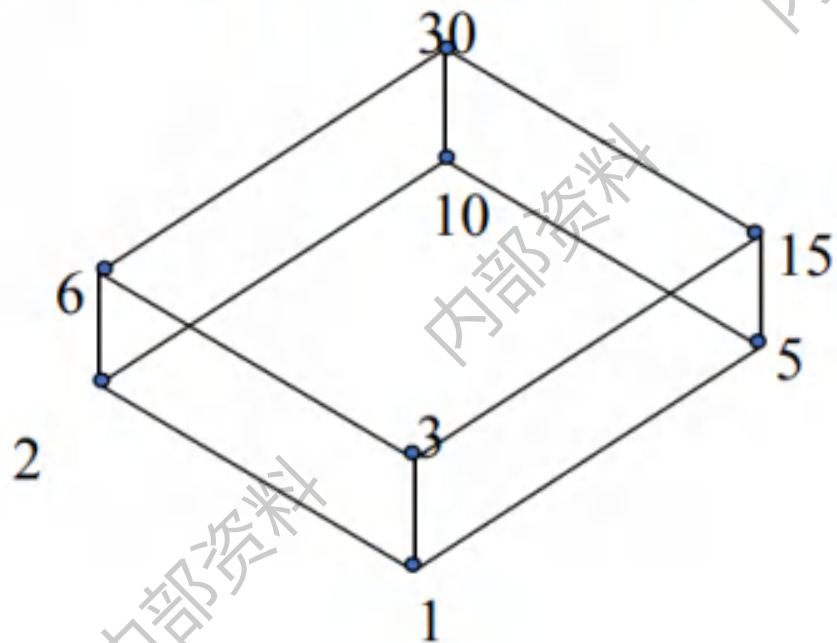
- $S = \{a, b, c\}$
- $P(S) = \{\emptyset, \{a\}, \{b\}, \{c\}, \dots, \{a, b, c\}\}$



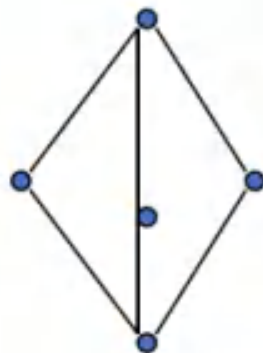


# Example

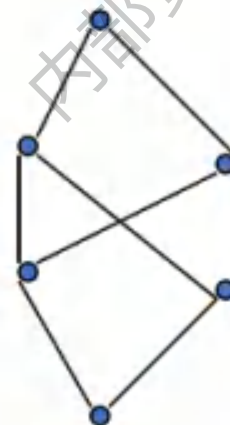
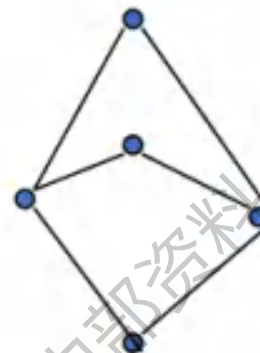
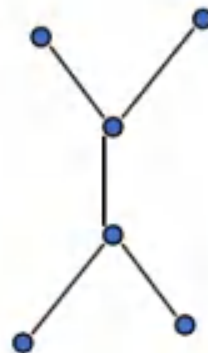
- Let  $n$  be a positive integer and  $D_n$  be the set of all positive divisors of  $n$ .
- $D_n$  表示正整数  $n$  的所有因子的集合，如下：
- $D_{30} = \{1, 2, 3, 5, 10, 15, 30\}$ 
  - $a \vee b = \text{LCM}(a, b)$
  - $a \wedge b = \text{GCD}(a, b)$
  - $(D_{30}, \vee)$  是格



例：判断是否为格



格



不是格

# Theorem-积偏序格

- If  $(L_1, \leq)$  and  $(L_2, \leq)$  are lattices, then  $(L, \leq)$  is a lattice, where
  - $L = L_1 \times L_2$
  - the partial order  $\leq$  of  $L$  is the **product partial order**.

- 格  $L_1$  的并  $\vee_1$  和交  $\wedge_1$
- 格  $L_2$  的并  $\vee_2$  和交  $\wedge_2$ ,
- 积偏序  $L = L_1 \times L_2$
- $(a_1, b_1)$  and  $(a_2, b_2) \in L$ ,
- 需要证明  $(a_1, b_1) \vee (a_2, b_2)$  and  $(a_1, b_1) \wedge (a_2, b_2) \in L$ ,
- 积偏序序偶运算如下
  - $(a_1, b_1) \vee (a_2, b_2) = (a_1 \vee_1 a_2, b_1 \vee_2 b_2)$
  - $(a_1, b_1) \wedge (a_2, b_2) = (a_1 \wedge_1 a_2, b_1 \wedge_2 b_2)$ .
- 显然  $L$  是格。



# Proof

- We denote the join and meet in  $L_1$  by  $\vee_1$  and  $\wedge_1$ , respectively, and the join and meet in  $L_2$  by  $\vee_2$  and  $\wedge_2$ , respectively. Then  $L$  is a poset.
- $L = L_1 \times L_2$  ,  $(a_1, b_1)$  and  $(a_2, b_2) \in L$ ,
- We now need to show that if  $(a_1, b_1)$  and  $(a_2, b_2) \in L$ , then  $(a_1, b_1) \vee (a_2, b_2)$  and  $(a_1, b_1) \wedge (a_2, b_2)$  exist in  $L$ . We leave it as an exercise to verify that
  - $(a_1, b_1) \vee (a_2, b_2) = (a_1 \vee_1 a_2, b_1 \vee_2 b_2)$
  - $(a_1, b_1) \wedge (a_2, b_2) = (a_1 \wedge_1 a_2, b_1 \wedge_2 b_2)$ .
- Thus  $L$  is a lattice.

# Example

$L_1$

$1_1$

$0_1$

$L_2' = L_1 \times L_1$

$(1_1, 1_1)$

$(0_1, 1_1)$

$(1_1, 0_1)$

$(0_1, 0_1)$

$L_2$

$1_2$

$0_2$

$a$

$b$

$L = L_1 \times L_2$

$(1_1, 1_2)$

$(1_1, a)$

$(1_1, b)$

$(0_1, 1_2)$

$(0_1, a)$

$(0_1, b)$

$(1_1, 0_2)$

$(0_1, 0_2)$

# Sublattice (子格)

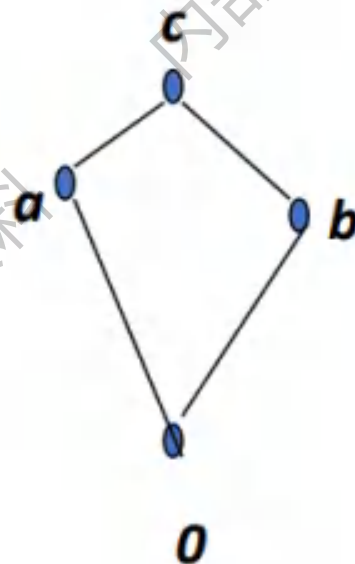
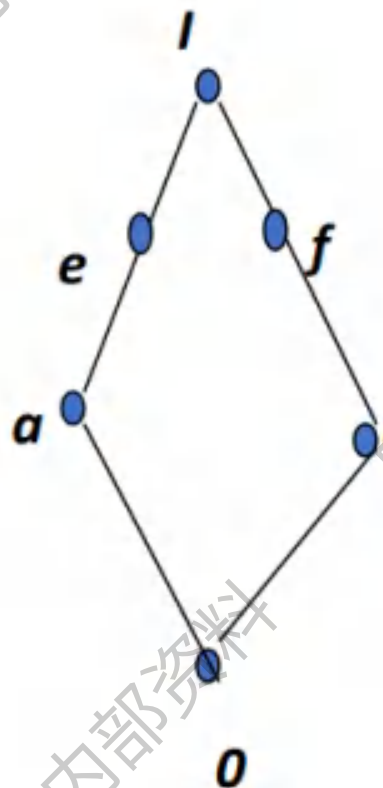
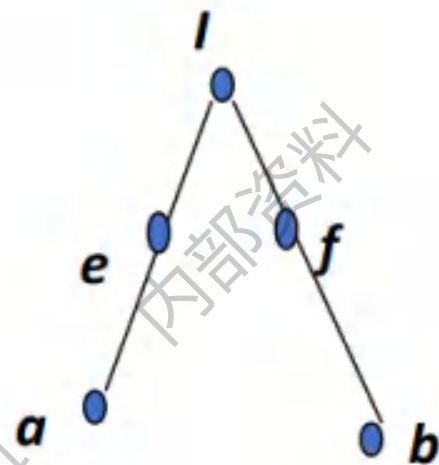
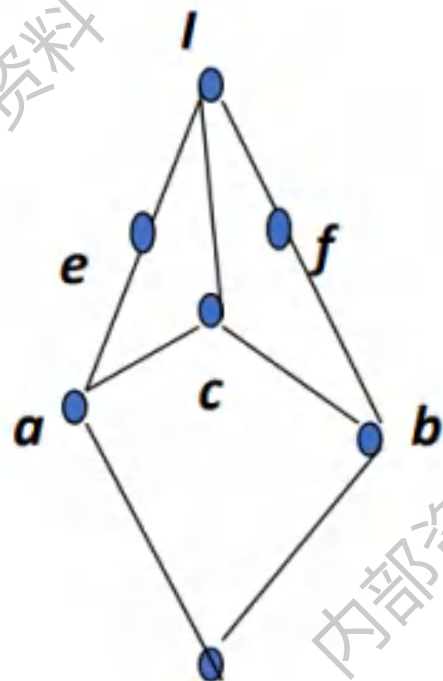
- Let  $(L, \leq)$  be a lattice. A nonempty subset  $S$  of  $L$  is called a *sublattice* of  $L$ 
  - if  $a \vee b \in S$  and  $a \wedge b \in S$  whenever  $a \in S$  and  $b \in S$ .

非空子集 $S$ ,  $S \subseteq L$ , if  $a \vee b \in S$  and  $a \wedge b \in S$ ,  $S$ 为子格

# Sublattice

1)  $(D_n, |)$  is a sublattice of  $(Z^+, |)$

2)





# Properties of Lattices

- Recall the meaning of  $a \vee b$  and  $a \wedge b$ :
  - $a \leq a \vee b$  and  $b \leq a \vee b$ ;  $a \vee b$  is an *upper bound* of  $a$  and  $b$ .
  - If  $a \leq c$  and  $b \leq c$ , then  $a \vee b \leq c$ ;  $a \vee b$  is the *least upper bound* of  $a$  and  $b$ .
  - $a \wedge b \leq a$  and  $a \wedge b \leq b$ ;  $a \wedge b$  is an *lower bound* of  $a$  and  $b$ .
  - If  $c \leq a$  and  $c \leq b$ , then  $c \leq a \wedge b$ ;  $a \wedge b$  is the *greatest lower bound* of  $a$  and  $b$ .

格是特殊的偏序，所以格具有偏序的性质

格特性：LUB，GLB唯一，

LUB:  $a \vee b$ ,      GLB:  $a \wedge b$

# 格的性质

## 格中的基本不等式和等式

$$a \leq a$$

$$a \leq b, b \leq c \Rightarrow a \leq c$$

自反, 传递

$$a \wedge b \leq a, a \wedge b \leq b$$

$$a \leq a \vee b, b \leq a \vee b$$

下界, 上界

$$a \leq b, a \leq c \Rightarrow a \leq b \wedge c$$

$$a \geq b, a \geq c \Rightarrow a \geq b \vee c$$

最大下界, 最小上界

$$a \leq b, b \leq a \Rightarrow a = b$$

## 格中的基本等价条件

$$a \leq b \Leftrightarrow a \wedge b = a \Leftrightarrow a \vee b = b$$

①

②

③

证：①  $\Rightarrow$  ②

$$\left. \begin{array}{l} a \leq a, a \leq b \Rightarrow a \leq a \wedge b \\ a \wedge b \leq a \end{array} \right\} \Rightarrow a \wedge b = a$$

②  $\Rightarrow$  ③

$$\left. \begin{array}{l} b \leq a \vee b \\ a = a \wedge b \leq b, b \leq b \Rightarrow a \vee b \leq b \end{array} \right\} \Rightarrow a \vee b = b$$

③  $\Rightarrow$  ①

$$a \leq a \vee b = b$$

# Theorem 2

---

- Let  $L$  be a lattice. Then for every  $a$  and  $b$  in  $L$ ,
  - $a \vee b = b$  if and only if  $a \leq b$ .
  - $a \wedge b = a$  if and only if  $a \leq b$ .
  - $a \wedge b = a$  if and only if  $a \vee b = b$



# Proof of $a \vee b = b$ if and only if $a \leq b$ .

---

- $\Rightarrow$  only if

- Suppose  $a \vee b = b$ , so  $a \vee b \leq b$  (reflexive)
- $a \vee b$  is an upper bound of  $a$  and  $b$ , so  $a \leq a \vee b$
- So,  $a \leq b$  (transitive)

- $\Leftarrow$  if

- Suppose  $a \leq b$
- $b \leq b$  (reflexive),  $b$  is an upper bound of  $a$  and  $b$
- So  $a \vee b \leq b$  (by definition of lub)
- But,  $b \leq a \vee b$  (by definition of lub)
- Therefore,  $a \vee b = b$  (antisymmetric)

# Theorem 3

- Let  $L$  be a lattice. Then
  - Idempotent Properties (幂等律)
    - (a)  $a \vee a = a$
    - (b)  $a \wedge a = a$
  - Commutative Properties (交换律)
    - (a)  $a \vee b = b \vee a$
    - (b)  $a \wedge b = b \wedge a$
  - Associative Properties (结合律)
    - (a)  $a \vee (b \vee c) = (a \vee b) \vee c$
    - (b)  $a \wedge (b \wedge c) = (a \wedge b) \wedge c$
  - Absorption Properties (吸收律)
    - (a)  $a \vee (a \wedge b) = a$
    - (b)  $a \wedge (a \vee b) = a$

# Proof of 4. (a)

---

- Since  $a \wedge b \leq a$  and  $a \leq a$ , we see that  $a$  is an upper bound of  $a \wedge b$  and  $a$ ; so  $a \vee (a \wedge b) \leq a$ . On the other hand, by the definition of LUB, we have  $a \leq a \vee (a \wedge b)$ , so  $a \vee (a \wedge b) = a$ .
  - Q.E.D
- It follows from property 3 that we can write  $a \vee (b \vee c)$  and  $(a \vee b) \vee c$  merely as  $a \vee b \vee c$ , and similarly for  $a \wedge b \wedge c$ .



# Theorem 4

---

- Let  $L$  be a lattice. Then for every  $a, b$ , and  $c$  in  $L$ ,
- If  $a \leq b$ , then
  - $a \vee c \leq b \vee c$ .
  - $a \wedge c \leq b \wedge c$ .
- $a \leq c$  and  $b \leq c$  if and only if  $a \vee b \leq c$ .
- $c \leq a$  and  $c \leq b$  if and only if  $c \leq a \wedge b$ .
- If  $a \leq b$  and  $c \leq d$ , then
  - $a \vee c \leq b \vee d$ .
  - $a \wedge c \leq b \wedge d$ .



内部资料

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# 几种特殊的格

# Special Types of Lattices

- A lattice  $L$  is said to be *bounded* (有界的) if it has a greatest element  $1$  and a least element  $0$

**最大元**: 格  $(L, \leq)$  中存在一个元素  $a$ , 如果任何元素  $x \in L$ , 都有  $x \leq a$ , 则  $a$  是  $L$  的最大元, 记为  $1$ .

**最小元**: 格  $(L, \leq)$  中存在一个元素  $b$ , 如果任何元素  $x \in L$ , 都有  $b \leq x$ , 则  $b$  是  $L$  的最小元, 记为  $0$ .

如果格存在一个最大元  $1$ , 一个最小元  $0$ , 则这个格是**有界格**。最大元和最小元唯一

# 性质

- If  $L$  is a bounded lattice, then for all  $a \in A$ ,
  - $-0 \leq a \leq 1$
  - $-a \vee 0 = a, \quad a \wedge 1 = a$  (同一律: )
  - $-a \wedge 0 = 0, \quad a \vee 1 = 1$  (零律: )

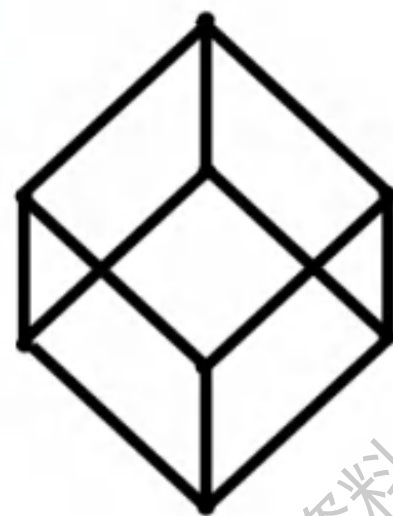
# Example 15

- The lattice  $P(S)$  of all subsets of a set  $S=\{a,b,c\}$  is bounded. Its greatest element is  $S$  and its least element is  $\emptyset$
- 幂集 $P(S)$ 构成的格，最大元 $S$ ，最小元 $\emptyset$

例：

$(\mathbb{Z}^+, |)$  ---不是有界格，最小元1，无最大元

$(\mathbb{Z}, \leq)$  ----不是有界格，无最大元和最小元





# Theorem 5

- Let  $L = \{a_1, a_2, \dots, a_n\}$  be a finite lattice. Then  $L$  is bounded.
- 所有有限格 都是有界格。
- **Proof**
  - The greatest element of  $L$  is  $a_1 \vee a_2 \vee \dots \vee a_n$ ,
  - and its least element is  $a_1 \wedge a_2 \wedge \dots \wedge a_n$

# Distributive lattice (分配格)

- A lattice is called *distributive* if for any elements  $a, b$ , and  $c$  in  $L$  we have the following distributive properties.
  - $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$
  - $a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$
- If  $L$  is not distributive, we say that  $L$  is *nondistributive*

满足上述分配律之一的格为分配格，否则不是分配格

# nondistributive lattices

- The lattice shown in Figure 6.44 are nondistributive.

$$\begin{aligned} & \blacksquare a \wedge (b \vee c) = a \\ & (a \wedge b) \vee (a \wedge c) = b \end{aligned}$$

$$\begin{aligned} & \blacksquare a \wedge (b \vee c) = a \\ & (a \wedge b) \vee (a \wedge c) = 0 \end{aligned}$$

都不是分配格

## Theorem 6

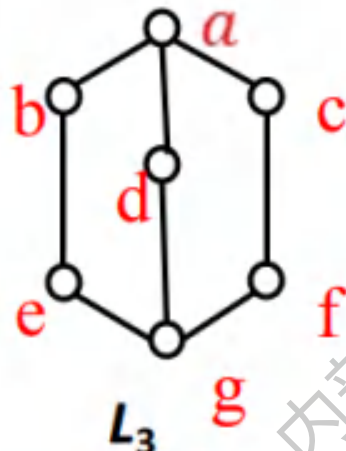
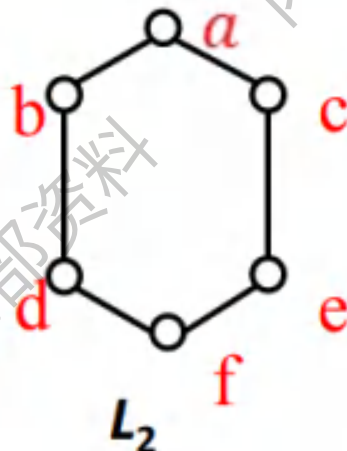
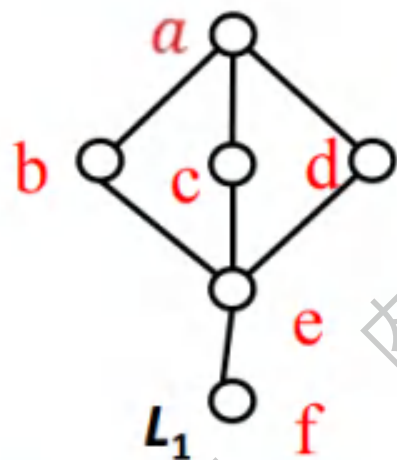
- A lattice  $L$  is nondistributive if and only if it contains a sublattice that is isomorphic to one of the above two lattices.
- Proof:
  - omitted

如果一个格的子格与上述2个特殊的格任何一个同构的，则不是分配格



例：判断图中的格是否为分配格。

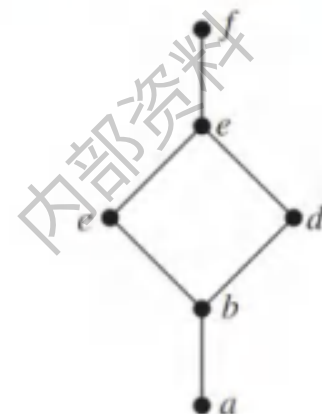
解：都不是分配格  
 $\{a, b, c, d, e\}$  是  $L_1$  的子格，同构于钻石格；  
 $\{a, b, c, e, f\}$  是  $L_2$  的子格，同构于五角格；  
 $\{a, b, c, d, g\}$  是  $L_3$  的子格，同构于钻石格。



- 小于5个元素的格都是分配格
- 任何一条链都是分配格

# Complement (补元)

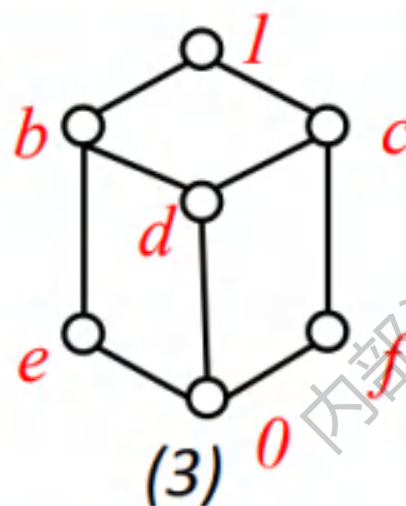
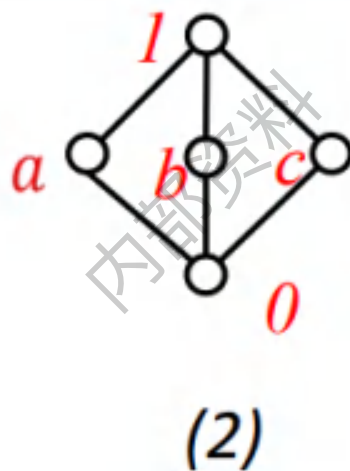
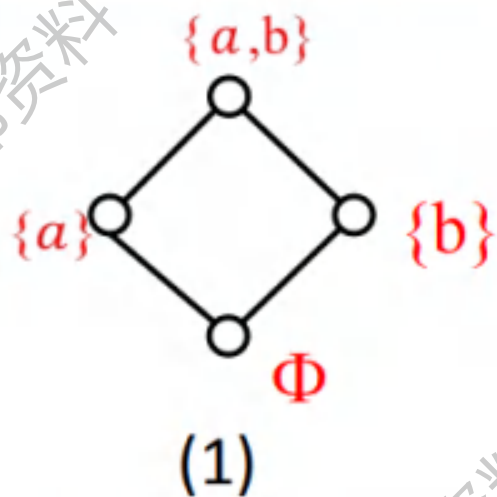
- Let  $L$  be a bounded lattice with greatest element  $1$  and least element  $0$ , and let  $a \in L$ . An element  $a' \in L$  is called a **complement of  $a$**  if
  - $a \vee a' = 1$  and  $a \wedge a' = 0$
- Note that
  - $0' = 1$  and  $1' = 0$ .
- An element  $a$  in a lattice need not have a complement, and it may have more than one complement.
- 有的元素可能没有补元，有的可能多个补元



# Complemented-有补格

A lattice is called *complemented* if it is bounded and if every element in  $L$  has a complement.

一个有界格中，如果每个元素都有补元，则称之为有补格。



- 上述有界格中，(1)和(2)是有补格。其中(2)中的 $a, b, c$ 的补元不唯一。(3)(4)不是有补格。

什么样的有补格中元素的补元唯一呢？——有补分配格。



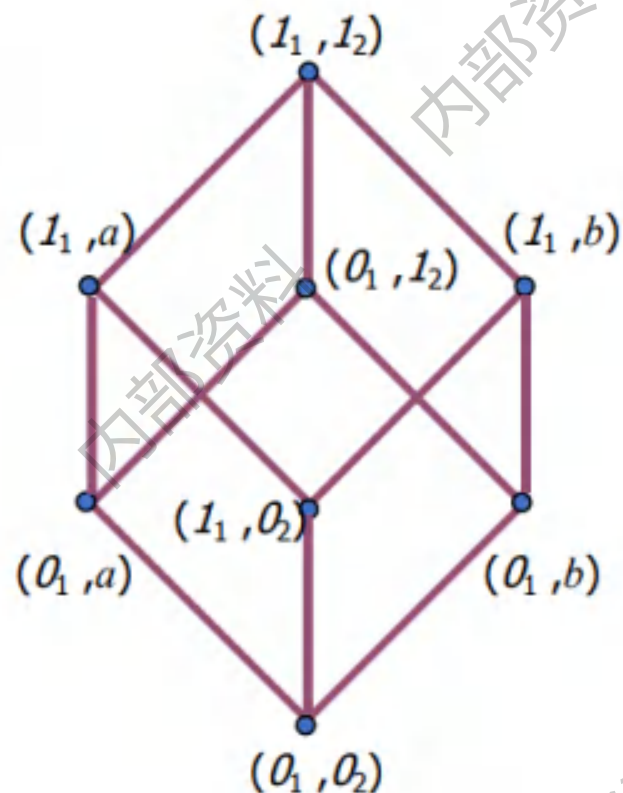
# Theorem 7

## 有补分配格

- Let  $L$  be a bounded distributive lattice. If a complement exists, it is unique.

有补分配格，如果有补元存在，则补元唯一

例  $(P(S), \subseteq)$  是有补分配格





# Proof 反证法

- Let  $a'$  and  $a''$  be complements of the element  $a \in L$ .  
Then

$$-a \vee a' = 1, \quad a \wedge a' = 0$$

$$-a \vee a'' = 1, \quad a \wedge a'' = 0.$$

- Using the distributive laws, we obtain

$$-a' = a' \vee 0 = a' \vee (a \wedge a'') = (a' \vee a) \wedge (a' \vee a'') = 1 \wedge (a' \vee a'') = a' \vee a''$$

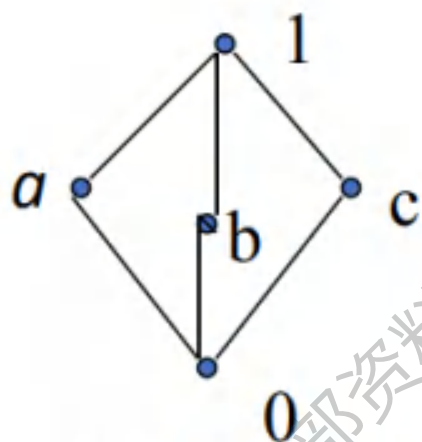
- Also

$$-a'' = a'' \vee 0 = a'' \vee (a \wedge a') = (a'' \vee a) \wedge (a'' \vee a') = 1 \wedge (a'' \vee a') = a' \vee a''$$

- Hence  $a' = a''$ .

# Modular 模格

定义：若  $\forall a, b, c \in (L, \leq)$ ,  
 $a \leq c \Rightarrow a \vee (b \wedge c) = (a \vee b) \wedge c$   
 称  $(L, \leq)$  是模格。



例：钻石格是模格

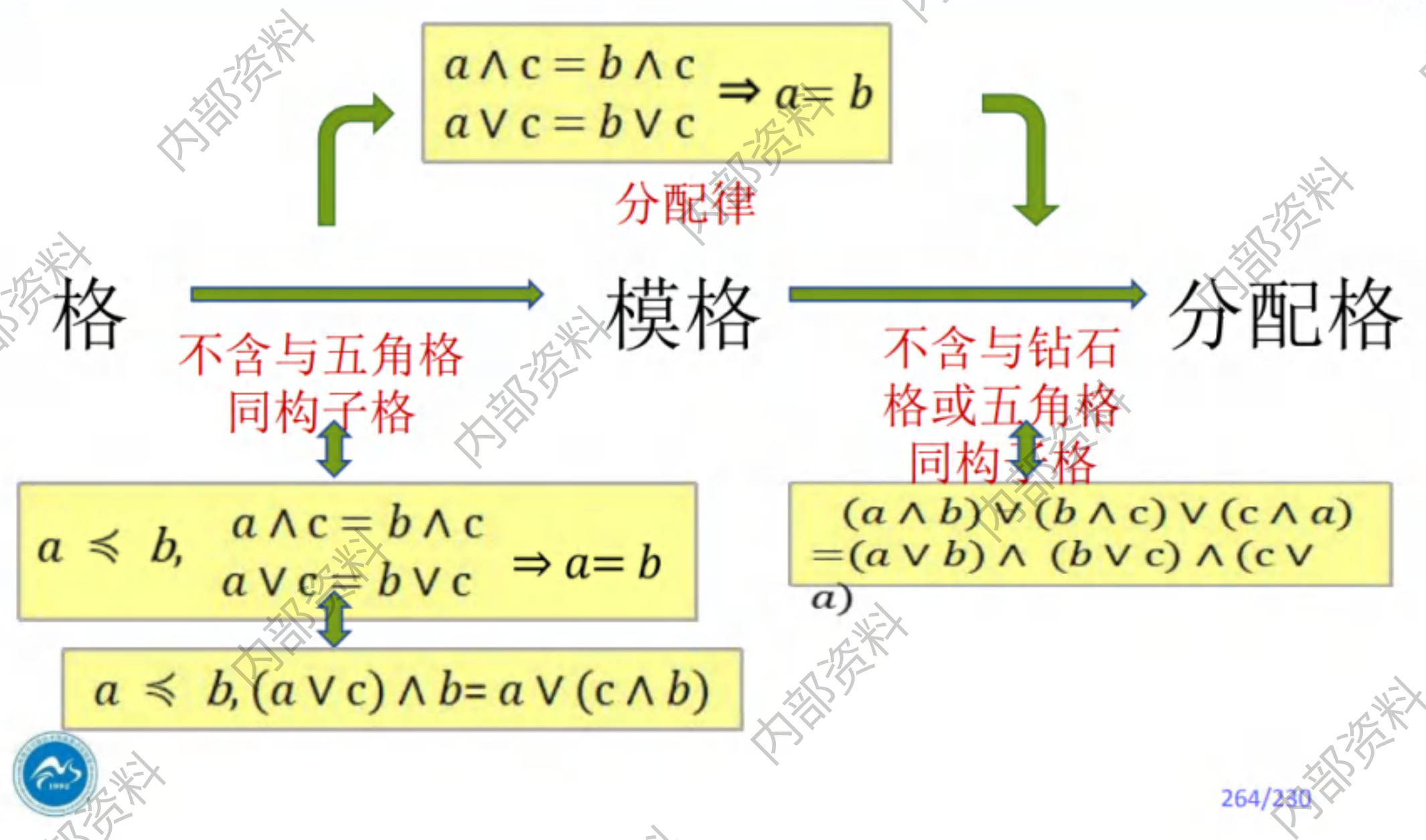
$$a \vee (b \wedge c) = a \vee 0 = a \quad (a \vee b) \wedge c = a \vee 0 = a$$

五角格不是模格

$$a \leq c, (a \vee b) \wedge c \leq a \vee (b \wedge c)$$

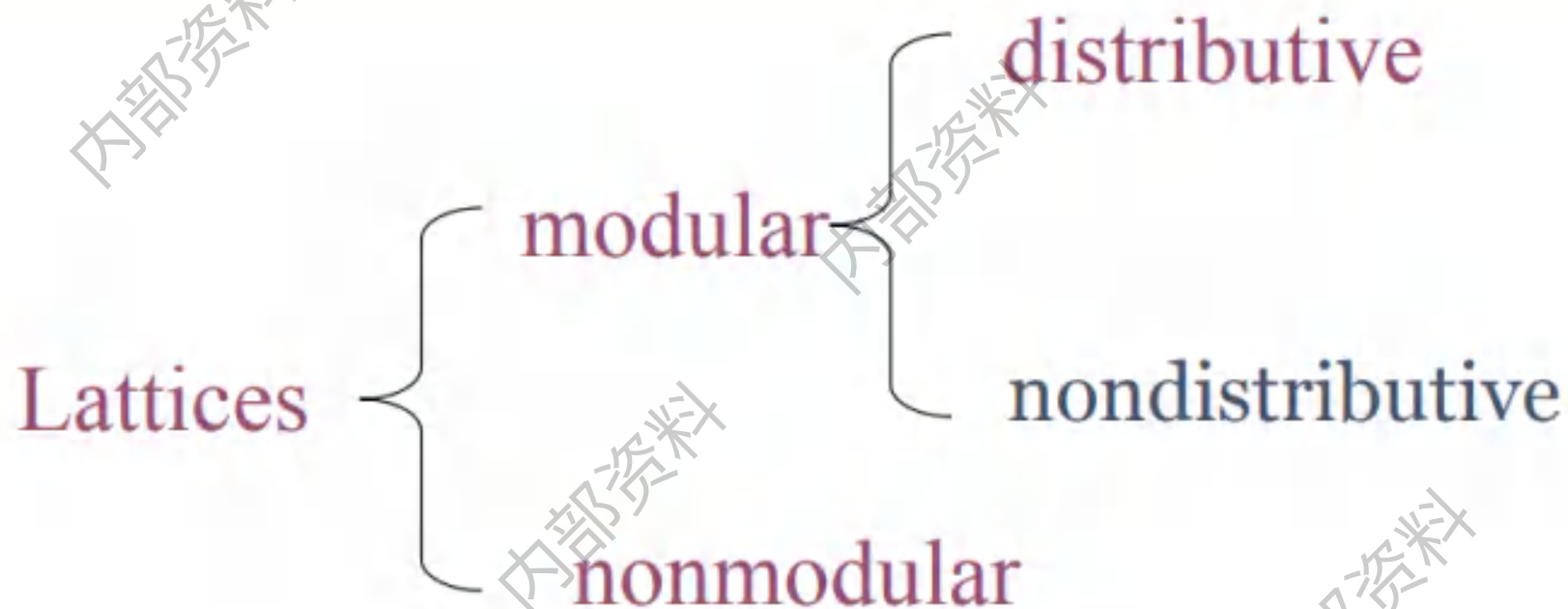
$L$  是模格，当且仅当不含与五角格同构的子格

# 模格、分配格之间关系

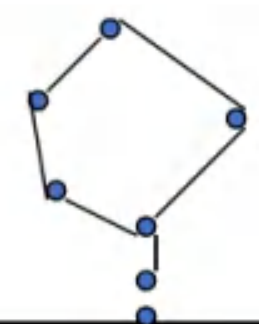
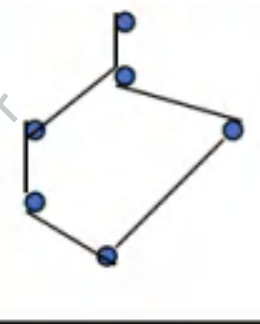
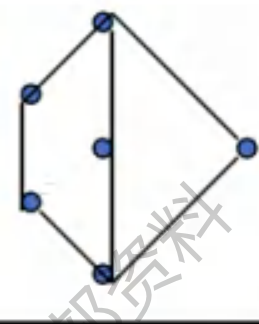
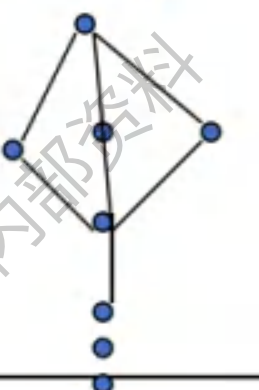
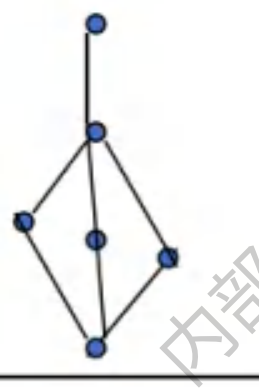
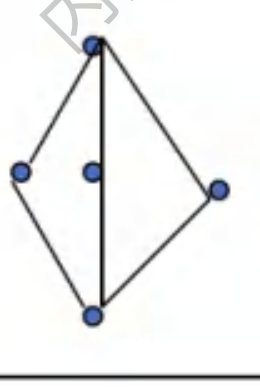
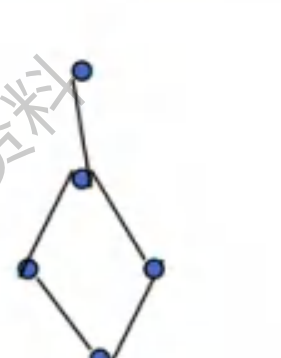
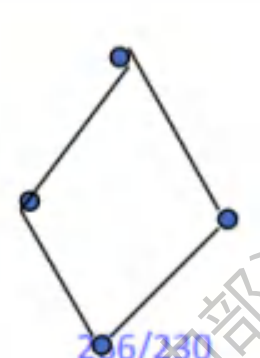


# lattices

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|     |      | 非有界格                                                                                | 有界格                                                                                   |                                                                                       |
|-----|------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|     |      |                                                                                     | 非有补格                                                                                  | 有补格                                                                                   |
| 非模格 |      |    |    |    |
| 模格  | 非分配格 |   |   |   |
|     | 分配格  |  |  |  |

# 作业

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- § 9.5 – 16, 56, 60, 64
- § 9.6 – 12, 28, 36, 44

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